

RPSC · RAS & RTS PRELIMINARY EXAMINATION 2026

Reasoning and Mental Ability

520 practice questions across 10 chapters, each chapter opening with a must-know fact sheet. Answers and explanations are at the back, so you can work through the paper first.

RM01	Statement & Assumptions / Arguments / Conclusions / Courses of Action	60
RM02	Analytical and Critical Reasoning; Syllogism	50
RM03	Number and Letter Series	50
RM04	Coding-Decoding	45
RM05	Blood Relations	40
RM06	Direction Sense	35
RM07	Ranking, Seating Arrangement and Logical Venn Diagrams	55
RM08	Shapes, Figure Counting, Mirror and Water Images, Embedded Figures	45
RM09	Numeracy I: Ratio, Proportion, Partnership, Percentage, Profit & Loss, Interest	70
RM10	Numeracy II: Mensuration, Averages, Statistics, Data Interpretation, Probability	70

520 questions · **267** fact-sheet points · **72** tagged as previously asked
Paper pattern: 150 questions · 200 marks · 3 hours · 1/3 negative marking

RM01

Statement & Assumptions / Arguments / Conclusions / Courses of Action

60 questions · 27 fact-sheet points

Must-know facts

1. ASSUMPTION = something the speaker/writer has TAKEN FOR GRANTED while making the statement. It is unstated but necessary for the statement to make sense.
2. An assumption is never something merely desirable, merely possible, or a future effect of the statement - those are conclusions or hopes, not assumptions.
3. Reject any assumption containing an extreme word - only, all, never, always, none, every, cannot - unless the statement itself is that extreme.
4. An assumption restating the statement itself is NOT implicit (no new taken-for-granted idea is added).
5. In advertisements, the implicit assumptions are: (i) the product has the advertised quality, (ii) people want that quality, (iii) the ad will influence buyers.
6. In notices/instructions/appeals, it is implicit that (i) the reader can read and understand it and (ii) people are likely to comply.
7. In 'In spite of X, Y happened', it is implicit that X normally prevents Y.
8. ARGUMENT test = RELEVANCE + SIGNIFICANCE. A strong argument must be directly related to the question AND deal with a major, not a trivial, aspect of it.
9. Weak-argument markers: appeal to tradition, mere imitation of other countries, personal ridicule, ambiguity, sweeping generalisation, circular restatement of the proposal, and 'it will not solve the problem completely'.
10. An argument saying 'a measure is useless because it does not remove the whole problem' is WEAK - a partial remedy can still be justified.
11. 'Yes/No' does not decide strength: a No-argument that gives a real, weighty objection is strong; a Yes-argument that gives a trivial gain is weak.
12. An argument based on an established fact or on a serious, foreseeable consequence is strong; one based on an unverifiable prediction or on emotion is weak.
13. CONCLUSION = what MUST be true given the statement. Test: could the statement be true and the conclusion false? If yes, it does not follow.
14. Percentages never give absolute numbers. A higher percentage in A than in B does NOT prove A has more persons than B (populations may differ).
15. 'All P are Q' is NOT convertible to 'All Q are P', and does not mean 'only P are Q' - so nothing follows about a non-P.
16. 'Some A are B' always yields 'Some B are A'. 'No A is B' always yields 'No B is A'. 'All A are B' yields only 'Some B are A'.
17. EITHER I OR II follows when neither follows singly, but the two conclusions are contradictory/complementary and together exhaust all possibilities.
18. NEITHER follows when both conclusions add outside information (causes, motives, comparisons, future behaviour) that the statement does not supply.
19. COURSE OF ACTION test: it must (i) address the problem stated, (ii) be practicable and within the authority of the agency, and (iii) be proportionate.
20. Reject a course of action that is extreme (ban everything, close it down, dismiss everybody), that punishes the victim, or that is a mere platitude ('people should be careful').
21. Both courses follow when one is an immediate remedy and the other is a preventive/long-term

measure aimed at the same problem.

22. In a course-of-action item, the statement's problem must actually exist - do not choose an action that assumes a different problem.
23. 'Should be investigated / responsibility should be fixed / relief should be given / awareness should be created through media' are the usual strong actions.
24. Do not import real-world knowledge into assumption and conclusion items - work only from the words given.
25. Standard option order used by RPSC/SSC: I only, II only, Either or Neither, Both - read the option list before keying, since papers reshuffle it.
26. Statement-Assumption, Statement-Argument, Statement-Conclusion and Statement-Course of Action have each appeared in every RAS Prelims paper - about 4-6 marks together.
27. Time discipline: these items need no calculation; 30-40 seconds each is enough. Read the statement twice, the options once.

1. Statement: The government has decided to increase the price of urea with effect from next month.

Assumptions:

- I. The government has the authority to fix the price of urea.
- II. Farmers will stop using urea from next month.

Which of the assumptions is/are implicit in the statement? RAS PRE 2024

- (A) Only assumption I is implicit
- (B) Only assumption II is implicit
- (C) Neither I nor II is implicit
- (D) Both I and II are implicit

2. Statement: 'Use Ujala detergent for whiter clothes.' - an advertisement.

Assumptions:

- I. People want their clothes to look whiter.
- II. Ujala is the only detergent available in the market.

Which of the assumptions is/are implicit in the statement?

- (A) Only assumption I is implicit
- (B) Only assumption II is implicit
- (C) Neither I nor II is implicit
- (D) Both I and II are implicit

3. Statement: 'In case of any difficulty regarding this scheme, please contact our toll-free helpline.' - instruction printed on a scheme pamphlet.

Assumptions:

- I. Some beneficiaries may face difficulty in understanding the scheme.
- II. The helpline is able to resolve the difficulties of beneficiaries.

Which of the assumptions is/are implicit in the statement?

- (A) Only assumption I is implicit
- (B) Only assumption II is implicit
- (C) Neither I nor II is implicit
- (D) Both I and II are implicit

4. Statement: 'All employees must reach the office by 9.30 a.m., failing which their casual leave will be deducted.' - office notice.

Assumptions:

- I. Some employees are likely to come late unless a penalty is prescribed.
- II. Deduction of casual leave is likely to deter late-coming.

Which of the assumptions is/are implicit in the statement?

- (A) Only assumption I is implicit
- (B) Only assumption II is implicit
- (C) Neither I nor II is implicit
- (D) Both I and II are implicit

5. Statement: Ravi says to Sunil, 'You should read the editorial page of the newspaper daily.'

Assumptions:

- I. Sunil does not read any newspaper.
- II. The editorial page carries material worth reading every day.

Which of the assumptions is/are implicit in the statement?

- (A) Only assumption I is implicit
- (B) Only assumption II is implicit
- (C) Neither I nor II is implicit
- (D) Both I and II are implicit

6. Statement: To improve its board results, the school has decided to hold extra classes for weak students on Sundays.

Assumptions:

- I. Weak students are likely to attend the classes held on Sundays.
- II. Additional teaching can improve the performance of weak students.

Which of the assumptions is/are implicit in the statement? RAS PRE 2021

- (A) Only assumption I is implicit
- (B) Only assumption II is implicit
- (C) Neither I nor II is implicit
- (D) Both I and II are implicit

7. Statement: 'Buy from our winter collection now - 40% discount till 31 December.' - a shop's notice.

Assumptions:

- I. A discount is likely to attract customers.
- II. Customers normally postpone their purchases to the last day.

Which of the assumptions is/are implicit in the statement?

- (A) Only assumption I is implicit
- (B) Only assumption II is implicit
- (C) Neither I nor II is implicit
- (D) Both I and II are implicit

8. Statement: The municipal corporation has advised citizens to boil drinking water during the monsoon months.

Assumptions:

- I. Drinking water is liable to get contaminated during the monsoon.
- II. Boiling destroys the harmful germs present in water.

Which of the assumptions is/are implicit in the statement?

- (A) Only assumption I is implicit
- (B) Only assumption II is implicit
- (C) Neither I nor II is implicit
- (D) Both I and II are implicit

9. Statement: The Chief Minister will inaugurate the new expressway on Sunday.

Assumptions:

- I. The construction of the expressway has been completed.
- II. The Chief Minister is fond of travelling by road.

Which of the assumptions is/are implicit in the statement?

- (A) Only assumption I is implicit
- (B) Only assumption II is implicit
- (C) Neither I nor II is implicit
- (D) Both I and II are implicit

10. Statement: Anil scored 92 per cent marks in the Class XII board examination.

Assumptions:

- I. Anil is the topper of his school.
- II. No other student of his school scored above 92 per cent.

Which of the assumptions is/are implicit in the statement?

- (A) Only assumption I is implicit
- (B) Only assumption II is implicit
- (C) Neither I nor II is implicit
- (D) Both I and II are implicit

11. Statement: The company has announced a bonus equal to one month's salary for all its employees this year.

Assumptions:

- I. The employees had gone on strike demanding a bonus.
- II. The company will not be able to pay a bonus next year.

Which of the assumptions is/are implicit in the statement?

- (A) Only assumption I is implicit
- (B) Only assumption II is implicit
- (C) Neither I nor II is implicit
- (D) Both I and II are implicit

12. Statement: In spite of the heavy rainfall yesterday, the cricket match was completed on schedule.

Assumptions:

- I. Heavy rainfall never affects the completion of a cricket match.
- II. Heavy rainfall usually disrupts the play in a cricket match.

Which of the assumptions is/are implicit in the statement?

- (A) Only assumption I is implicit
- (B) Only assumption II is implicit
- (C) Neither I nor II is implicit
- (D) Both I and II are implicit

13. Statement: The invigilator instructed the candidates to reach the examination hall at least thirty minutes before the paper begins.

Assumptions:

- I. Candidates always reach the examination hall late.
- II. The candidates are likely to follow the instruction.

Which of the assumptions is/are implicit in the statement?

- (A) Only assumption I is implicit
- (B) Only assumption II is implicit
- (C) Neither I nor II is implicit
- (D) Both I and II are implicit

14. Statement: Meena told her colleague, 'I could not attend yesterday's meeting as I was unwell.'

Assumptions:

- I. Meena remains unwell very often.
- II. Meena was expected to attend yesterday's meeting.

Which of the assumptions is/are implicit in the statement?

- (A) Only assumption I is implicit
- (B) Only assumption II is implicit
- (C) Neither I nor II is implicit
- (D) Both I and II are implicit

15. Statement: The state government has decided to make the wearing of helmets compulsory for pillion riders on two-wheelers.

Assumptions:

- I. Helmets reduce the risk of head injury to pillion riders.
- II. The rule can be enforced by the traffic police.

Which of the assumptions is/are implicit in the statement? RAS PRE 2023

- (A) Only assumption I is implicit
- (B) Only assumption II is implicit
- (C) Neither I nor II is implicit
- (D) Both I and II are implicit

16. Statement: Should private cars be barred from the central business district of big cities during peak hours?

Arguments:

- I. Yes, it would substantially reduce traffic congestion and air pollution in the most crowded part of the city.
- II. No, car owners have spent a lot of money on buying their vehicles.

Which of the arguments is/are strong? RAS PRE 2024

- (A) Only argument I is strong
- (B) Only argument II is strong
- (C) Neither I nor II is strong
- (D) Both I and II are strong

17. Statement: Should education up to Class XII be made completely free in all government schools?

Arguments:

- I. Yes, it would enable children of poor families to complete their schooling.
- II. No, things that are given free are never valued by people.

Which of the arguments is/are strong?

- (A) Only argument I is strong
- (B) Only argument II is strong
- (C) Neither I nor II is strong
- (D) Both I and II are strong

18. Statement: Should the minimum age for obtaining a driving licence for a motor car be raised to 20 years?

Arguments:

- I. Yes, young people are always careless.
- II. No, many people need to drive for work or study soon after school; road safety is better secured by strict testing and enforcement than by raising the age.

Which of the arguments is/are strong?

- (A) Only argument I is strong
- (B) Only argument II is strong
- (C) Neither I nor II is strong
- (D) Both I and II are strong

19. Statement: Should coaching institutes preparing students for competitive examinations be banned in the country?

Arguments:

I. Yes, all successful candidates are products of coaching institutes.

II. No, a ban would only push coaching underground and deprive small-town students of guidance; the real need is to regulate fees, hours and advertising.

Which of the arguments is/are strong?

- (A) Only argument I is strong
- (B) Only argument II is strong
- (C) Neither I nor II is strong
- (D) Both I and II are strong

20. Statement: Should there be a complete ban on the manufacture and sale of firecrackers in India?

Arguments:

I. Yes, firecrackers cause severe air and noise pollution and a large number of burn injuries every year.

II. No, lakhs of workers depend on the firecracker industry for their livelihood, and regulated green crackers can cut pollution substantially.

Which of the arguments is/are strong?

- (A) Only argument I is strong
- (B) Only argument II is strong
- (C) Neither I nor II is strong
- (D) Both I and II are strong

21. Statement: Should the age of superannuation of government employees be raised from 60 to 62 years?

Arguments:

I. Yes, experienced employees would remain available for two more years and pension liability would be deferred.

II. No, it would postpone fresh recruitment and reduce job opportunities for unemployed youth.

Which of the arguments is/are strong?

- (A) Only argument I is strong
- (B) Only argument II is strong
- (C) Neither I nor II is strong
- (D) Both I and II are strong

22. Statement: Should the study of Sanskrit be made compulsory in all schools of the country?

Arguments:

I. Yes, Sanskrit is the oldest language of the world.

II. No, students of today are not interested in old languages.

Which of the arguments is/are strong?

- (A) Only argument I is strong
- (B) Only argument II is strong
- (C) Neither I nor II is strong
- (D) Both I and II are strong

23. Statement: Should the examination system in schools be abolished altogether?

Arguments:

I. Yes, several other countries follow a different system.

II. No, examinations have been a part of our education system since ancient times.

Which of the arguments is/are strong?

- (A) Only argument I is strong
- (B) Only argument II is strong
- (C) Neither I nor II is strong
- (D) Both I and II are strong

24. Statement: Should all beggars found in big cities be sent back to their native villages?

Arguments:

I. Yes, this will reduce the population of big cities.

II. No, they will come back to the cities again.

Which of the arguments is/are strong? **RAS PRE 2021**

- (A) Only argument I is strong
- (B) Only argument II is strong
- (C) Neither I nor II is strong
- (D) Both I and II are strong

25. Statement: Should the subsidy on domestic LPG cylinders be withdrawn for high-income households?

Arguments:

I. Yes, targeting the subsidy at poor households releases scarce public resources for essential welfare expenditure.

II. No, the government should never interfere with the market.

Which of the arguments is/are strong? **RAS PRE 2021**

- (A) Only argument I is strong
- (B) Only argument II is strong
- (C) Neither I nor II is strong
- (D) Both I and II are strong

26. Statement: Should mobile phones be prohibited inside examination halls?

Arguments:

I. Yes, a mobile phone can be used to receive answers and thus destroys the fairness of the examination.

II. No, prohibiting mobile phones will not put an end to every form of cheating.

Which of the arguments is/are strong?

- (A) Only argument I is strong
- (B) Only argument II is strong
- (C) Neither I nor II is strong
- (D) Both I and II are strong

27. Statement: Should all the major rivers of India be interlinked through canals?

Arguments:

I. Yes, it will solve the water problem of the country forever.

II. No, large-scale interlinking would submerge forests and displace people, and its ecological consequences have not been fully assessed.

Which of the arguments is/are strong?

- (A) Only argument I is strong
- (B) Only argument II is strong
- (C) Neither I nor II is strong
- (D) Both I and II are strong

28. Statement: Should all public recruitment examinations in the country be conducted only in the online mode?

Arguments:

I. Yes, online examinations are the fashion of the day.

II. No, a large number of rural candidates lack reliable internet connectivity and devices, and would effectively be excluded.

Which of the arguments is/are strong?

- (A) Only argument I is strong
- (B) Only argument II is strong
- (C) Neither I nor II is strong
- (D) Both I and II are strong

29. Statement: Should agricultural income above a high threshold be brought within the ambit of income tax?

Arguments:

- I. Yes, it would check the routing of non-farm income through the agricultural exemption and would widen the tax base.
- II. No, the income of the vast majority of farmers is low and uncertain, and the cost of assessing farm income may exceed the revenue collected.

Which of the arguments is/are strong? **HARD**

- (A) Only argument I is strong
- (B) Only argument II is strong
- (C) Neither I nor II is strong
- (D) Both I and II are strong

30. Statement: Should the use of plastic carry bags be completely banned in the state?

Arguments:

- I. Yes, plastic bags choke drains, are swallowed by stray cattle and take centuries to decompose.
- II. No, unless affordable alternatives are made widely available, the ban will mainly burden small shopkeepers and poor consumers.

Which of the arguments is/are strong?

- (A) Only argument I is strong
- (B) Only argument II is strong
- (C) Neither I nor II is strong
- (D) Both I and II are strong

31. Statement: The literacy rate of District A is 78 per cent, while that of District B is 62 per cent.

Conclusions:

- I. District A has a larger number of literate persons than District B.
- II. Some persons in District B are illiterate.

Which of the conclusions logically follows/follow from the statement? **RAS PRE 2023**

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

32. Statement: All trains running between Jaipur and Delhi were cancelled today because of dense fog.

Conclusions:

- I. No train left Jaipur for Delhi today.
- II. Dense fog can disrupt railway operations.

Which of the conclusions logically follows/follow from the statement?

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

33. Statement: Every student who scored above 90 marks was awarded a scholarship. Ramesh scored 85 marks.

Conclusions:

- I. Ramesh scored less than 90 marks.
- II. Ramesh was not awarded a scholarship.

Which of the conclusions logically follows/follow from the statement? **RAS PRE 2023**

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

34. Statement: The price of petrol has been raised for the fourth time in this calendar year.

Conclusions:

- I. The price of petrol has been raised more than once this year.
- II. People will shift from private vehicles to public transport.

Which of the conclusions logically follows/follow from the statement?

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

35. Statement: Of the 500 employees of a company, 300 are graduates, and 150 of these graduates are women.

Conclusions:

- I. Most of the women employees of the company are graduates.
- II. At least 150 employees of the company are women.

Which of the conclusions logically follows/follow from the statement?

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

36. Statement: The Food Safety Commissioner has warned that anyone found selling adulterated milk will be prosecuted.

Conclusions:

- I. Nobody will now sell adulterated milk.
- II. Selling adulterated milk is punishable under the law.

Which of the conclusions logically follows/follow from the statement?

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

37. Statement: A large number of tourists visited Rajasthan during the winter months this year.

Conclusions:

- I. Tourists do not visit Rajasthan during the summer.
- II. Rajasthan is the most visited state in India.

Which of the conclusions logically follows/follow from the statement?

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

38. Statement: The sale of electric two-wheelers in the state doubled last year.

Conclusions:

- I. Electric two-wheelers are cheaper than petrol two-wheelers.
- II. The sale of petrol two-wheelers fell last year.

Which of the conclusions logically follows/follow from the statement?

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

39. Statement: In a class test the highest marks were obtained by exactly one student, and it is known that this student is either Anil or Bharat.

Conclusions:

I. Anil obtained the highest marks.

II. Bharat obtained the highest marks.

Which of the conclusions logically follows/follow from the statement?

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Either I or II follows
- (D) Neither I nor II follows

40. Statement: The school will remain closed on Monday and Tuesday on account of the festival.

Conclusions:

I. The school will not function on Monday.

II. The school observes holidays on the occasion of festivals.

Which of the conclusions logically follows/follow from the statement?

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

41. In a survey of 200 households it was found that 120 own a two-wheeler, 60 own a car and 20 own both. Which one of the following conclusions is definitely correct?

- (A) 80 households own only a two-wheeler
- (B) 40 households own neither a two-wheeler nor a car
- (C) 50 households own only a car
- (D) 30 households own both a two-wheeler and a car

42. Statement: Every member of the drama club is also a member of the music club. Some members of the music club are members of the debating club.

Which one of the following conclusions definitely follows?

- (A) Some members of the drama club are members of the debating club
- (B) All members of the music club are members of the drama club
- (C) Some members of the music club are members of the drama club
- (D) No member of the drama club is a member of the debating club

43. Statement: The morning bus reaches Jaipur on time only if it leaves Jodhpur before 6 a.m. Today the morning bus reached Jaipur on time.

Which one of the following conclusions definitely follows? HARD

- (A) The bus left Jodhpur before 6 a.m. today
- (B) The bus left Jodhpur after 6 a.m. today
- (C) The bus leaves Jodhpur before 6 a.m. every day
- (D) Every bus that leaves Jodhpur before 6 a.m. reaches Jaipur on time

44. Statement: If it rains, the match will be cancelled. The match was not cancelled.

Which one of the following conclusions definitely follows?

- (A) It rained
- (B) It did not rain
- (C) The match was played in the rain
- (D) Nothing can be said about the rain

45. Statement: Only those candidates who have passed the written test are called for the interview. Mohit has been called for the interview.

Which one of the following conclusions definitely follows?

- (A) Mohit will be selected for the post
- (B) Mohit obtained the highest marks in the written test
- (C) Every candidate who passed the written test has been called for the interview
- (D) Mohit has passed the written test

46. Statement: Several passengers were injured when an overcrowded bus overturned on a state highway.

Courses of action:

I. The transport authority should immediately check overloading of buses on this route.

II. All bus services on this route should be stopped permanently.

Which of the courses of action logically follows/follow? RAS PRE 2024

- (A) Only I follows
- (B) Only II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

47. Statement: A large number of students of a school failed in mathematics in the annual examination.

Courses of action:

I. The school should hold remedial classes in mathematics and review its method of teaching the subject.

II. The school should expel all the students who failed.

Which of the courses of action logically follows/follow?

- (A) Only I follows
- (B) Only II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

48. Statement: The number of dengue cases reported in the city has risen sharply in the last two weeks.

Courses of action:

I. The municipal body should carry out fogging and remove stagnant water in the affected localities.

II. The health department should advise residents through the media to use mosquito nets and to report fever early.

Which of the courses of action logically follows/follow?

- (A) Only I follows
- (B) Only II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

49. Statement: In several villages of a district the drinking water has been found to contain fluoride above the permissible limit.

Courses of action:

I. Defluoridation units should be installed and alternative safe sources of drinking water arranged in the affected villages.

II. The health department should screen the residents for fluorosis and treat those affected.

Which of the courses of action logically follows/follow?

- (A) Only I follows
- (B) Only II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

50. Statement: A newly constructed flyover developed wide cracks within six months of being opened to traffic.

Courses of action:

I. The flyover should be demolished immediately.

II. A technical inspection should be ordered and responsibility for the substandard work fixed.

Which of the courses of action logically follows/follow?

- (A) Only I follows
- (B) Only II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

51. Statement: Some candidates were found using unfair means during a state recruitment examination.

Courses of action:

I. The system of holding recruitment examinations should be abolished.

II. The candidates found guilty should be debarred and proceeded against under the law relating to unfair means in recruitment examinations.

Which of the courses of action logically follows/follow? RAS PRE 2023

- (A) Only I follows
- (B) Only II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

52. Statement: The air quality index of the city has remained in the 'very poor' category for the whole of last week.

Courses of action:

I. All the residents of the city should be advised to leave the city till the air improves.

II. The state pollution control board should enforce restrictions on construction dust and on the open burning of waste in the city.

Which of the courses of action logically follows/follow? RAS PRE 2024

- (A) Only I follows
- (B) Only II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

53. Statement: A private school has been charging fees far in excess of the limit prescribed under the state fee-regulation law.

Courses of action:

I. Parents should stop sending their children to that school.

II. The education department should hold an inquiry and take action against the school under the Act.

Which of the courses of action logically follows/follow?

- (A) Only I follows
- (B) Only II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

54. Statement: Farmers of the district have suffered heavy crop losses because of unseasonal hailstorms.

Courses of action:

I. The administration should order a special crop-loss survey (girdawari) and disburse relief to the affected farmers.

II. The cultivation of the affected crops should be banned in the district.

Which of the courses of action logically follows/follow?

- (A) Only I follows
- (B) Only II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

55. Statement: Very long queues are seen every morning at the registration counter of the district hospital.

Courses of action:

- I. The hospital should introduce token or online appointments and open additional counters during peak hours.
- II. The hospital should reduce the number of patients it registers each day.

Which of the courses of action logically follows/follow?

- (A) Only I follows
- (B) Only II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

56. Statement: The number of road accidents caused by drunken driving has increased sharply in the state.

Courses of action:

- I. The sale of motor vehicles in the state should be banned.
- II. All persons who consume liquor should be arrested.

Which of the courses of action logically follows/follow?

- (A) Only I follows
- (B) Only II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

57. Statement: The board examination results of a government school have been poor for three consecutive years.

Courses of action:

- I. The school should be closed down.
- II. The students of the school should be asked to appear as private candidates.

Which of the courses of action logically follows/follow?

- (A) Only I follows
- (B) Only II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

58. Statement: A number of complaints have been received about the poor quality of mid-day meals served in some schools of the district.

Courses of action:

- I. The mid-day meal scheme should be discontinued in the district.
- II. The parents who complained should be warned against spreading rumours.

Which of the courses of action logically follows/follow?

- (A) Only I follows
- (B) Only II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

59. Statement: Several old buildings in the walled city have been found to be in a dangerously dilapidated condition.

Courses of action:

- I. The municipal corporation should get the buildings surveyed and issue notices to vacate those declared unsafe.
- II. Arrangements should be made for the temporary rehabilitation of the families required to vacate.

Which of the courses of action logically follows/follow?

- (A) Only I follows
- (B) Only II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

60. Statement: A large number of young people in the state remain unemployed even after completing graduation.

Courses of action:

I. Skill-development and employability training should be integrated with degree courses.

II. Industry-institute linkages and campus placement arrangements should be strengthened.

Which of the courses of action logically follows/follow?

- (A) Only I follows
- (B) Only II follows
- (C) Neither I nor II follows
- (D) Both I and II follow

RM02

Analytical and Critical Reasoning; Syllogism

50 questions · 27 fact-sheet points

Must-know facts

- Four standard propositions: A = All A are B (universal affirmative), E = No A is B (universal negative), I = Some A are B (particular affirmative), O = Some A are not B (particular negative).
- Valid conversions: 'Some A are B' gives 'Some B are A'; 'No A is B' gives 'No B is A'; 'All A are B' gives only 'Some B are A' (never 'All B are A').
- 'Some A are not B' has NO valid conversion - 'Some B are not A' does not follow from it.
- Two particular premises (Some + Some) give no definite conclusion. Two negative premises give no definite conclusion.
- If one premise is particular, the conclusion must be particular; if one premise is negative, the conclusion must be negative.
- The middle term (the term common to both statements) must be distributed at least once, and must not appear in the conclusion.
- All + All = All (A + A = A); All + No = No (A + E = E); Some + All = Some (I + A = I); Some + No = Some-not (I + E = O).
- No + All in that order (E + A) gives 'Some ... are not ...' after conversion; check the direction of the terms carefully before keying.
- EITHER-OR case: when neither conclusion follows separately but the two are contradictory (All-Some not; Some-No; A pair covering every possibility) and share the same subject and predicate, the answer is 'Either I or II follows'.
- POSSIBILITY conclusions ('All A being B is a possibility') follow whenever at least one consistent Venn diagram allows it; they fail only when the statements definitely forbid it.
- A definite conclusion and its possibility-negation cannot both hold: if 'No A is B' definitely follows, 'Some A being B is a possibility' is false.
- Existential convention followed in Indian competitive exams: every class named in a statement is taken to be non-empty, so 'All A are B' yields 'Some B are A'.
- In cause-and-effect items, the earlier and independently occurring event is the cause; if both events are consequences of a third, unmentioned event, the answer is 'effects of a common cause'.
- Two events that are merely simultaneous, or belong to unrelated spheres (world gold price and local vegetable price), are effects of independent causes, not cause and effect.
- FACT = something already known/observed or verifiable from published data. INFERENCE = a conclusion drawn from given facts. JUDGEMENT = an opinion, approval, disapproval or recommendation ('should', 'better', 'ought').
- Words like 'must have', 'therefore', 'is likely to' signal an inference; 'should', 'must be made', 'better than' signal a judgement.
- In critical reasoning, an argument = premises + conclusion. To WEAKEN, supply an alternative explanation for the evidence; to STRENGTHEN, rule out alternative explanations (a controlled or randomised comparison is the strongest support).
- Correlation is not causation: if a third factor accompanies both the supposed cause and the effect, the causal claim collapses.
- Common flaws: affirming the consequent, confusing a necessary condition with a sufficient one, hasty generalisation from a small or self-selected sample, and appeal to authority or tradition.
- In 'A only if B', B is necessary for A - so A implies B, never B implies A.

21. Modus ponens: If P then Q; P is true, therefore Q. Modus tollens: If P then Q; Q is false, therefore P is false. 'Q is true therefore P' and 'P is false therefore not Q' are both fallacies.
22. Two-set formula: $n(A \text{ or } B) = n(A) + n(B) - n(\text{both})$; neither = Total - $n(A \text{ or } B)$; exactly one = $n(A) + n(B) - 2 \times n(\text{both})$.
23. Logical Venn representation: fully included classes (Rajasthan-India-Asia) = concentric circles; mutually exclusive classes inside a wider class (Aunt-Uncle-Relative) = two separate circles inside a bigger one; partly overlapping classes = intersecting circles.
24. In ordering puzzles, convert every clue into a single chain with '>' before answering; count positions only after the whole chain is fixed.
25. In truth-teller puzzles, test each possibility one by one against every statement and keep the case that satisfies the given count of true statements.
26. In input-output items, identify what moves at each step (largest/smallest number, alphabetically first/last word) and to which end it moves; the untouched elements keep their relative order.
27. Syllogism and logical Venn diagram items have appeared in RAS Prelims 2021, 2023 and 2024 - always draw the diagram, never rely on the 'sound' of the sentence.

1. Statements:

All pens are books.

All books are papers.

Conclusions:

I. All pens are papers.

II. Some papers are pens.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.)

- (A) Both I and II follow
- (B) Neither I nor II follows
- (C) Only conclusion I follows
- (D) Only conclusion II follows

2. Statements:

Some cats are dogs.

All dogs are animals.

Conclusions:

I. Some cats are animals.

II. All animals are cats.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.)

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Both I and II follow
- (D) Neither I nor II follows

3. Statements:

All flowers are roses.

No rose is a leaf.

Conclusions:

I. No flower is a leaf.

II. Some leaves are flowers.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.)

- (A) Neither I nor II follows
- (B) Only conclusion I follows
- (C) Only conclusion II follows
- (D) Both I and II follow

4. Statements:

Some books are pens.

No pen is a pencil.

Conclusions:

I. No book is a pencil.

II. Some books are not pencils.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.)

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Both I and II follow
- (D) Neither I nor II follows

5. Statements:

All singers are dancers.

Some dancers are poets.

Conclusions:

I. Some singers are poets.

II. Some poets are dancers.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.)

- (A) Neither I nor II follows
- (B) Only conclusion I follows
- (C) Only conclusion II follows
- (D) Both I and II follow

6. Statements:

Some pens are books.

All books are copies.

Conclusions:

I. All pens are copies.

II. Some pens are not copies.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.) **HARD**

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Either I or II follows
- (D) Neither I nor II follows

7. Statements:

All cups are plates.

Some plates are spoons.

Conclusions:

I. Some cups are spoons.

II. No cup is a spoon.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.) **HARD**

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Either I or II follows
- (D) Neither I nor II follows

8. Statements:

Some mangoes are apples.

Some apples are grapes.

Conclusions:

I. Some mangoes are grapes.

II. All apples are mangoes.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.)

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Both I and II follow
- (D) Neither I nor II follows

9. Statements:

All doctors are engineers.

No engineer is a teacher.

Conclusions:

I. Some doctors are teachers.

II. Some teachers are engineers.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.)

- (A) Neither I nor II follows
- (B) Only conclusion I follows
- (C) Only conclusion II follows
- (D) Both I and II follow

10. Statements:

No student is lazy.

All lazy persons are sleepy.

Conclusions:

I. Some sleepy persons are not students.

II. Some sleepy persons are lazy.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.)

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Both I and II follow
- (D) Neither I nor II follows

11. Statements:

All teachers are graduates.

Some graduates are lawyers.

Conclusions:

I. Some graduates are teachers.

II. All lawyers being teachers is a possibility.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.) **HARD**

- (A) Neither I nor II follows
- (B) Only conclusion I follows
- (C) Only conclusion II follows
- (D) Both I and II follow

12. Statements:

All apples are fruits.

No fruit is a vegetable.

Conclusions:

I. Some apples being vegetables is a possibility.

II. No apple is a vegetable.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.)

- (A) Both I and II follow
- (B) Neither I nor II follows
- (C) Only conclusion I follows
- (D) Only conclusion II follows

13. Statements:

Some chairs are tables.

All tables are of wood.

Conclusions:

I. All chairs being of wood is a possibility.

II. Some wooden things are chairs.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.)

- (A) Both I and II follow
- (B) Neither I nor II follows
- (C) Only conclusion I follows
- (D) Only conclusion II follows

14. Statements:

All keys are locks.

No lock is a door.

Some doors are windows.

Conclusions:

I. No key is a door.

II. Some windows are locks.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.)

- (A) Both I and II follow
- (B) Neither I nor II follows
- (C) Only conclusion I follows
- (D) Only conclusion II follows

15. Statements:

Some pencils are erasers.

All erasers are sharpeners.

No sharpener is a scale.

Conclusions:

I. Some pencils are not scales.

II. No eraser is a scale.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.)

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Both I and II follow
- (D) Neither I nor II follows

16. Statements:

All roads are paths.

Some paths are lanes.

No lane is a street.

Conclusions:

I. Some paths are not streets.

II. Some roads are lanes.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.)

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Both I and II follow
- (D) Neither I nor II follows

17. Statements:

No book is a magazine.

Some magazines are journals.

Conclusions:

I. Some journals are not books.

II. All journals being books is a possibility.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.) HARD

- (A) Neither I nor II follows
- (B) Only conclusion I follows
- (C) Only conclusion II follows
- (D) Both I and II follow

18. Statements:

Some cars are buses.

No bus is a truck.

All trucks are vehicles.

Conclusions:

I. Some cars are not trucks.

II. Some vehicles are not buses.

Which of the conclusions logically follow(s) from the statements? (Take the statements to be true even if they seem at variance with commonly known facts.)

- (A) Neither I nor II follows
- (B) Only conclusion I follows
- (C) Only conclusion II follows
- (D) Both I and II follow

19. Below are given two statements. Read them and mark the correct relationship between them.

I. The state government has raised the tariff of electricity for domestic consumers.

II. The state electricity distribution companies reported heavy losses in the last financial year.

- (A) Statement I is the cause and statement II is its effect
- (B) Statement II is the cause and statement I is its effect
- (C) Both the statements are effects of a common cause
- (D) Both the statements are effects of independent causes

20. Below are given two statements. Read them and mark the correct relationship between them.

I. Very heavy rainfall was recorded in the catchment area of the Bisalpur dam this monsoon.

II. The water level in the Bisalpur dam rose sharply during the same period.

- (A) Statement I is the cause and statement II is its effect
- (B) Statement II is the cause and statement I is its effect
- (C) Both the statements are effects of a common cause
- (D) Both the statements are effects of independent causes

21. Below are given two statements. Read them and mark the correct relationship between them.

I. Many flights from the city airport were cancelled this morning.

II. All schools in the city were ordered to remain closed this morning.

- (A) Statement I is the cause and statement II is its effect
- (B) Statement II is the cause and statement I is its effect
- (C) Both the statements are effects of a common cause
- (D) Both the statements are effects of independent causes

22. Below are given two statements. Read them and mark the correct relationship between them.

I. The price of gold in the international market rose last week.

II. The price of green vegetables in the local mandi rose last week.

- (A) Statement I is the cause and statement II is its effect
- (B) Statement II is the cause and statement I is its effect
- (C) Both the statements are effects of a common cause
- (D) Both the statements are effects of independent causes

23. Below are given two statements. Read them and mark the correct relationship between them.

I. A large number of farmers in the district sowed mustard this year in place of wheat.

II. The government announced a substantially higher minimum support price for mustard before the sowing season.

- (A) Statement I is the cause and statement II is its effect
- (B) Statement II is the cause and statement I is its effect
- (C) Both the statements are effects of a common cause
- (D) Both the statements are effects of independent causes

24. Classify the following statement:

'As per the Census of 2011, the sex ratio of Rajasthan was 928 females per 1000 males.'

- (A) A statement of fact
- (B) An inference
- (C) A judgement
- (D) An assumption

25. Classify the following statement:

'Road traffic on this route has become noticeably lighter since the metro service started; the metro must have taken a large number of commuters off the road.'

- (A) A statement of fact
- (B) An inference
- (C) A judgement
- (D) An assumption

26. Classify the following statement:

'The state government must ban the sale of soft drinks within 100 metres of school gates.'

- (A) A statement of fact
- (B) An inference
- (C) A judgement
- (D) An assumption

27. Classify the following statement:

'Rajasthan shares its longest inter-state boundary with Madhya Pradesh.'

- (A) An inference
- (B) A judgement
- (C) An assumption
- (D) A statement of fact

28. Classify the following statement:

'Village life in India is far more peaceful and satisfying than city life.'

- (A) A judgement
- (B) A statement of fact
- (C) An inference
- (D) An assumption

29. Statement: Although the state harvested a record wheat crop this year, wheat prices in the local mandis did not fall.

Which one of the following can most reasonably be inferred from the statement?

- (A) Wheat production has no relation whatever with wheat prices
- (B) Factors other than local production also influence local wheat prices
- (C) The entire record crop was exported out of the state
- (D) Traders always manipulate the prices of foodgrains

30. Statement: Only 30 per cent of the households in the block have a piped water connection, although 95 per cent of the villages in the block have been declared covered under the water supply scheme.

Which one of the following can most reasonably be inferred from the statement?

- (A) Declaring a village 'covered' does not ensure that its households actually get connections
- (B) The water supply scheme has failed completely in the block
- (C) The remaining 70 per cent of the households have no source of water at all
- (D) The coverage figures reported by the block officials are false

31. Statement: A company has decided to move from a five-day to a four-day working week without reducing the total number of working hours in a week.

Which one of the following must be true?

- (A) The salaries of the employees will be reduced
- (B) The total weekly output of the company will fall
- (C) Employees will have to work longer on each working day
- (D) Employees will work fewer hours in a week

32. Statement: All the candidates who were finally selected had cleared the physical test; however, some candidates who cleared the physical test were not selected.

Which one of the following can be inferred? HARD

- (A) Clearing the physical test is sufficient for selection
- (B) Some selected candidates did not clear the physical test
- (C) The physical test plays no part in the selection process
- (D) Clearing the physical test is necessary but not sufficient for selection

33. A city introduced a congestion fee on cars entering its central zone. In the following year the number of cars entering the zone fell by 20 per cent, and the mayor concluded that the fee had reduced car use in the city.

Which one of the following, if true, most weakens the mayor's conclusion?

- (A) Many drivers merely shifted to roads just outside the central zone, and the total number of car trips in the city was unchanged
- (B) The congestion fee was collected electronically without stopping vehicles
- (C) The revenue from the fee was spent on improving public transport
- (D) A section of car owners protested against the fee when it was introduced

34. Doctors at a hospital observed that patients who walked for thirty minutes a day recovered faster after surgery than those who did not, and concluded that walking speeds up recovery.

Which one of the following, if true, would most strengthen the conclusion? HARD

- (A) Walking is a popular form of exercise among patients of all ages
- (B) In a trial where patients were assigned at random either to daily walking or to bed rest, the walking group recovered faster
- (C) Walking costs the hospital much less than physiotherapy
- (D) Patients who walked daily also ate a more nutritious diet

35. 'The state will certainly meet its target of generating 50 per cent of its electricity from renewable sources by 2030, because contracts for all the required solar capacity have already been awarded.'

The above argument assumes that:

- (A) Solar power is cheaper to produce than coal-based power
- (B) The awarded capacity will actually be built and will generate power by 2030
- (C) No other state has set a comparable renewable energy target
- (D) The total demand for electricity in the state will fall by 2030

36. 'Every successful entrepreneur I know takes risks. So if you take risks, you will become a successful entrepreneur.'

The reasoning in the above argument is flawed mainly because:

- (A) It appeals to emotion instead of offering evidence
- (B) It assumes that all entrepreneurs are honest
- (C) It treats a characteristic shared by successful entrepreneurs as if it were enough to guarantee success
- (D) It denies that risk-taking has any role in success

37. The number of accidents on a highway fell sharply in the year after speed cameras were installed on it. The transport department concluded that the cameras had made the highway safer.

Which one of the following, if true, most weakens this conclusion?

- (A) The cameras were installed at only three points on the highway
- (B) Speed cameras are used on highways in many other countries as well
- (C) Speed cameras are expensive to install and maintain
- (D) During the same year the highway was closed for repairs for four months and the traffic on it fell by half

38. A study of 10,000 school children found that children who ate breakfast scored higher in mathematics tests than those who skipped it. The study also recorded that the children who ate breakfast slept, on average, an hour longer each night.

The passage most supports which one of the following? HARD

- (A) Eating breakfast causes an improvement in mathematics scores
- (B) Skipping breakfast causes children to sleep less
- (C) The association between breakfast and higher scores may be explained by another factor
- (D) Sleep has no effect on the test scores of school children

39. In a group of 60 students, 35 play cricket, 30 play football and 12 play both the games. How many students of the group play neither of the two games? RAS PRE 2023

- (A) 5
- (B) 7
- (C) 10
- (D) 12

40. In an examination 70 per cent of the candidates passed in English and 65 per cent passed in Hindi, while 15 per cent failed in both the subjects. What percentage of the candidates passed in both the subjects? RAS PRE 2021

- (A) 55 per cent
- (B) 60 per cent
- (C) 50 per cent
- (D) 45 per cent

41. Which one of the following diagrams correctly represents the relationship among Rajasthan, India and Asia? RAS PRE 2024

- (A) Three circles that are entirely separate from one another
- (B) Two overlapping circles enclosed within a third circle
- (C) One circle inside another, with the third circle lying entirely outside both
- (D) Three concentric circles, the innermost representing Rajasthan and the outermost representing Asia

42. Which one of the following diagrams best represents the relationship among Aunt, Uncle and Relative?

- (A) Three concentric circles
- (B) Two separate circles, both lying inside a larger third circle
- (C) Three circles overlapping one another
- (D) Two overlapping circles lying outside a third circle

43. Five friends A, B, C, D and E have different weights. A is heavier than B but lighter than C. D is heavier than C, and E is lighter than B. Who among them is the heaviest?

- (A) A
- (B) B
- (C) C
- (D) D

44. Ravi is taller than Mohan but shorter than Suresh. Kamal is taller than Suresh, and Naveen is shorter than Mohan. Who among the five is the second tallest?

- (A) Suresh (B) Ravi
(C) Kamal (D) Mohan

45. Anita, Bina and Chitra are a doctor, an engineer and a teacher, in some order. Anita is not the teacher, Bina is not the doctor, the engineer is the youngest of the three, and Chitra is the eldest while Anita is older than Bina. Who is the teacher?

- (A) Anita (B) Bina
(C) Chitra (D) It cannot be determined

46. There are three boxes A, B and C, and exactly one of them contains a prize. Box A carries the label 'The prize is not in this box', Box B carries the label 'The prize is in this box', and Box C carries the label 'The prize is not in Box B'. If exactly one of the three labels is true, in which box is the prize? **HARD**

- (A) Box A (B) Box B
(C) Box C (D) It cannot be determined

47. In a colony of 100 families, 60 subscribe to newspaper A, 45 subscribe to newspaper B and 20 subscribe to both. How many families subscribe to exactly one of the two newspapers?

- (A) 55 (B) 85
(C) 75 (D) 65

48. Six books are placed one above another in a pile. U is at the top and R is at the bottom. P is immediately above Q, S is immediately below T, and Q is somewhere above S. Counting from the top, what is the position of S? **HARD**

- (A) Second (B) Third
(C) Fifth (D) Fourth

49. Four students - Pooja, Qadir, Ravi and Sunita - study four different subjects: History, Geography, Physics and Economics, not necessarily in that order. Pooja studies neither Physics nor Economics, Ravi studies Geography and Sunita does not study Economics. Who studies Economics?

- (A) Pooja (B) Qadir
(C) Sunita (D) It cannot be determined

50. A machine rearranges a given input in a fixed manner. Its working is shown below:

Input: 68 32 51 09 24

Step I: 09 68 32 51 24

Step II: 24 09 68 32 51

What will Step III be?

- (A) 32 24 09 68 51
(B) 24 32 09 68 51
(C) 32 09 24 68 51
(D) 09 24 32 68 51

RM03

Number and Letter Series

50 questions · 26 fact-sheet points

Must-know facts

1. Alphabet positions (learn cold): A1 B2 C3 D4 E5 F6 G7 H8 I9 J10 K11 L12 M13 N14 O15 P16 Q17 R18 S19 T20 U21 V22 W23 X24 Y25 Z26.
2. EJOY mnemonic: E=5, J=10, O=15, T=20, Y=25 — count a few steps forward or back from the nearest anchor instead of counting from A.
3. Reverse (opposite) position = 27 minus the forward position. Opposite pairs: A-Z, B-Y, C-X, D-W, E-V, F-U, G-T, H-S, I-R, J-Q, K-P, L-O, M-N.
4. Squares to remember: 1,4,9,16,25,36,49,64,81,100,121,144,169,196,225,256,289,324,361,400.
5. Cubes to remember: 1,8,27,64,125,216,343,512,729,1000,1331,1728.
6. Primes under 60: 2,3,5,7,11,13,17,19,23,29,31,37,41,43,47,53,59. (1 is NOT prime; 2 is the only even prime.)
7. Triangular numbers: 1,3,6,10,15,21,28,36,45,55 — differences are 2,3,4,5,...
8. Fibonacci: 1,1,2,3,5,8,13,21,34,55,89 — each term = sum of the two before it.
9. Standard attack order on a number series: (1) first differences, (2) second differences, (3) ratios, (4) squares/cubes plus-or-minus a constant, (5) alternate terms, (6) primes.
10. Constant difference = arithmetic series; constant ratio = geometric series; differences forming an AP = a quadratic (n-squared type) series.
11. If differences themselves double, the rule is usually 'previous $\times 2$ plus-or-minus k' — test k on two consecutive pairs.
12. Rising multipliers pattern: $\times 2$, $\times 3$, $\times 4$, $\times 5$ (e.g. 3, 6, 18, 72, 360). Falling divisors: $/2$, $/3$, $/4$, $/5$ (e.g. 240, 120, 40, 10, 2).
13. Common families: n^2+1 (2,5,10,17,26), n^2-1 (0,3,8,15,24), n^3+1 (2,9,28,65), n^3-1 (0,7,26,63,124), $n(n+1)$ (2,6,12,20,30).
14. Alternate / interleaved series: if no single rule fits, split the series into terms at odd places and terms at even places and solve each separately.
15. Wrong-term questions: fix the rule from THREE consecutive terms that agree, then run it forward; the first term that breaks is the answer.
16. Letter series: convert letters to numbers, solve as a number series, convert back. Never guess by 'feel'.
17. Frequent letter gaps: +1, +2, +3; alternating +2/+3; mirrored pairs (AZ, BY, CX, DW); squares as positions (A, D, I, P, Y).
18. Letter groups: watch the group length (2, 3, 4, 5 letters) and how many letters are skipped between groups.
19. Alphanumeric series: treat the letter track and the number track as two completely independent series.
20. Number analogy checks, in order: difference, ratio, square, cube, $n(n+1)$, n^2+or-1 , and the digit-reversal relation.
21. Letter analogy: compute the positional shift for each letter of the first pair, then apply exactly the same shifts to the second.
22. Common word-analogy relations: worker-workplace, tool-function, specialist-organ, product-raw material, part-whole, cause-effect, animal-young one.

23. Odd-one-out for numbers: test primes, perfect squares, perfect cubes, divisibility by a common factor, and digit sum.
24. Odd-one-out for letter groups: compute the internal gaps of each group; the group whose gap pattern differs is the answer.
25. Odd-one-out for words: identify the common category first (metals, flowers, alloys, birds that fly), then find the member outside it.
26. In RAS, series and analogy items are almost always easy scoring questions — do them first and bank the marks, since there is 1/3 negative marking.

1. Find the missing term in the following series:

3, 7, 11, 15, ?, 23 RAS PRE 2016

- (A) 17
- (B) 18
- (C) 19
- (D) 21

2. Find the missing term in the following series:

5, 11, 19, 29, 41, ?

- (A) 55
- (B) 51
- (C) 53
- (D) 57

3. Find the missing term in the following series:

2, 6, 12, 20, 30, ?

- (A) 40
- (B) 44
- (C) 46
- (D) 42

4. Find the missing term in the following series:

4, 9, 19, 39, 79, ?

- (A) 149
- (B) 158
- (C) 159
- (D) 169

5. Find the missing term in the following series:

7, 10, 16, 28, 52, ?

- (A) 96
- (B) 100
- (C) 98
- (D) 104

6. Find the missing term in the following series:

1, 4, 9, 16, 25, 36, ? RAS PRE 2013

- (A) 42
- (B) 45
- (C) 49
- (D) 48

7. Find the missing term in the following series:

1, 8, 27, 64, 125, ?

- (A) 216
- (B) 196
- (C) 200
- (D) 225

8. Find the missing term in the following series:

2, 3, 5, 7, 11, 13, ? RAS PRE 2021

- (A) 15
- (B) 17
- (C) 19
- (D) 21

9. Find the missing term in the following series:

1, 1, 2, 3, 5, 8, 13, ?

- (A) 18
- (B) 20
- (C) 26
- (D) 21

10. Find the missing term in the following series:

3, 6, 18, 72, ?

- (A) 144
- (B) 216
- (C) 360
- (D) 288

11. Find the missing term in the following series:

625, 125, 25, 5, ?

- (A) 0
- (B) 1
- (C) 2
- (D) 5

12. Find the missing term in the following series:

2, 5, 10, 17, 26, ?

- (A) 37
- (B) 35
- (C) 36
- (D) 38

13. Find the missing term in the following series:

0, 7, 26, 63, 124, ?

- (A) 195
- (B) 215
- (C) 210
- (D) 216

14. Find the missing term in the following series:

6, 11, 21, 36, 56, ?

- (A) 76
- (B) 78
- (C) 81
- (D) 86

15. Find the missing term in the following series:

120, 99, 80, 63, 48, ? HARD

- (A) 33
- (B) 36
- (C) 39
- (D) 35

16. Find the missing term in the following series:

4, 9, 7, 18, 10, 27, 13, ?

- (A) 30
- (B) 32
- (C) 40
- (D) 36

17. Find the missing term in the following series:

1, 2, 4, 6, 9, 12, 16, ? HARD

- (A) 18
- (B) 19
- (C) 20
- (D) 22

18. Find the missing term in the following series:

3, 4, 7, 11, 18, 29, ?

- (A) 47
- (B) 40
- (C) 45
- (D) 48

19. Find the missing term in the following series:

1, 3, 6, 10, 15, 21, ?

- (A) 28
- (B) 26
- (C) 27
- (D) 30

20. Find the missing term in the following series:

2, 12, 36, 80, 150, ? HARD

- (A) 240
- (B) 245
- (C) 256
- (D) 252

21. Find the missing term in the following series:

5, 15, 12, 36, 33, 99, ?

- (A) 96
- (B) 93
- (C) 102
- (D) 297

22. Find the missing term in the following series:

7, 26, 63, 124, 215, ?

- (A) 336
- (B) 340
- (C) 342
- (D) 344

23. One term in the following series is wrong. Identify the wrong term:

3, 8, 15, 24, 34, 48, 63

- (A) 34
- (B) 24
- (C) 48
- (D) 15

24. One term in the following series is wrong. Identify the wrong term:

2, 5, 10, 17, 26, 38, 50

- (A) 38
- (B) 17
- (C) 26
- (D) 50

25. One term in the following series is wrong. Identify the wrong term:

5, 10, 20, 40, 75, 160

- (A) 20
- (B) 40
- (C) 160
- (D) 75

26. One term in the following series is wrong. Identify the wrong term:

1, 4, 9, 16, 24, 36, 49

- (A) 9
- (B) 16
- (C) 24
- (D) 36

27. Find the missing term in the following letter series:

A, C, F, J, O, ? RAS PRE 2018

- (A) S
- (B) U
- (C) T
- (D) V

28. Find the missing term in the following letter series:

Z, X, V, T, ?

- (A) Q
- (B) S
- (C) P
- (D) R

29. Find the missing term in the following letter series:

A, D, I, P, ?

- (A) Y
- (B) W
- (C) X
- (D) Z

30. Find the missing term in the following letter series:

C, F, I, L, ?

- (A) M
- (B) N
- (C) O
- (D) P

31. Find the missing term in the following series:

AZ, BY, CX, DW, ? RAS PRE 2023

- (A) EV
- (B) EU
- (C) FV
- (D) EW

32. Find the missing term in the following series:

ACE, BDF, CEG, ?

- (A) DEG
- (B) DFH
- (C) EGI
- (D) DGI

33. Find the missing term in the following series:

AB, DEF, HIJK, ? HARD

- (A) LMNOP
- (B) MNOPQ
- (C) MNOP
- (D) NOPQR

34. Find the missing term in the following letter series:

B, E, J, Q, ?

- (A) X
- (B) Z
- (C) Y
- (D) W

35. Find the missing term in the following series:

A1, C3, E5, G7, ?

- (A) H8
- (B) I9
- (C) I8
- (D) J9

36. Find the missing term in the following series:

Z1, Y2, X3, W4, ?

- (A) V4
- (B) U5
- (C) W5
- (D) V5

37. Find the missing term in the following series:

A2, D4, G8, J16, ?

- (A) M24
- (B) K32
- (C) M30
- (D) M32

38. Find the missing term in the following series:

2A, 5D, 10G, 17J, ? HARD

- (A) 24M
- (B) 25M
- (C) 26N
- (D) 26M

39. 8 : 27 :: 64 : ?

- | | |
|---------|---------|
| (A) 81 | (B) 100 |
| (C) 216 | (D) 125 |

40. 25 : 36 :: 49 : ?

- | | |
|--------|--------|
| (A) 56 | (B) 64 |
| (C) 63 | (D) 81 |

41. 6 : 42 :: 8 : ?

- | | |
|--------|--------|
| (A) 56 | (B) 72 |
| (C) 64 | (D) 80 |

42. BDF : CEG :: MOQ : ?

- | | |
|---------|---------|
| (A) NPS | (B) NOQ |
| (C) LNP | (D) NPR |

43. ACE : BDF :: PRT : ?

- | | |
|---------|---------|
| (A) QSU | (B) QRT |
| (C) QSV | (D) OQS |

44. Pen : Write :: Knife : ?

- | | |
|-----------|-------------|
| (A) Sharp | (B) Blade |
| (C) Cut | (D) Kitchen |

- 45.** Ophthalmologist : Eye :: Cardiologist : ? RAS PRE 2024
- (A) Kidney (B) Brain
(C) Heart (D) Lung
- 46.** Which one of the following numbers is different from the rest? RAS PRE 2016
- (A) 100 (B) 27
(C) 64 (D) 125
- 47.** Which one of the following numbers does NOT belong to the group?
- (A) 3 (B) 9
(C) 5 (D) 11
- 48.** Which one of the following is different from the other three?
- (A) Rose (B) Mango
(C) Lotus (D) Lily
- 49.** Which one of the following does NOT belong to the group?
- (A) Brass (B) Copper
(C) Zinc (D) Iron
- 50.** Which one of the following letter groups is different from the rest?
- (A) ACE (B) GIK
(C) RTW (D) MOQ

RM04

Coding-Decoding

45 questions · 26 fact-sheet points

Must-know facts

1. Alphabet positions: A1 B2 C3 D4 E5 F6 G7 H8 I9 J10 K11 L12 M13 N14 O15 P16 Q17 R18 S19 T20 U21 V22 W23 X24 Y25 Z26. Learn this before anything else.
2. EJOY mnemonic (E=5, J=10, O=15, T=20, Y=25) and CFILORUX (C=3, F=6, I=9, L=12, O=15, R=18, U=21, X=24) let you find any position in one or two steps.
3. Opposite (reverse) letter = 27 minus the position. Pairs: A-Z, B-Y, C-X, D-W, E-V, F-U, G-T, H-S, I-R, J-Q, K-P, L-O, M-N.
4. First test in any letter-coding item: is the code the SAME LENGTH as the word? If yes, it is a letter-to-letter rule; if no, suspect number, substitution or compression coding.
5. Rule family 1 - constant forward shift (+1, +2, +3, +4). Rule family 2 - constant backward shift (-1, -2, -3).
6. Rule family 3 - alternating shift, e.g. +1, -1, +1, -1 (CHAIR to DGBHS). Check letter by letter; two letters are never enough.
7. Rule family 4 - shift by position in the word: first letter +1, second +2, third +3 (ABC to BDF).
8. Rule family 5 - structural rules with NO shift: full reversal (DELHI to IHLED), interchange of successive pairs (MONKEY to OMKNYE), interchange of only the first and last letters (SYSTEM to MYSTES).
9. Rule family 6 - opposite-letter (complement) coding: DOG to WLT.
10. Rule family 7 - combination: reverse the word first and then shift (CAT to UBD).
11. Shifts wrap around the alphabet: after Z comes A, before A comes Z. $A-3 = X$, $Z+2 = B$.
12. Number coding forms: (a) position written out (CAT = 3-1-20), (b) sum of positions (CAT = 24), (c) a multiple of the position (A=2, B=4, ...), (d) reverse position (Z=1 ... A=26), (e) position plus or minus a constant.
13. Sum-of-positions coding is confirmed by testing a SECOND given word; one word alone can fit several rules.
14. Mixed letter-number coding (PALE = 2134, EARTH = 41590): build a letter-to-digit dictionary from the given words; shared letters must get the same digit, which is your consistency check.
15. In mixed coding, if two given words share a letter and the digits do not agree, you have mis-aligned the words - realign before answering.
16. Substitution coding ('if X is called Y'): first answer the question in ordinary language, then translate that answer into the new name. Never answer with the real word.
17. Coded-sentence problems: any two sentences that share exactly ONE English word and exactly ONE code word fix that pair immediately.
18. With three sentences that pairwise share one word each, all three shared words are fixed and the remaining word of each sentence is then forced - this is the standard RAS pattern.
19. Cross out each code word as you fix it; the answer is usually the single word left unmatched.
20. Conditional coding: write the plain digit code first, then apply EVERY condition in the stated order, then re-check the first and the last letter of the group.
21. Most conditional sets hinge on whether the first and last letters are vowels (A, E, I, O, U) or consonants - underline those two letters before you start.
22. If two conditions could both apply, the question normally states an order of precedence; apply condition (i) first unless told otherwise.

23. Matrix coding: a letter is given by row number then column number; note which letter is left out of the matrix (often Z) and whether rows or columns are numbered first.
24. Coding by analogy (BOOK : CPPL :: LOCK : ?): derive the rule from the first pair, verify it on EVERY letter, then apply it to the second word.
25. Quick elimination: check only the first and last letters of each option against your rule; usually three options die at once.
26. Coding-decoding appears in every RAS Prelims paper, usually 1-3 questions, and is pure method - it should never be left unattempted.

1. In a certain code language, MADRAS is written as NBESBT. How will BOMBAY be written in that code? RAS PRE 2016

- (A) CPNBAY (B) CPNCAZ
(C) CPNCBZ (D) CPMCBZ

2. In a certain code language, FRIEND is written as HTKGPF. How is CANDLE written in that code? RAS PRE 2021

- (A) ECPFNH (B) ECPFNG
(C) DBOEMF (D) ECPFOG

3. In a certain code, TEACHER is written as WHDFKHU. How is STUDENT written in the same code?

- (A) VWXGHQV (B) VWXGIQW
(C) VWXGHQW (D) UVWFGPV

4. In a certain code, TRIPPLE is written as SQHOOKD. How is DISPOSE written in that code?

- (A) CHRONRE (B) CHRONRD
(C) EJTQPTF (D) CHROMRD

5. If WATER is coded as TXQBO, then how is FIRE coded in the same language?

- (A) CFOA (B) CFOB
(C) CEOB (D) DFOB

6. In a certain code language, DOG is written as WLT. How is CAT written in that code? RAS PRE 2023

- (A) XZH (B) YZG
(C) XAG (D) XZG

7. In a certain code, DELHI is written as IHLED. How will MUMBAI be written in that code?

- (A) IABMUM (B) IABMMU
(C) IABUMM (D) MUMIAB

8. In a certain code, CAT is written as UBD. How is DOG written in the same code? HARD

- (A) HPF (B) HPE
(C) GPE (D) HQE

9. In a certain code, CHAIR is written as DGBHS. How is TABLE written in that code?

- (A) UZCKF (B) UZBKF
(C) SBAMD (D) UCCKF

10. If ABC is coded as BDF, how is PQR coded in the same language?

- (A) QRS (B) QSU
(C) QST (D) RSU

- 11.** In a certain code, MONKEY is written as OMKNYE. How is CHAIRS written in that code? **HARD**
- (A) HCAIRS (B) HCIARS
(C) CHIASR (D) HCIASR
- 12.** If SYSTEM is coded as MYSTES, then how is NUMBER coded in the same language?
- (A) RUMBEN (B) PUMBEN
(C) REBMUN (D) RUMBNE
- 13.** If the letters of the English alphabet are numbered A = 1, B = 2, ... , Z = 26, and CAB is coded as 312, how will FED be coded? **RAS PRE 2018**
- (A) 645 (B) 456
(C) 654 (D) 664
- 14.** In a certain code, SUN is written as 54. On the same basis, how will MOON be written?
- (A) 55 (B) 57
(C) 60 (D) 48
- 15.** If CAT is coded as 24, then DOG is coded as which number in the same system?
- (A) 25 (B) 27
(C) 26 (D) 30
- 16.** In a certain code, LOVE is written as 54. How will LIFE be written in that code?
- (A) 32 (B) 30
(C) 34 (D) 36
- 17.** If A = 2, B = 4, C = 6 and so on, what is the code for the word FACE?
- (A) 6-1-3-5 (B) 12-2-6-10
(C) 12-2-6-8 (D) 10-2-6-12
- 18.** In a certain code, the letters are numbered from the end of the alphabet, i.e. Z = 1, Y = 2, ... , A = 26. What is the code for the word CAB? **HARD**
- (A) 3-1-2 (B) 24-26-25
(C) 24-25-26 (D) 23-26-25
- 19.** In a certain code, PALE is written as 2134 and EARTH is written as 41590. How is PEARL written in that code?
- (A) 21435 (B) 24135
(C) 24153 (D) 25143
- 20.** In a certain code, ROSE is written as 6821, CHAIR is written as 73456 and PREACH is written as 961473. How is SEARCH written in that code? **HARD**
- (A) 216473 (B) 214763
(C) 214673 (D) 241673
- 21.** If the code for MANGO is 13-1-14-7-15, what is the code for APPLE?
- (A) 1-16-16-11-5 (B) 1-15-15-12-5
(C) 1-16-16-12-5 (D) 1-16-16-12-4
- 22.** If CAT is coded as 3120 and DOG is coded as 4157, how is BAT coded in the same system?
- (A) 2119 (B) 2220
(C) 2121 (D) 2120

23. If 'sky' is called 'sea', 'sea' is called 'water', 'water' is called 'air' and 'air' is called 'cloud', then where do fish live? RAS PRE

2013

- | | |
|-----------|---------|
| (A) Water | (B) Sea |
| (C) Cloud | (D) Air |

24. If 'white' is called 'blue', 'blue' is called 'red', 'red' is called 'yellow' and 'yellow' is called 'green', what is the colour of human blood in this language?

- | | |
|-----------|------------|
| (A) Red | (B) Blue |
| (C) Green | (D) Yellow |

25. If 'pen' is called 'book', 'book' is called 'table', 'table' is called 'chair' and 'chair' is called 'board', then with which of these do we write?

- | | |
|-----------|-----------|
| (A) Book | (B) Pen |
| (C) Table | (D) Board |

26. If 'green' means 'sing', 'blue' means 'dance', 'red' means 'jump' and 'yellow' means 'run', then what will a person normally do to music?

- | | |
|----------|------------|
| (A) Blue | (B) Green |
| (C) Red | (D) Yellow |

27. In a certain code language, 'gel hai jar' means 'apples are ripe' and 'hai tir sup' means 'children are playing'. Which word in that language means 'are'? RAS PRE 2024

- | | |
|---------|---------|
| (A) gel | (B) jar |
| (C) hai | (D) tir |

28. In a certain code language:

'tim nap sok' means 'she is intelligent',

'nap dor lin' means 'is very smart',

'sok lin bap' means 'intelligent smart boy'.

Which word in that language means 'boy'?

- (A) sok
(B) lin
(C) dor
(D) bap

29. In a certain code language:

'pit na tod' means 'you are good',

'na rok tim' means 'they are bad',

'tod rok sim' means 'good bad boys'.

Which word in that language means 'you'?

- (A) pit
(B) na
(C) tod
(D) tim

30. In a certain code language:

'nik sud pal' means 'hard work pays',

'sud tor mip' means 'work is worship',

'pal jos tor' means 'pays money is'.

Which word in that language means 'worship'?

- (A) sud
- (B) tor
- (C) mip
- (D) jos

31. In a certain code language:

'ma po ri' means 'cats eat fish',

'po lin sa' means 'dogs eat bones',

'ri sa dun' means 'fish bones white'.

Which word in that language means 'dogs'?

- (A) lin
- (B) po
- (C) sa
- (D) ma

32. In a certain code, HELP is written as GDKO. In the same code, what does RTMCZX stand for?

- | | |
|------------|-------------|
| (A) SUNDAY | (B) MONDAY |
| (C) FRIDAY | (D) TUESDAY |

33. Study the letter-digit table and the conditions given below and answer the question.

Letter : P M A K R E T

Digit : 4 7 2 9 5 3 8

Conditions:

(i) If the first letter of the group is a vowel, both the first and the last letter are coded as \$.

(ii) If the last letter is a vowel and the first letter is not a vowel, the last letter is coded as #.

What is the code for the group AMKRT?

- (A) 27958
- (B) \$795\$
- (C) \$7958
- (D) 2795\$

34. Study the letter-digit table and the conditions given below and answer the question.

Letter : P M A K R E T

Digit : 4 7 2 9 5 3 8

Conditions:

(i) If the first letter of the group is a vowel, both the first and the last letter are coded as \$.

(ii) If the last letter is a vowel and the first letter is not a vowel, the last letter is coded as #.

What is the code for the group PMAKE?

- (A) 47293
- (B) \$729\$
- (C) #7294
- (D) 4729#

35. Study the letter-digit table and the conditions given below and answer the question.

Letter : P M A K R E T

Digit : 4 7 2 9 5 3 8

Conditions:

(i) If the first letter of the group is a vowel, both the first and the last letter are coded as \$.

(ii) If the last letter is a vowel and the first letter is not a vowel, the last letter is coded as #.

What is the code for the group KRAMP?

- (A) 95274
- (B) \$5274
- (C) 9527#
- (D) 95278

36. Study the letter-digit table and the conditions given below and answer the question.

Letter : B J N S U D I

Digit : 6 1 4 9 3 7 2

Conditions:

(i) If both the first and the last letter are consonants, the first letter is coded as *.

(ii) If the first letter is a vowel and the last letter is a consonant, the codes of the first and the last letter are interchanged.

What is the code for the group UDNSB? HARD

- (A) 67493
- (B) 37496
- (C) *7496
- (D) 67496

37. Study the letter-digit table and the conditions given below and answer the question.

Letter : B J N S U D I

Digit : 6 1 4 9 3 7 2

Conditions:

(i) If both the first and the last letter are consonants, the first letter is coded as *.

(ii) If the first letter is a vowel and the last letter is a consonant, the codes of the first and the last letter are interchanged.

What is the code for the group SNBDJ?

- (A) 94671
- (B) *4671
- (C) 14679
- (D) *4679

38. In the matrix given below, a letter is coded by its row number followed by its column number (the letter Z is not used).

C0 C1 C2 C3 C4

R0 : A B C D E

R1 : F G H I J

R2 : K L M N O

R3 : P Q R S T

R4 : U V W X Y

What is the code for the word CAT?

- (A) 20-00-43
- (B) 02-00-43
- (C) 02-00-34
- (D) 03-01-34

39. In the matrix given below, a letter is coded by its row number followed by its column number (the letter Z is not used).

C0 C1 C2 C3 C4

R0 : A B C D E

R1 : F G H I J

R2 : K L M N O

R3 : P Q R S T

R4 : U V W X Y

Which word is represented by the code 10-13-32-04?

- (A) FILE
- (B) FIRM
- (C) HIRE
- (D) FIRE

40. In the matrix given below, a letter is coded by its row number followed by its column number (the letter Z is not used).

C0 C1 C2 C3 C4

R0 : A B C D E

R1 : F G H I J

R2 : K L M N O

R3 : P Q R S T

R4 : U V W X Y

What is the code for the word MOON?

- (A) 22-42-42-32
- (B) 22-24-24-32
- (C) 22-24-24-23
- (D) 23-24-24-22

41. BOOK : CPPL :: LOCK : ?

- | | |
|----------|----------|
| (A) MPDJ | (B) KNBJ |
| (C) MQDL | (D) MPDL |

42. GATE : HBUF :: DOOR : ?

- | | |
|----------|----------|
| (A) EPPR | (B) EPPT |
| (C) CNNQ | (D) EPPS |

43. CAT : XZG :: DOG : ?

- | | |
|---------|---------|
| (A) WMT | (B) WLT |
| (C) WLU | (D) VLT |

44. MOTHER : PRWKHU :: FATHER : ?

- | | |
|------------|------------|
| (A) IDWKGU | (B) IDWJHU |
| (C) HCVJGT | (D) IDWKHU |

45. LIGHT : JGEFR :: SOUND : ?

- | | |
|-----------|-----------|
| (A) QMSLB | (B) QMSLD |
| (C) QNSLB | (D) UQWPF |

RM05

Blood Relations

40 questions · 24 fact-sheet points

Must-know facts

1. METHOD: draw the tree top-down. Put the eldest generation on the top line, the next generation below it, and never mix two generations on one line.
2. SYMBOLS to use while solving: + for male, - for female, = or a double line for a married couple, a vertical line for parent-to-child, a horizontal line for siblings.
3. Read a chain left to right and add one link at a time; never try to jump from the first name to the last name in one step.
4. 'X is the son/daughter of Y' -> Y goes one line ABOVE X. 'X is the father/mother of Y' -> X goes one line above Y.
5. 'X is the brother/sister of Y' -> same line, so X automatically inherits every parent of Y.
6. Marriage assumption used in every exam puzzle: a married couple are the parents of the same children, so the spouse of a parent is also a parent.
7. Only son / only daughter / only child are the strongest clues - they force a unique person; solve those clues first.
8. 'My mother's only son' = the speaker himself (if male). 'My father's only child' = the speaker.
9. MATERNAL means through the mother; PATERNAL means through the father. Mama = mother's brother (maternal uncle); Chacha = father's brother (paternal uncle).
10. Mami = wife of mother's brother = maternal aunt. Chachi = wife of father's brother = paternal aunt. Bua = father's sister; Mausi = mother's sister.
11. Grandfather via father = paternal grandfather (dada); via mother = maternal grandfather (nana).
12. Child of an uncle or aunt = cousin. Child of a brother or sister = nephew (male) or niece (female).
13. IN-LAWS: parent of spouse = father/mother-in-law; spouse of child = son/daughter-in-law; sibling of spouse OR spouse of sibling = brother/sister-in-law.
14. GENERATION COUNTING: each 'father/mother of' or 'son/daughter of' step changes the generation by one; 'brother/sister/husband/wife of' keeps the same generation.
15. Two parent-steps up = grandparent; three parent-steps up = great-grandparent. Count the steps, do not guess the name.
16. PHOTOGRAPH TYPE: always start from the SPEAKER, mark the speaker on the tree first, then move outward exactly as the sentence reads.
17. In a photograph question the gender of the person in the photograph is given by 'he/she/his/her' in the stem - use it, because it decides son vs daughter, nephew vs niece.
18. CODED RELATIONS: decode the expression pair by pair, left to right, i.e. for 'P + Q - R' first apply P+Q, then Q-R.
19. In a coded expression the FIRST letter of each pair is the one that holds the stated relation; a common error is reversing the direction.
20. For 'which expression means X is the ... of Y', build the tree for each option separately rather than reasoning verbally.
21. The gender of a middle letter in a coded chain is often not determined - if the answer depends on it, the correct choice is 'cannot be determined'.
22. Counting members: count each named person once; a married-in spouse is a member of the family but not a blood relation.

23. Remember that a person can be both a son and a father, so the same node may carry two relations at once - label the node, not the sentence.
24. Fastest check: after building the tree, re-read every sentence of the question and tick it off against the tree; if all tick, the tree is right.

1. Pointing to a man in a photograph, Reena said, "His mother's only daughter is my mother." How is the man related to Reena? **RAS PRE 2023**

- (A) Paternal uncle (B) Father
(C) Maternal grandfather (D) Maternal uncle

2. Introducing a boy, a girl said, "He is the son of my grandfather's only child." How is the boy related to the girl?

- (A) Uncle (B) Nephew
(C) Cousin brother (D) Brother

3. A is the father of B. B is the mother of C. C is the sister of D. If D is a male, how is A related to D?

- (A) Father (B) Maternal uncle
(C) Paternal grandfather (D) Maternal grandfather

4. P is the brother of Q. Q is the son of R. R is the wife of S. How is P related to S? **RAS PRE 2018**

- (A) Nephew (B) Son-in-law
(C) Brother (D) Son

5. In a certain code, 'A + B' means 'A is the father of B', 'A - B' means 'A is the sister of B', 'A × B' means 'A is the mother of B' and 'A ÷ B' means 'A is the brother of B'. Which of the following means 'P is the paternal grandfather of Q'? **HARD**

- (A) $P \times R + Q$ (B) $P + R + Q$
(C) $P + R \times Q$ (D) $P \div R + Q$

6. Using the code 'A + B' = 'A is the father of B', 'A - B' = 'A is the sister of B', 'A × B' = 'A is the mother of B' and 'A ÷ B' = 'A is the brother of B', how is M related to O in 'M × N - O'?

- (A) Grandmother (B) Sister
(C) Aunt (D) Mother

7. Using the code 'A + B' = 'A is the father of B', 'A - B' = 'A is the sister of B', 'A × B' = 'A is the mother of B' and 'A ÷ B' = 'A is the brother of B', how is K related to M in 'K ÷ L × M'? **RAS PRE 2021**

- (A) Paternal uncle (B) Father
(C) Brother (D) Maternal uncle

8. In a code, 'A + B' means 'A is the brother of B', 'A - B' means 'A is the sister of B', 'A × B' means 'A is the father of B' and 'A ÷ B' means 'A is the mother of B'. How is D related to F in 'D + E ÷ F'? **HARD**

- (A) Maternal uncle (B) Brother
(C) Son (D) Father

9. Using the code 'A + B' = 'A is the brother of B', 'A - B' = 'A is the sister of B', 'A × B' = 'A is the father of B' and 'A ÷ B' = 'A is the mother of B', how is X related to Z in 'X × Y + Z'?

- (A) Grandfather (B) Brother
(C) Uncle (D) Father

- 10.** In a code, 'A + B' means 'A is the husband of B', 'A - B' means 'A is the mother of B', 'A × B' means 'A is the son of B' and 'A ÷ B' means 'A is the daughter of B'. How is S related to U in 'S - T × U'?
- (A) Wife (B) Sister
(C) Mother (D) Daughter
- 11.** Using the code 'A + B' = 'A is the husband of B', 'A - B' = 'A is the mother of B', 'A × B' = 'A is the son of B' and 'A ÷ B' = 'A is the daughter of B', how is P related to R in 'P + Q ÷ R'?
- (A) Brother-in-law (B) Father
(C) Son (D) Son-in-law
- 12.** In a code, 'A + B' means 'A is the son of B', 'A - B' means 'A is the daughter of B', 'A × B' means 'A is the wife of B' and 'A ÷ B' means 'A is the father of B'. How is A related to C in 'A - B + C'?
- (A) Granddaughter (B) Niece
(C) Daughter (D) Sister
- 13.** Pointing to a lady, a man said, "She is the daughter of the only sister of my father." How is the lady related to the man?
- (A) Sister (B) Niece
(C) Cousin sister (D) Aunt
- 14.** Pointing to a photograph, Meena said, "He is the son of the only son of my grandmother." How is the boy in the photograph related to Meena? RAS PRE 2016
- (A) Uncle (B) Cousin brother
(C) Nephew (D) Brother
- 15.** Pointing to a girl, a man said, "She is the daughter of my wife's only son." How is the girl related to the man?
- (A) Granddaughter (B) Sister
(C) Daughter (D) Niece
- 16.** Pointing to a boy, a woman said, "He is the son of the only brother of my mother's daughter." How is the woman related to the boy, if the woman has only one brother?
- (A) Maternal aunt (B) Mother
(C) Paternal aunt (D) Sister
- 17.** Pointing to a woman in a photograph, a man said, "She is the wife of the only son of my mother other than myself." How is the woman related to the man?
- (A) Mother (B) Wife
(C) Sister (D) Sister-in-law
- 18.** Pointing to a man, a lady said, "He is the son of the only brother of my mother." How is the man related to the lady?
- (A) Cousin brother (B) Nephew
(C) Maternal uncle (D) Brother
- 19.** A is the father of B, B is the mother of C and C is the father of D. How is A related to D?
- (A) Maternal uncle (B) Grandfather
(C) Great-grandfather (D) Father
- 20.** R is married to S. T is the son of R. T is married to U and V is the son of T. How is S related to V?
- (A) Aunt (B) Mother-in-law
(C) Mother (D) Grandmother

21. A and B are a married couple. C and D are their son and daughter respectively. E is married to D. How is E related to B?

RAS PRE 2024

- | | |
|----------------|--------------------|
| (A) Son-in-law | (B) Nephew |
| (C) Son | (D) Brother-in-law |

22. P is the husband of Q. R is the son of P. S is the sister of R. T is the husband of S. How is T related to R?

- | | |
|--------------------|----------------|
| (A) Father-in-law | (B) Brother |
| (C) Brother-in-law | (D) Son-in-law |

23. X is married to Y and Z is their daughter. Z is married to W, and V is the daughter of W. How is Y related to V?

- | | |
|-------------------|-----------------|
| (A) Mother-in-law | (B) Mother |
| (C) Aunt | (D) Grandmother |

24. A and B are husband and wife. C is the son of A. D is the brother of A and E is the daughter of D. How is E related to C?

- | | |
|-------------------|------------|
| (A) Cousin sister | (B) Niece |
| (C) Aunt | (D) Sister |

25. A and B are husband and wife. C is the son of A. D is the brother of A and E is the daughter of D. How is B related to E?

- | | |
|-------------------|-------------------|
| (A) Mother | (B) Maternal aunt |
| (C) Paternal aunt | (D) Grandmother |

26. In a family, K is married to L. M and N are the son and daughter of K. O is the son of M and P is the daughter of N. How is O related to P?

- | | |
|-------------|--------------------|
| (A) Uncle | (B) Nephew |
| (C) Brother | (D) Cousin brother |

27. In a family, K is married to L. M and N are the son and daughter of K. O is the son of M and P is the daughter of N. How is L related to O?

- | | |
|-----------------|------------|
| (A) Grandmother | (B) Aunt |
| (C) Sister | (D) Mother |

28. In a family of six members A, B, C, D, E and F, A and B are a married couple, A being the male member. D is the only son of C, who is the brother of A. E is the sister of D. B is the daughter-in-law of F, whose husband has died. How is E related to A? **HARD**

- | | |
|-------------------|--------------|
| (A) Cousin sister | (B) Daughter |
| (C) Niece | (D) Sister |

29. In a family of six members A, B, C, D, E and F, A and B are a married couple, A being the male member. D is the only son of C, who is the brother of A. E is the sister of D. B is the daughter-in-law of F, whose husband has died. How is F related to D? **HARD**

- | | |
|------------|-------------------|
| (A) Aunt | (B) Mother-in-law |
| (C) Mother | (D) Grandmother |

30. A family has five members P, Q, R, S and T. Q is the son of R, but R is not the mother of Q. P and R are a married couple. T is the brother of R. S is the daughter of P. How is Q related to S? **RAS PRE 2013**

- | | |
|-------------|------------|
| (A) Brother | (B) Cousin |
| (C) Uncle | (D) Nephew |

31. A family has five members P, Q, R, S and T. Q is the son of R, but R is not the mother of Q. P and R are a married couple. T is the brother of R. S is the daughter of P. How is T related to S?

- | | |
|--------------------|--------------------|
| (A) Father | (B) Maternal uncle |
| (C) Paternal uncle | (D) Brother |

- 32.** A is married to B. C and D are the son and daughter of A. C is married to E and G is the son of C. How is D related to G?
- (A) Maternal aunt (B) Mother
(C) Sister (D) Paternal aunt
- 33.** A is married to B. C and D are the son and daughter of A. C is married to E and G is the son of C. How is E related to B?
- (A) Daughter-in-law (B) Sister-in-law
(C) Granddaughter (D) Daughter
- 34.** A is the sister of B. B is the father of C. C is the mother of D. How is A related to C?
- (A) Mother (B) Maternal aunt
(C) Paternal aunt (D) Grandmother
- 35.** A is the mother of B, B is the father of C, C is the father of D and D is the mother of E. How many generations does this family span?
- (A) Three (B) Four
(C) Five (D) Six
- 36.** A is the mother of B. B is the brother of C. C is the mother of D. D is the brother of E. How many generations are there in this family?
- (A) Two (B) Three
(C) Four (D) Five
- 37.** P is married to Q and R is their only son. R is married to S; T and U are the son and daughter of R. Which of the following statements is correct?
- (A) S is the daughter-in-law of Q (B) T is the son of Q
(C) U is the sister of S (D) P is the father of S
- 38.** Pointing to a woman, a man said, "She is the wife of the only brother of my mother." How is the woman related to the man?
- (A) Paternal aunt (B) Mother-in-law
(C) Sister-in-law (D) Maternal aunt
- 39.** In a family, A is the husband of B. C and D are their son and daughter. C is married to G and E is the son of C. Excluding no one, how many male members are there in this family?
- (A) Two (B) Three
(C) Four (D) Five
- 40.** A man said, "That man's mother's mother is my mother's mother." Given that the two men are not brothers, how is that man related to the speaker?
- (A) Cousin brother (B) Maternal uncle
(C) Nephew (D) Brother

RM06

Direction Sense

35 questions · 23 fact-sheet points

Must-know facts

1. STANDARD MAP CONVENTION: North at the top, South at the bottom, East to the right, West to the left of the page. Draw it once in the corner of the rough sheet.
2. Clockwise order of the four cardinal directions: North → East → South → West → North. Anti-clockwise is the reverse.
3. A RIGHT turn moves the facing one step clockwise; a LEFT turn moves it one step anti-clockwise.
4. Facing North: right = East, left = West. Facing South: right = West, left = East. Facing East: right = South, left = North. Facing West: right = North, left = South.
5. The single most tested trap: when the person turns around and faces the opposite direction, his left and right INTERCHANGE - a left turn now points where a right turn pointed before.
6. 45 degrees = one step on the eight-point compass (N, NE, E, SE, S, SW, W, NW). 90 degrees = two steps, 135 degrees = three steps, 180 degrees = four steps.
7. The eight-point compass in clockwise order: N, NE, E, SE, S, SW, W, NW - memorise it as a ring for angle-turn questions.
8. Always convert the walk into coordinates: North +y, South -y, East +x, West -x. Add the signed lengths; the sums give the final position at once.
9. Net displacement, not total distance walked, decides 'how far is he from the starting point'.
10. If the North and South legs cancel and the East and West legs cancel, the person is back at the starting point.
11. SHORTEST DISTANCE between start and finish when the net movement has both an x and a y part = $\sqrt{x^2 + y^2}$.
12. Memorise the Pythagorean triples that examiners use: 3-4-5, 6-8-10, 5-12-13, 9-12-15, 8-15-17, 7-24-25, 20-21-29, 12-16-20, 15-20-25.
13. If x and y are equal, the distance is $x * \sqrt{2}$ (about 1.414x) and the direction is exactly NE, NW, SE or SW.
14. Direction of the end point from the start: +x +y = North-East, -x +y = North-West, +x -y = South-East, -x -y = South-West.
15. Direction of A from B is exactly the opposite of the direction of B from A - reverse both letters (North-East becomes South-West).
16. SUN RULE: the Sun rises in the East, so in the MORNING every shadow falls towards the WEST; it sets in the West, so in the EVENING every shadow falls towards the EAST.
17. At noon (12 o'clock) the Sun is overhead and the shadow is negligible - questions on 'shadow at noon' are usually invalid or answer 'no shadow'.
18. Shadow in front of you in the morning means you face West; behind you means you face East; on your right means you face South; on your left means you face North.
19. In the evening every one of those answers is reversed, because the shadow now points East.
20. Two persons facing each other always face opposite directions - if one faces North the other faces South.
21. For a question combining direction with blood relations, solve the family tree and the coordinate diagram separately, then answer each part.

22. Sketch every leg roughly to scale; a picture that is even roughly to scale prevents sign errors in the final subtraction.
23. In 'minimum distance to return', the answer is the straight-line distance, not the retraced path.

1. Ramesh walks 10 km towards the North, then turns right and walks 6 km, then turns right again and walks 10 km. How far is he now from his starting point? RAS PRE 2024
- (A) 4 km (B) 6 km
(C) 10 km (D) 16 km
2. Ramesh walks 10 km towards the North, then turns right and walks 6 km, then turns right again and walks 10 km. In which direction is he now from his starting point?
- (A) North (B) South
(C) East (D) West
3. A man walks 4 km towards the North and then 3 km towards the East. How far is he from his starting point? RAS PRE 2023
- (A) 7 km (B) 6 km
(C) 12 km (D) 5 km
4. Vijay walks 5 km towards the South, turns left and walks 3 km, then turns left again and walks 5 km. How far and in which direction is he from the starting point?
- (A) 3 km East (B) 3 km West
(C) 5 km East (D) 8 km North
5. A cyclist rides 12 km towards the East and then 5 km towards the North. What is the shortest distance between his starting point and his present position?
- (A) 11 km (B) 13 km
(C) 17 km (D) 15 km
6. Sunita walks 8 km towards the South and then 6 km towards the West. How far is she from her starting point and in which direction?
- (A) 10 km North-East (B) 12 km South-West
(C) 10 km South-West (D) 14 km South-West
7. Mohan goes 15 metres towards the North, then turns right and walks 20 metres. How far is he from his starting point?
- (A) 35 metres (B) 20 metres
(C) 30 metres (D) 25 metres
8. A man walks 25 km East, then 10 km North, then 15 km West. How far is he from his starting point?
- (A) 14.14 km (B) 15 km
(C) 20 km (D) 10 km
9. A man walks 25 km East, then 10 km North, then 15 km West. In which direction is he from his starting point?
- (A) East (B) North-East
(C) North-West (D) South-East
10. Amit walks 3 km North, turns right and walks 4 km, turns right and walks 3 km, and finally turns right and walks 2 km. How far is he from the starting point?
- (A) 6 km (B) 1 km
(C) 2 km (D) 4 km

- 11.** A man is facing North. He turns 90 degrees in the clockwise direction and then 180 degrees in the anti-clockwise direction. Which direction is he facing now?
- (A) North (B) South
(C) East (D) West
- 12.** Ravi is facing South-West. He turns 135 degrees in the clockwise direction and then 180 degrees in the anti-clockwise direction. Which direction is he facing now? **HARD**
- (A) South (B) North-West
(C) West (D) North-East
- 13.** Deepak was facing East. He turned to his left and walked, then turned to his right and walked, and finally turned right again. In which direction is he facing now?
- (A) North (B) South
(C) East (D) West
- 14.** If South-East becomes North, North-East becomes West and so on (i.e. the whole compass is rotated), what will West become?
- (A) North-East (B) South-East
(C) South (D) North-West
- 15.** One morning, just after sunrise, Suresh was standing facing a pole. The shadow of the pole fell exactly to his right. Which direction was Suresh facing? **RAS PRE 2013**
- (A) West (B) North
(C) South (D) East
- 16.** Kavita, walking in the evening just before sunset, found that her own shadow fell exactly to her left. In which direction was she walking?
- (A) East (B) West
(C) North (D) South
- 17.** At sunrise two friends stand facing each other. The shadow of the first friend falls exactly behind him. Which direction is the second friend facing?
- (A) West (B) North
(C) South (D) East
- 18.** A boy walks 7 km North, turns right and walks 5 km, turns right and walks 9 km, then turns left and walks 3 km. In which direction is he from the starting point?
- (A) North-East (B) South-East
(C) South-West (D) North-West
- 19.** A jeep travels 9 km towards the West and then 12 km towards the North. What is the shortest distance from the starting point? **RAS PRE 2016**
- (A) 18 km (B) 13 km
(C) 15 km (D) 21 km
- 20.** A man goes 24 km South and then 7 km East. What is the minimum distance he must travel to return to his starting point?
- (A) 31 km (B) 28 km
(C) 17 km (D) 25 km
- 21.** A man walks 4 km East, 3 km North, 4 km East and again 3 km North. How far is he from his starting point?
- (A) 10 km (B) 14 km
(C) 12 km (D) 8 km

- 22.** Starting from a point, a man walks 20 m North, 10 m East, 5 m South, 10 m West and 15 m South. Where is he with respect to the starting point?
- (A) 5 m South of it (B) At the starting point
(C) 5 m North of it (D) 10 m East of it
- 23.** Anita walks 6 km North, then 8 km West, then 6 km South. How far is she from the starting point and in which direction?
- (A) 8 km East (B) 14 km West
(C) 8 km West (D) 10 km North-West
- 24.** A man starts walking towards the North. After 5 km he turns right and walks 5 km, then turns right and walks 5 km, then turns left and walks 5 km. In which direction is he finally moving?
- (A) West (B) North
(C) South (D) East
- 25.** A man starts walking towards the North. After 5 km he turns right and walks 5 km, then turns right and walks 5 km, then turns left and walks 5 km. How far is he from the starting point?
- (A) 10 km (B) 5 km
(C) 20 km (D) 15 km
- 26.** Village A is 5 km North of village B, village C is 5 km East of village B, and village D is 5 km North of village C. In which direction is village D from village A?
- (A) North (B) East
(C) South (D) West
- 27.** A bus goes 30 km East and then 40 km North. If it now returns to the starting point by the shortest route, what total distance has it covered in the whole journey?
- (A) 100 km (B) 110 km
(C) 120 km (D) 70 km
- 28.** Priya walks 10 km South, turns left and walks 5 km, turns left again and walks 10 km, then turns right and walks 4 km. How far is she from her starting point?
- (A) 5 km (B) 1 km
(C) 14 km (D) 9 km
- 29.** If a person walks towards the North-East and then turns 90 degrees clockwise, in which direction is he moving?
- (A) South-East (B) North-West
(C) South-West (D) East
- 30.** Facing North-West, a soldier turns 45 degrees anti-clockwise, then 180 degrees clockwise, and then 90 degrees anti-clockwise. Which direction is he facing?
- (A) West (B) North
(C) South (D) East
- 31.** A's house is 3 km North of B's house and C's house is 4 km East of B's house. A is the father of B and B is the father of C. What is the distance between A's house and C's house, and how is C related to A? **HARD**
- (A) 5 km; nephew (B) 4 km; grandson
(C) 5 km; grandson (D) 7 km; son
- 32.** P's house is 8 km West of Q's house, and R's house is 6 km North of Q's house. P is the brother of Q and Q is the mother of R. What is the shortest distance between P's house and R's house, and how is P related to R? **HARD**
- (A) 14 km; paternal uncle (B) 10 km; father
(C) 12 km; maternal uncle (D) 10 km; maternal uncle

33. Starting from her home, Z walks 5 km North to reach the home of X, who is her father, and then 5 km East to reach a temple. Y, the wife of X, is Z's ____ and the temple lies in which direction from Z's home? **HARD**

- (A) mother; North-East
- (B) aunt; North-East
- (C) mother; South-East
- (D) sister; North

34. A man walks 12 km North, turns right and walks 9 km, turns right and walks 12 km, and turns right again and walks 15 km. How far and in which direction is he from the starting point?

- (A) 24 km West
- (B) 6 km West
- (C) 6 km East
- (D) 9 km West

35. From a point, a man walks 5 km South, 5 km East, 5 km North and 5 km East. What is his distance and direction from the starting point? **RAS PRE 2021**

- (A) 10 km West
- (B) 20 km East
- (C) 10 km East
- (D) 5 km East

RM07

Ranking, Seating Arrangement and Logical Venn Diagrams

55 questions · 29 fact-sheet points

Must-know facts

1. RANKING: rank from the bottom = (total) - (rank from the top) + 1. The +1 is because the person is counted at both ends.
2. When both ranks are of the SAME person, total = (rank from one end) + (rank from the other end) - 1.
3. When the two ranks are of DIFFERENT persons and they do not overlap, number between them = total - (rank of first from left) - (rank of second from right).
4. Shifting: if a person moves k places to the right, the new position from the left = old position + k ; from the right it decreases by k .
5. After two persons interchange seats, each takes over the other's old position - convert the new rank into the old person's rank first.
6. COMPARISON PUZZLES: write one single inequality chain ($A > B > C \dots$) instead of separate pairs; the tallest is at one end, the shortest at the other.
7. LINEAR ROW FACING NORTH: the persons' right is towards your right as you read the row; left is towards your left.
8. LINEAR ROW FACING SOUTH: everything reverses - the persons' right is towards your LEFT. Write the row once and mentally flip it.
9. 'X sits third to the left of Y' means exactly two persons sit between them; 'n persons sit between X and Y' means the gap is $n+1$ seats.
10. Start a row puzzle from the clue that fixes an END seat or the exact MIDDLE seat; those clues have the fewest possibilities.
11. Use a fixed number of blank dashes equal to the number of seats and slide the largest block of clues along it.
12. DOUBLE ROW: row 1 facing North and row 2 facing South sit opposite each other, so the two rows' left-right senses are mirror images.
13. In a double row, if X faces Y then the person to X's right faces the person to Y's left.
14. CIRCULAR TABLE, ALL FACING THE CENTRE: the neighbour in the clockwise direction is on a person's LEFT and the anti-clockwise neighbour is on the RIGHT.
15. CIRCULAR TABLE, ALL FACING OUTWARD: exactly the reverse - clockwise is the RIGHT and anti-clockwise is the LEFT.
16. Fix one person's seat as the reference point and place everyone else relative to it; a circle has no absolute position, only relative order.
17. In a circle of n persons the seat exactly opposite is $n/2$ seats away (defined only for even n).
18. Number of persons between X and Y counted clockwise = (clockwise gap) - 1; counted the other way it is $n - (\text{clockwise gap}) - 1$.
19. SQUARE TABLE, one person per corner facing the centre: 'diagonally opposite' means two corners away, which is also 'second to the left' and 'second to the right'.
20. FLOOR PUZZLES: always draw floor 1 at the bottom and the top floor at the top; 'above' means a larger floor number.
21. 'Only k persons live between X and Y' fixes the gap but not which of the two is higher - test both cases before rejecting.

22. DAY / MONTH PUZZLES: treat the days or months as numbered slots 1 to n and convert every 'before/after' clue into a difference of slot numbers.
23. Always verify a finished arrangement by reading every clue once more; a puzzle with a valid but second solution means a clue was mis-read.
24. LOGICAL VENN: if every A is a B, draw A wholly inside B. If some A are B, draw two partly overlapping circles. If no A is a B, draw two separate circles.
25. Three fully nested circles (e.g. Jaipur - Rajasthan - India) is the standard 'nested' pattern; two separate circles inside a bigger one (Table - Chair - Furniture) is the standard 'co-ordinate' pattern.
26. TWO-SET FORMULA: $n(A \text{ or } B) = n(A) + n(B) - n(A \text{ and } B)$; those in neither = Total - $n(A \text{ or } B)$.
27. THREE-SET FORMULA: $n(A \text{ or } B \text{ or } C) = n(A) + n(B) + n(C) - n(A \text{ and } B) - n(B \text{ and } C) - n(A \text{ and } C) + n(A \text{ and } B \text{ and } C)$.
28. Exactly one of three = sum of singles - 2(sum of pairwise) + 3(all three). Exactly two of three = sum of pairwise - 3(all three).
29. In set questions the pairwise figures normally INCLUDE the people who take all three, unless the paper says 'only'.

1. In a class of 40 students, Ravi's rank from the top is 12th. What is his rank from the bottom? RAS PRE 2018

- (A) 27th (B) 30th
(C) 28th (D) 29th

2. In a queue, Meena is 10th from the front and 15th from the back. How many persons are there in the queue? RAS PRE 2023

- (A) 23 (B) 24
(C) 25 (D) 26

3. In a row of 40 boys, Suresh is 12th from the left end and Rakesh is 18th from the right end. How many boys sit between Suresh and Rakesh?

- (A) 9 (B) 10
(C) 11 (D) 12

4. In a row of 30 children, Aman was shifted 4 places to the right, after which he became 12th from the right end. What was his original position from the left end?

- (A) 14th (B) 15th
(C) 16th (D) 19th

5. Rahul is 15th from the top and 20th from the bottom in a class. How many students are there in the class?

- (A) 33 (B) 34
(C) 35 (D) 36

6. In a row of children, Nita is 7th from the left end and Radha is 9th from the right end. When they interchange their places, Nita becomes 11th from the left end. How many children are there in the row?

- (A) 17 (B) 18
(C) 19 (D) 20

7. In a class of 45 students, Kamal's rank is 12th from the top. Naveen's rank is 6 ranks below Kamal's. What is Naveen's rank from the bottom?

- (A) 28th (B) 29th
(C) 26th (D) 27th

- 8.** In a row of 21 girls, when Seema was shifted two places towards the left she became 8th from the left end. What was her earlier position from the right end?
- (A) 10th (B) 11th
(C) 12th (D) 14th
- 9.** Among five friends A, B, C, D and E: A is taller than B but shorter than C; D is taller than C; E is shorter than B. Who is the tallest of them all?
- (A) C (B) D
(C) E (D) A
- 10.** Five boxes P, Q, R, S and T have different weights. P is heavier than S, R is heavier than P, Q is heavier than R and T is lighter than S. Which box is the third heaviest?
- (A) S (B) T
(C) P (D) R
- 11.** Seven friends A, B, C, D, E, F and G are sitting in a straight row facing North. The following information is known about their seating:
D sits third to the right of A.
E sits second to the left of G.
C and E sit next to each other.
F sits second to the right of B.
Who is sitting at the extreme right end of the row? RAS PRE 2024
- (A) B
(B) A
(C) C
(D) F
- 12.** Seven friends A, B, C, D, E, F and G are sitting in a straight row facing North. The following information is known about their seating:
D sits third to the right of A.
E sits second to the left of G.
C and E sit next to each other.
F sits second to the right of B.
Who sits third to the left of B?
- (A) E
(B) G
(C) A
(D) C
- 13.** Six persons P, Q, R, S, T and U sit in a straight row facing South. (For a person facing South, the printed row runs from that person's right to that person's left.) The following facts are known:
U sits immediately to the right of P.
S sits third to the left of Q.
R is not an immediate neighbour of T.
R sits at the extreme left end of the row.
Who sits at the extreme right end of the row?
- (A) S
(B) T
(C) P
(D) Q

14. Six persons P, Q, R, S, T and U sit in a straight row facing South. (For a person facing South, the printed row runs from that person's right to that person's left.) The following facts are known:

U sits immediately to the right of P.

S sits third to the left of Q.

R is not an immediate neighbour of T.

R sits at the extreme left end of the row.

How many persons sit between S and Q?

- (A) One
- (B) Two
- (C) Three
- (D) Four

15. Eight students A, B, C, D, E, F, G and H are seated in a straight row facing North. It is known that:

C sits third to the right of H.

G is not an immediate neighbour of E.

H is not an immediate neighbour of F.

Only one person sits between G and H.

D sits at the extreme left end of the row.

C is not an immediate neighbour of G.

Two persons sit between B and F.

G sits third to the left of A.

Who is sitting exactly between H and F? HARD

- (A) D
- (B) G
- (C) A
- (D) E

16. Eight students A, B, C, D, E, F, G and H are seated in a straight row facing North. It is known that:

C sits third to the right of H.

G is not an immediate neighbour of E.

H is not an immediate neighbour of F.

Only one person sits between G and H.

D sits at the extreme left end of the row.

C is not an immediate neighbour of G.

Two persons sit between B and F.

G sits third to the left of A.

Which of the following pairs occupies the two extreme ends of the row?

- (A) D and E
- (B) D and C
- (C) G and E
- (D) B and E

17. Five persons V, W, X, Y and Z sit in a row facing North, and the following is known:

W sits second to the right of V.

X and Z sit next to each other.

Only one person sits between X and V.

Who sits exactly in the middle of the row?

- (A) Z
- (B) W
- (C) Y
- (D) V

18. Five persons V, W, X, Y and Z sit in a row facing North, and the following is known:

W sits second to the right of V.

X and Z sit next to each other.

Only one person sits between X and V.

Who sits immediately to the right of Y?

- (A) W
- (B) V
- (C) Z
- (D) X

19. Ten persons are seated in two parallel rows of five each. A, B, C, D and E sit in the first row facing North, while P, Q, R, S and T sit in the second row facing South, and each member of the first row faces exactly one member of the second row. The following is known:

A sits immediately to the right of C.

Three persons sit between C and D.

Only one person sits between A and B.

Only one person sits between T and R.

C faces S.

R sits third to the left of Q.

Who faces B? **HARD**

- (A) T
- (B) P
- (C) Q
- (D) S

20. Ten persons are seated in two parallel rows of five each. A, B, C, D and E sit in the first row facing North, while P, Q, R, S and T sit in the second row facing South, and each member of the first row faces exactly one member of the second row. The following is known:

A sits immediately to the right of C.

Three persons sit between C and D.

Only one person sits between A and B.

Only one person sits between T and R.

C faces S.

R sits third to the left of Q.

Who sits at the extreme right end of the second row? **HARD**

- (A) T
- (B) R
- (C) S
- (D) P

21. Ten persons are seated in two parallel rows of five each. A, B, C, D and E sit in the first row facing North, while P, Q, R, S and T sit in the second row facing South, and each member of the first row faces exactly one member of the second row. The following is known:

A sits immediately to the right of C.

Three persons sit between C and D.

Only one person sits between A and B.

Only one person sits between T and R.

C faces S.

R sits third to the left of Q.

How many persons sit between C and D in the first row?

- (A) One
- (B) Two
- (C) Three
- (D) Four

22. Ten persons are seated in two parallel rows of five each. A, B, C, D and E sit in the first row facing North, while P, Q, R, S and T sit in the second row facing South, and each member of the first row faces exactly one member of the second row. The following is known:

A sits immediately to the right of C.

Three persons sit between C and D.

Only one person sits between A and B.

Only one person sits between T and R.

C faces S.

R sits third to the left of Q.

Who sits immediately to the right of Q in the second row? **HARD**

- (A) R
- (B) T
- (C) P
- (D) S

23. Eight friends P, Q, R, S, T, U, V and W are sitting around a circular table facing the centre. The following information is known:

S sits immediately to the right of P.

W sits second to the right of Q.

V sits third to the left of W.

T sits fourth to the left of U.

T sits third to the left of S.

Who sits third to the right of Q?

- (A) P
- (B) T
- (C) V
- (D) S

24. Eight friends P, Q, R, S, T, U, V and W are sitting around a circular table facing the centre. The following information is known:

S sits immediately to the right of P.

W sits second to the right of Q.

V sits third to the left of W.

T sits fourth to the left of U.

T sits third to the left of S.

Who is sitting exactly opposite W? RAS PRE 2021

(A) V

(B) R

(C) T

(D) S

25. Eight friends P, Q, R, S, T, U, V and W are sitting around a circular table facing the centre. The following information is known:

S sits immediately to the right of P.

W sits second to the right of Q.

V sits third to the left of W.

T sits fourth to the left of U.

T sits third to the left of S.

How many persons sit between T and V when counted clockwise starting from T?

(A) One

(B) Two

(C) Three

(D) Four

26. Six persons A, B, C, D, E and F are seated around a circular table facing the centre. It is known that:

B is not an immediate neighbour of C.

B sits third to the left of A.

C sits second to the right of D.

E sits fourth to the right of F.

Who sits immediately to the left of E?

(A) A

(B) D

(C) C

(D) F

27. Six persons A, B, C, D, E and F are seated around a circular table facing the centre. It is known that:

B is not an immediate neighbour of C.

B sits third to the left of A.

C sits second to the right of D.

E sits fourth to the right of F.

Who is sitting exactly opposite B?

(A) D

(B) C

(C) F

(D) A

28. Five persons K, L, M, N and O are sitting around a circular table facing the centre. The following is known:

N sits second to the right of O.

M sits immediately to the right of K.

Who sits second to the left of L?

- (A) M
- (B) O
- (C) N
- (D) K

29. Five persons K, L, M, N and O are sitting around a circular table facing the centre. The following is known:

N sits second to the right of O.

M sits immediately to the right of K.

Who sits immediately to the right of O?

- (A) N
- (B) L
- (C) K
- (D) M

30. Six persons P, Q, R, S, T and U are seated around a circular table but all of them are facing outward, away from the centre. (For a person facing outward the clockwise neighbour is on the right and the anti-clockwise neighbour is on the left.) It is known that:

T sits fourth to the left of S.

T sits third to the left of Q.

U sits immediately to the right of R.

Who sits third to the left of R? HARD

- (A) Q
- (B) T
- (C) S
- (D) P

31. Six persons P, Q, R, S, T and U are seated around a circular table but all of them are facing outward, away from the centre. (For a person facing outward the clockwise neighbour is on the right and the anti-clockwise neighbour is on the left.) It is known that:

T sits fourth to the left of S.

T sits third to the left of Q.

U sits immediately to the right of R.

Who sits second to the right of T?

- (A) R
- (B) S
- (C) Q
- (D) U

32. Six persons P, Q, R, S, T and U are seated around a circular table but all of them are facing outward, away from the centre. (For a person facing outward the clockwise neighbour is on the right and the anti-clockwise neighbour is on the left.) It is known that:

T sits fourth to the left of S.

T sits third to the left of Q.

U sits immediately to the right of R.

Who is sitting exactly opposite P?

- (A) U
- (B) S
- (C) R
- (D) T

33. Four persons A, B, C and D are sitting at the four corners of a square table, one at each corner, all facing the centre of the table. It is known that:

B sits immediately to the right of D.

B sits third to the right of C.

Which of the following pairs sits diagonally opposite each other?

- (A) B and C
- (B) A and B
- (C) A and C
- (D) A and D

34. Four persons A, B, C and D are sitting at the four corners of a square table, one at each corner, all facing the centre of the table. It is known that:

B sits immediately to the right of D.

B sits third to the right of C.

Which of the following gives the correct clockwise order of the four persons?

- (A) A, B, C, D
- (B) A, C, B, D
- (C) A, D, C, B
- (D) A, B, D, C

35. Six persons L, M, N, O, P and Q are seated around a rectangular table, all facing the centre, with two persons on each of the longer sides and one person on each of the shorter sides. Treating the seats as going round the table, the following is known:

O sits third to the right of N.

Q sits immediately to the right of M.

P sits second to the left of O.

Who sits immediately to the right of M?

- (A) P
- (B) O
- (C) Q
- (D) L

36. Five persons A, B, C, D and E live on five different floors of a building. The lowest floor is numbered 1 and the topmost floor is numbered 5. The following is known:

B and D live on adjacent floors.

E lives two floors above C.

B lives two floors above A.

Who lives on the topmost floor?

- (A) A
- (B) B
- (C) C
- (D) D

37. Five persons A, B, C, D and E live on five different floors of a building. The lowest floor is numbered 1 and the topmost floor is numbered 5. The following is known:

B and D live on adjacent floors.

E lives two floors above C.

B lives two floors above A.

Who lives on the third floor?

- (A) E
- (B) C
- (C) D
- (D) A

38. Six persons P, Q, R, S, T and U live on six different floors of a building, the lowest floor being numbered 1 and the topmost floor 6. The following is known:

P lives three floors below S.

Only one person lives between P and Q.

R lives two floors below P.

P lives immediately below T.

How many persons live between P and Q?

- (A) One
- (B) Two
- (C) Three
- (D) Four

39. Six persons P, Q, R, S, T and U live on six different floors of a building, the lowest floor being numbered 1 and the topmost floor 6. The following is known:

P lives three floors below S.

Only one person lives between P and Q.

R lives two floors below P.

P lives immediately below T.

Who lives immediately above T?

- (A) R
- (B) Q
- (C) P
- (D) S

40. Seven persons A, B, C, D, E, F and G live on seven different floors of a building. The lowest floor is numbered 1 and the topmost floor is numbered 7. It is known that:

B and G live on adjacent floors.

A lives two floors below C.

Two persons live between G and E.

G lives immediately below D.

F lives two floors below G.

Who lives on floor number 5?

- (A) C
- (B) F
- (C) A
- (D) D

41. Seven persons A, B, C, D, E, F and G live on seven different floors of a building. The lowest floor is numbered 1 and the topmost floor is numbered 7. It is known that:

B and G live on adjacent floors.

A lives two floors below C.

Two persons live between G and E.

G lives immediately below D.

F lives two floors below G.

How many persons live above D?

- (A) Two
- (B) Three
- (C) Four
- (D) Five

42. Five persons P, Q, R, S and T visit a museum on five different days of the same week, from Monday to Friday, one person per day. It is known that:

R visits the museum two days after P.

T visits the museum two days after S.

Q visits the museum immediately after R.

Who visits the museum on Wednesday?

- (A) P
- (B) Q
- (C) S
- (D) T

43. Five persons P, Q, R, S and T visit a museum on five different days of the same week, from Monday to Friday, one person per day. It is known that:

R visits the museum two days after P.

T visits the museum two days after S.

Q visits the museum immediately after R.

On which day does R visit the museum?

- (A) Thursday
- (B) Friday
- (C) Monday
- (D) Tuesday

44. Six candidates A, B, C, D, E and F have their examinations in six different months of the same year - January, February, March, April, May and June - one candidate per month. It is known that:

C has the examination two months before B.

A has the examination three months before E.

A has the examination immediately after D.

In which month does C have the examination?

- (A) March
- (B) April
- (C) May
- (D) February

45. Six candidates A, B, C, D, E and F have their examinations in six different months of the same year - January, February, March, April, May and June - one candidate per month. It is known that:

C has the examination two months before B.

A has the examination three months before E.

A has the examination immediately after D.

How many candidates have their examination between the examinations of A and E? HARD

- (A) One
- (B) Two
- (C) Three
- (D) Four

46. Which of the following best represents the relationship between the three classes: Doctors, Men and Human beings? RAS PRE 2023

- (A) Three circles that do not touch one another
- (B) Two intersecting circles for Doctors and Men, both lying wholly inside a larger circle for Human beings
- (C) A circle for Men lying inside a circle for Doctors, and that inside a circle for Human beings
- (D) A circle for Doctors lying inside a circle for Men, and that inside a circle for Human beings

47. Which of the following best represents the relationship between Pigeon, Bird and Dog?

- (A) A circle for Pigeon inside a circle for Bird, with a separate circle for Dog outside both
- (B) Three separate circles that intersect one another
- (C) A circle for Bird inside a circle for Pigeon, with Dog intersecting Bird
- (D) A circle for Dog inside a circle for Bird, with Pigeon separate

48. Which of the following best represents the relationship between Jaipur, Rajasthan and India? RAS PRE 2024

- (A) Three separate circles
- (B) Jaipur and Rajasthan intersecting, both inside India
- (C) A circle for Jaipur inside a circle for Rajasthan, which in turn lies inside a circle for India
- (D) A circle for Rajasthan inside a circle for Jaipur, which in turn lies inside a circle for India

49. Which of the following best represents the relationship between Table, Chair and Furniture?

- (A) Two separate circles for Table and Chair, both lying wholly inside a larger circle for Furniture
- (B) Two intersecting circles for Table and Chair, both inside Furniture
- (C) A circle for Chair inside a circle for Table, both inside Furniture
- (D) Three circles intersecting one another

50. Which of the following best represents the relationship between Boys, Girls and Students? RAS PRE 2021

- (A) A circle for Students inside both Boys and Girls
- (B) Three circles all intersecting one another
- (C) Two separate circles for Boys and Girls, each of which partly overlaps a third circle for Students
- (D) Two separate circles for Boys and Girls, both lying wholly inside Students

51. In a class of 60 students, 35 play cricket, 25 play hockey and 10 play both games. How many students play neither game?

RAS PRE 2016

- | | |
|--------|--------|
| (A) 5 | (B) 10 |
| (C) 15 | (D) 20 |

52. In a survey of 100 persons, 65 read newspaper A and 45 read newspaper B, while 20 read both. How many persons read at least one of the two newspapers?

- | | |
|--------|---------|
| (A) 80 | (B) 85 |
| (C) 90 | (D) 110 |

53. In a group of 80 persons, 50 like tea and 40 like coffee. If every person likes at least one of the two drinks, how many like both?

- | | |
|--------|--------|
| (A) 8 | (B) 10 |
| (C) 12 | (D) 15 |

54. Out of 200 students, 100 study Mathematics, 90 study Physics and 80 study Chemistry. 40 study both Mathematics and Physics, 35 both Physics and Chemistry, 30 both Mathematics and Chemistry, and 20 study all three subjects. How many students study none of these three subjects? RAS PRE 2024

- | | |
|--------|--------|
| (A) 10 | (B) 15 |
| (C) 20 | (D) 25 |

55. In a survey of 120 households, 60 use brand X, 55 use brand Y and 50 use brand Z. 25 use both X and Y, 20 use both Y and Z, 22 use both X and Z, and 10 use all three brands. How many households use exactly one of the three brands?

- | | |
|--------|--------|
| (A) 37 | (B) 48 |
| (C) 61 | (D) 71 |

RM08

Shapes, Figure Counting, Mirror and Water Images, Embedded Figures

45 questions · 28 fact-sheet points

Must-know facts

1. Mirror image (lateral inversion) = reflection about a VERTICAL axis; water image = reflection about a HORIZONTAL axis.
2. Capitals unchanged in a MIRROR image: A, H, I, M, O, T, U, V, W, X, Y (11 letters).
3. Capitals unchanged in a WATER image: B, C, D, E, H, I, K, O, X (9 letters).
4. Unchanged in BOTH mirror and water image: H, I, O, X. Digits unchanged in both: 0 and 8.
5. Mirror image of a word = write the word backwards, then mirror each letter; the letter ORDER always reverses.
6. Mirror image of a clock time = 11:60 minus the given time (use 12:00 if the time is exactly 12:00).
7. Water image of a clock time = 18:30 minus the given time (subtract 12 hours if the answer exceeds 12:00).
8. Rectangle (or square) + both diagonals = 8 triangles. Square + both diagonals + both mid-lines = 16 triangles.
9. Rectangle + both diagonals + ONE mid-line = 12 triangles. Triangle + all three medians = 16 triangles.
10. Triangle with n cevians drawn from ONE vertex to the opposite side: number of triangles = $C(n+2, 2)$.
11. Joining the midpoints of a triangle's sides gives 4 small triangles; total in the figure = 5.
12. Equilateral triangle with each side divided into n parts and lines drawn parallel to all sides: $n=2$ gives 5 triangles, $n=3$ gives 13, $n=4$ gives 27.
13. Squares in an $n \times n$ grid = $1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$. For $3 \times 3 = 14$, $4 \times 4 = 30$, $8 \times 8 = 204$.
14. Rectangles in an $m \times n$ grid = $C(m+1, 2) \times C(n+1, 2)$. Chessboard (8×8) = $36 \times 36 = 1296$ rectangles.
15. Counting method for triangles: count the smallest regions first, then triangles made of 2, 3, 4 ... of them, and finally the whole figure.
16. Regular polygon: number of diagonals = $\frac{n(n-3)}{2}$; sum of interior angles = $(n-2) \times 180$ degrees; each exterior angle of a REGULAR polygon = $\frac{360}{n}$.
17. Given each interior angle A of a regular polygon, $n = \frac{360}{(180 - A)}$.
18. Dice: a cube face touches 4 faces and is opposite the 5th. If a number is adjacent to four given numbers, the sixth number is opposite it.
19. Standard dice: numbers on opposite faces add up to 7, i.e. the pairs are 1-6, 2-5, 3-4.
20. Two views of a dice with the SAME face on top: the other four numbers lie around it; a quarter turn matching one pair also matches the other, giving both pairs of opposites.
21. Two views with ONE face common but in different positions: the two faces that never appear together in any single view are opposite each other.
22. Painted cube cut into n^3 smaller cubes (all six faces painted): 3 faces painted = 8 (always the corners); 2 faces = $12(n-2)$; 1 face = $6(n-2)^2$; 0 faces = $(n-2)^3$.
23. Check: $8 + 12(n-2) + 6(n-2)^2 + (n-2)^3 = n^3$. At least one face painted = $n^3 - (n-2)^3$.
24. If only two OPPOSITE faces of a cube are painted and it is cut into n^3 cubes: exactly one face painted = $2n^2$, two faces = 0.

25. Paper folding: n folds double the layers each time, so one punch after n folds gives 2^n holes; k punches give $k \times 2^n$ holes.
26. A punch made ON a fold line gives fewer holes than 2^n , because the layers meet at that line.
27. Embedded / hidden figure questions: count the number of line segments and closed regions of the target shape and match them piece by piece.
28. Classification of quadrilaterals: square, rectangle and rhombus are parallelograms; a trapezium has only one pair of parallel sides and is usually the odd one out.

1. A rectangle ABCD is drawn and both of its diagonals AC and BD are drawn. How many triangles are there in the figure in all? RAS PRE 2023, 2024
- (A) 4 (B) 8
(C) 6 (D) 10
2. A square is drawn. Both its diagonals are drawn, and also the two lines joining the midpoints of opposite sides. How many triangles are there in the figure in all?
- (A) 8 (B) 12
(C) 16 (D) 20
3. A rectangle is divided by both its diagonals and also by the straight line joining the midpoints of its two longer sides. How many triangles are formed in all? RAS PRE 2024
- (A) 8 (B) 12
(C) 10 (D) 14
4. In a triangle all three medians are drawn. How many triangles are there in the resulting figure? HARD
- (A) 12 (B) 16
(C) 14 (D) 18
5. A square is drawn with one diagonal and with both the lines joining the midpoints of opposite sides. How many triangles are there in the figure?
- (A) 4 (B) 8
(C) 10 (D) 6
6. A square is drawn with both its diagonals and with the line joining the midpoints of one pair of opposite sides. The total number of triangles in the figure is
- (A) 8 (B) 10
(C) 16 (D) 12
7. From one vertex of a triangle, two straight lines are drawn to two different points on the opposite side. How many triangles are formed in all?
- (A) 6 (B) 3
(C) 4 (D) 5
8. From one vertex of a triangle, three straight lines are drawn to three different points on the opposite side. The number of triangles in the figure is
- (A) 10 (B) 6
(C) 8 (D) 12

- 9.** The midpoints of the three sides of a triangle are joined to one another. How many triangles are there in the figure so obtained?
- (A) 5 (B) 4
(C) 6 (D) 8
- 10.** Each side of an equilateral triangle is divided into three equal parts and lines are drawn through these points parallel to the three sides. How many triangles are there in the whole figure? **HARD**
- (A) 9 (B) 13
(C) 11 (D) 15
- 11.** A square is divided into 9 equal smaller squares by two horizontal and two vertical lines. How many squares of all sizes are there in the figure? **RAS PRE 2023**
- (A) 14 (B) 9
(C) 10 (D) 13
- 12.** A square is divided into 16 equal smaller squares. The total number of squares of all sizes in the figure is
- (A) 16 (B) 20
(C) 30 (D) 25
- 13.** How many squares of all sizes are there on a standard 8 x 8 chessboard?
- (A) 64 (B) 128
(C) 204 (D) 1296
- 14.** A rectangle is divided into 6 equal parts by two lines parallel to its length and one line parallel to its breadth, forming a 3 x 2 arrangement of small rectangles. How many rectangles of all sizes are there in the figure?
- (A) 6 (B) 18
(C) 12 (D) 24
- 15.** How many rectangles (including squares) of all sizes are there on a standard 8 x 8 chessboard? **HARD**
- (A) 204 (B) 784
(C) 1024 (D) 1296
- 16.** Which one of the following capital letters looks exactly the same as itself in a plane mirror placed vertically beside it? **RAS PRE 2023, 2024**
- (A) F (B) G
(C) T (D) J
- 17.** Which one of the following capital letters remains unchanged in its mirror image?
- (A) B (B) W
(C) C (D) E
- 18.** In the word MOUNTAIN, how many letters remain unchanged in their mirror image?
- (A) 4 (B) 5
(C) 6 (D) 7
- 19.** Choose the correct mirror image of the word REASONING when the mirror is held vertically to its right.
- (A) REASONING (B) GNINOSEAR
(C) GNINOSAER (D) GNINSOAER
- 20.** A clock seen in a mirror shows the time as 3:40. What is the correct time? **RAS PRE 2024**
- (A) 8:20 (B) 9:20
(C) 8:40 (D) 7:20

- 21.** A wall clock actually shows 4:50. What time will its image in a plane mirror show?
- (A) 8:10 (B) 7:10
(C) 7:50 (D) 6:10
- 22.** Which one of the following words reads exactly the same in its mirror image?
- (A) TOOT (B) MOTOR
(C) HIM (D) AXE
- 23.** Which one of the following capital letters remains unchanged in its water image?
- (A) D (B) A
(C) V (D) T
- 24.** In the word CHOICE, how many letters remain unchanged in their water image?
- (A) 3 (B) 4
(C) 5 (D) 6
- 25.** Out of the capital letters A, B, H, M, O and X, how many remain unchanged in their water image?
- (A) 2 (B) 3
(C) 5 (D) 4
- 26.** Which group of capital letters remains unchanged in BOTH the mirror image and the water image?
- (A) A, M, T, V (B) B, C, D, E
(C) H, I, O, X (D) N, S, Z, Y
- 27.** A square sheet of paper is folded exactly in half, and the folded sheet is folded exactly in half once more. A single hole is then punched through all the layers at a point not lying on any fold. On unfolding, how many holes will the sheet have?
- (A) 2 (B) 3
(C) 8 (D) 4
- 28.** A sheet of paper is folded in half three times in succession and one hole is punched through the folded sheet. How many holes will there be when the paper is unfolded?
- (A) 3 (B) 6
(C) 16 (D) 8
- 29.** A square sheet is folded in half twice and then two separate holes are punched through the folded sheet. On unfolding, the number of holes will be
- (A) 4 (B) 6
(C) 8 (D) 16
- 30.** A cubical dice has its six faces numbered 1 to 6. If the face bearing 1 is adjacent to the faces bearing 2, 3, 4 and 5, which number lies on the face opposite to 1?
- (A) 6 (B) 2
(C) 4 (D) 5
- 31.** On a standard dice the numbers on any two opposite faces add up to 7. If the face showing 2 is on top, what number is on the face at the bottom?
- (A) 5 (B) 3
(C) 4 (D) 6

32. Two positions of the same dice are shown below.

Position I: top face 1, front face 2, right face 3.

Position II: top face 1, front face 4, right face 5.

Which number lies opposite the face bearing 2? RAS PRE 2024

- (A) 3
- (B) 5
- (C) 6
- (D) 4

33. Two positions of the same dice are shown below.

Position I: top face 1, front face 2, right face 3.

Position II: top face 2, front face 4, right face 5.

Which number lies opposite the face bearing 3?

- (A) 1
- (B) 5
- (C) 6
- (D) 4

34. Two positions of the same dice are shown below.

Position I: top face 1, front face 2, right face 3.

Position II: top face 3, front face 5, right face 4.

Which number lies opposite the face bearing 1?

- (A) 2
- (B) 4
- (C) 6
- (D) 5

35. Two positions of a cube whose faces are marked A, B, C, D, E and F are given below.

Position I: top face A, front face B, right face C.

Position II: top face A, front face D, right face E.

Which letter is on the face opposite to C?

- (A) B
- (B) E
- (C) D
- (D) F

36. A cube painted on all its six faces is cut into 64 identical smaller cubes. How many of the smaller cubes have exactly two faces painted?

- | | |
|--------|--------|
| (A) 8 | (B) 24 |
| (C) 12 | (D) 32 |

37. A cube is painted on all faces and then cut into 125 identical smaller cubes. How many smaller cubes have exactly one face painted?

- | | |
|--------|--------|
| (A) 36 | (B) 48 |
| (C) 60 | (D) 54 |

38. A painted cube is cut into 27 identical smaller cubes. How many of them have three faces painted?

- | | |
|--------|-------|
| (A) 8 | (B) 6 |
| (C) 12 | (D) 1 |

- 39.** A cube painted on all its faces is cut into 64 identical smaller cubes. How many of the smaller cubes have no face painted at all?
- (A) 4 (B) 12
(C) 8 (D) 16
- 40.** A cube painted red on all faces is cut into 27 identical smaller cubes. How many of these have at least one face painted red?
- (A) 18 (B) 20
(C) 26 (D) 24
- 41.** A cube is painted only on its top and bottom faces, the four side faces being left unpainted. It is then cut into 64 identical smaller cubes. How many smaller cubes have exactly one face painted? **HARD**
- (A) 32 (B) 16
(C) 24 (D) 48
- 42.** How many diagonals does a regular octagon have?
- (A) 16 (B) 20
(C) 24 (D) 28
- 43.** What is the sum of all the interior angles of a hexagon?
- (A) 540 degrees (B) 720 degrees
(C) 900 degrees (D) 1080 degrees
- 44.** Each interior angle of a regular polygon measures 144 degrees. How many sides does the polygon have?
- (A) 10 (B) 8
(C) 9 (D) 12
- 45.** Three of the following four figures are alike in a certain way and one is different. Which one is different?
- (A) Rhombus (B) Square
(C) Trapezium (D) Rectangle

RM09

Numeracy I: Ratio, Proportion, Partnership, Percentage, Profit & Loss, Interest

70 questions · 28 fact-sheet points

Must-know facts

- Simple interest: $SI = P \times R \times T / 100$; Amount = $P + SI$. Compound interest: $A = P(1 + R/100)^n$ and $CI = A - P$.
- Half-yearly compounding: rate $R/2$, periods $2n$. Quarterly: rate $R/4$, periods $4n$. CI always exceeds SI for $n > 1$.
- DIFFERENCE between CI and SI for 2 YEARS = $P(R/100)^2 = P \times R^2 / 10000$. (Asked in RAS Pre 2018, 2021, 2023 and 2024.)
- DIFFERENCE between CI and SI for 3 YEARS = $P \times R^2 \times (300 + R) / 10^6 = P(R/100)^2 \times (3 + R/100)$.
- From the 2-year difference D : $P = D \times (100/R)^2$, and $R = 100 \times \text{square root of } (D/P)$.
- SI doubles a sum when $SI = P$, so rate = $100/T$; triples it when $SI = 2P$, so rate = $200/T$. At CI, if a sum doubles in t years it becomes 4 times in $2t$ and 8 times in $3t$ years.
- Population after n years = $P(1 + r/100)^n$; depreciation = $P(1 - r/100)^n$. Growth or decline over unequal years multiplies the separate factors.
- Instalments at CI: each instalment x satisfies $x/(1+r)^1 + x/(1+r)^2 + \dots = \text{debt}$ (present-value method).
- Successive percentage change of $a\%$ then $b\%$: net = $a + b + ab/100$ (use a negative sign for a fall).
- An $x\%$ rise followed by an $x\%$ fall always gives a NET FALL of $x^2/100$ per cent, never zero.
- If price rises by $x\%$, consumption must fall by $100x/(100 + x)$ per cent to keep the expenditure unchanged.
- If A is $x\%$ more than B , then B is $100x/(100 + x)$ per cent less than A . If A is $x\%$ less than B , B is $100x/(100 - x)$ per cent more than A .
- Profit % and loss % are ALWAYS calculated on the cost price unless stated otherwise: Profit% = $(SP - CP)/CP \times 100$.
- $SP = CP(100 + p)/100$; $CP = 100 \times SP/(100 + p)$; for a loss replace $(100 + p)$ by $(100 - l)$.
- Marked price and discount: $SP = MP(100 - d)/100$. Two successive discounts d_1 and d_2 equal a single discount of $d_1 + d_2 - d_1d_2/100$.
- Marking $m\%$ above cost and giving $d\%$ discount gives profit % = $m - d - md/100$.
- Selling two articles at the SAME price, one at $+x\%$ and one at $-x\%$, always gives a net LOSS of $x^2/100$ per cent.
- If CP of x articles = SP of y articles, then profit % = $(x - y)/y \times 100$ (a loss if $y > x$).
- Dishonest dealer selling at cost price with a false weight: gain % = $\text{error}/(\text{true value} - \text{error}) \times 100$.
- A profit of $p\%$ on the cost price equals $100p/(100 + p)$ per cent of the selling price.
- Ratio $a : b : c$ -- divide a total in the ratio by giving each share (its part / sum of parts) $\times$ total.
- Proportion $a : b :: c : d$ means $ad = bc$. Mean proportional between a and b = square root of ab ; third proportional to a , $b = b^2/a$; fourth proportional to a , b , $c = bc/a$.
- To combine $a : b$ and $b : c$, scale both so that b matches, then read off $a : b : c$.

24. Rule of ALLIGATION: $(\text{quantity of cheaper})/(\text{quantity of dearer}) = (\text{dearer price} - \text{mean price})/(\text{mean price} - \text{cheaper price})$. Water is taken at zero cost.
25. Repeated replacement: if y litres are drawn from x litres of pure liquid and replaced with water, n times, the liquid left $= x(1 - y/x)^n$.
26. Mixing equal quantities of two mixtures: add the component fractions directly (e.g. $4/5$ and $3/5$ milk give 7 parts milk to 3 parts water).
27. To change a mixture from $a : b$ to $a : c$ by adding only the second liquid, keep the first quantity fixed and compute the new second quantity.
28. Partnership: profit is divided in the ratio of (capital $\times$ time) for each partner; a working partner's salary or commission is deducted first and the balance shared.

1. What is the difference between the compound interest and the simple interest on Rs 10,000 for 2 years at 5% per annum, interest being compounded annually? RAS PRE 2023, 2024

- (A) Rs 25 (B) Rs 50
(C) Rs 20 (D) Rs 100

2. The difference between compound interest and simple interest on a sum of Rs 8,000 for 2 years at 10% per annum is RAS PRE 2023, 2024

- (A) Rs 100 (B) Rs 80
(C) Rs 160 (D) Rs 40

3. Find the difference between the compound interest and the simple interest on Rs 16,000 for 3 years at 5% per annum. RAS PRE 2023, 2024

- (A) Rs 120 (B) Rs 125
(C) Rs 122 (D) Rs 244

4. The difference between compound interest and simple interest on Rs 10,000 for 3 years at 10% per annum is RAS PRE 2023, 2024

- (A) Rs 300 (B) Rs 331
(C) Rs 210 (D) Rs 310

5. The difference between the compound interest and the simple interest on a certain sum for 2 years at 4% per annum is Rs 32. What is the sum? RAS PRE 2023, 2024

- (A) Rs 16,000 (B) Rs 25,000
(C) Rs 20,000 (D) Rs 18,000

6. The difference between compound interest and simple interest on Rs 10,000 for 2 years is Rs 144. The rate of interest per annum is RAS PRE 2023, 2024 HARD

- (A) 10% (B) 14%
(C) 15% (D) 12%

7. The difference between the compound interest and the simple interest on a sum for 3 years at 10% per annum is Rs 62. The sum is RAS PRE 2023, 2024

- (A) Rs 2,200 (B) Rs 2,000
(C) Rs 1,800 (D) Rs 2,500

8. What is the difference between compound interest and simple interest on Rs 25,000 for 2 years at 8% per annum? RAS PRE 2023, 2024

- (A) Rs 200 (B) Rs 128
(C) Rs 320 (D) Rs 160

- 9.** A vessel contains 100 litres of pure milk. 20 litres of milk is drawn out and replaced by water. This operation is performed once more. How much pure milk is left in the vessel? RAS PRE 2016, 2024
- (A) 60 litres (B) 64 litres
(C) 68 litres (D) 72 litres
- 10.** A vessel holds 64 litres of pure milk. 16 litres is taken out and replaced by water; the same operation is repeated once more. What is the ratio of milk to water in the vessel finally? RAS PRE 2016, 2024 HARD
- (A) 9 : 7 (B) 7 : 9
(C) 3 : 1 (D) 4 : 3
- 11.** In what ratio must water be mixed with milk costing Rs 32 per litre so that the mixture is worth Rs 24 per litre? RAS PRE 2016, 2024
- (A) 1 : 4 (B) 3 : 1
(C) 2 : 3 (D) 1 : 3
- 12.** A milkman buys milk at Rs 40 per litre, mixes water with it and sells the mixture at Rs 40 per litre, thereby gaining 25%. Find the ratio of water to milk in the mixture. RAS PRE 2016, 2024 HARD
- (A) 1 : 5 (B) 1 : 3
(C) 1 : 4 (D) 2 : 5
- 13.** Vessel A contains milk and water in the ratio 4 : 1 and vessel B contains them in the ratio 3 : 2. Equal quantities from the two vessels are mixed. What is the ratio of milk to water in the new mixture?
- (A) 7 : 3 (B) 3 : 7
(C) 1 : 1 (D) 5 : 3
- 14.** A mixture of 60 litres contains milk and water in the ratio 2 : 1. How much water must be added to make the ratio of milk to water 1 : 1?
- (A) 20 litres (B) 10 litres
(C) 15 litres (D) 30 litres
- 15.** 40 litres of a mixture contains milk and water in the ratio 3 : 1. How much water should be added so that the ratio becomes 3 : 2?
- (A) 10 litres (B) 5 litres
(C) 8 litres (D) 20 litres
- 16.** In what ratio must rice costing Rs 30 per kg be mixed with rice costing Rs 45 per kg so that the mixture is worth Rs 36 per kg?
- (A) 2 : 3 (B) 5 : 2
(C) 4 : 3 (D) 3 : 2
- 17.** Two numbers are in the ratio 3 : 5. If 4 is added to each of them, the ratio becomes 5 : 7. What is the smaller number?
- (A) 8 (B) 10
(C) 6 (D) 12
- 18.** Rs 1,200 is divided among A, B and C in the ratio 2 : 3 : 7. What is C's share?
- (A) Rs 600 (B) Rs 500
(C) Rs 700 (D) Rs 300
- 19.** Two numbers are in the ratio 5 : 8 and their difference is 36. What is the larger number?
- (A) 60 (B) 96
(C) 108 (D) 84

- 20.** What is the mean proportional between 9 and 64?
- (A) 36 (B) 18
(C) 32 (D) 24
- 21.** What is the third proportional to 8 and 12?
- (A) 18 (B) 16
(C) 24 (D) 20
- 22.** What is the fourth proportional to 4, 9 and 12?
- (A) 24 (B) 36
(C) 27 (D) 18
- 23.** If $a : b = 2 : 3$ and $b : c = 4 : 5$, then $a : c$ is
- (A) $2 : 5$ (B) $8 : 5$
(C) $4 : 15$ (D) $8 : 15$
- 24.** Rs 5,900 is divided among A, B and C in such a way that $A : B = 3 : 4$ and $B : C = 5 : 6$. What is A's share?
- (A) Rs 2,000 (B) Rs 1,800
(C) Rs 1,500 (D) Rs 2,400
- 25.** A and B invest Rs 12,000 and Rs 16,000 respectively in a business for the same period. If the total profit at the end of the year is Rs 7,000, what is B's share?
- (A) Rs 4,000 (B) Rs 3,000
(C) Rs 3,500 (D) Rs 4,500
- 26.** A invests Rs 5,000 for 12 months and B invests Rs 6,000 for 8 months in the same business. If the profit is Rs 18,000, what is A's share?
- (A) Rs 8,000 (B) Rs 9,000
(C) Rs 10,000 (D) Rs 12,000
- 27.** A and B start a business with capitals in the ratio $3 : 2$. A withdraws his entire capital at the end of 6 months while B continues for the whole year. If the annual profit is Rs 14,000, what is B's share?
- (A) Rs 6,000 (B) Rs 8,000
(C) Rs 7,000 (D) Rs 8,400
- 28.** A, B and C invest Rs 20,000, Rs 15,000 and Rs 25,000 respectively in a partnership for one year. If the total profit is Rs 24,000, what is C's share?
- (A) Rs 10,000 (B) Rs 8,000
(C) Rs 6,000 (D) Rs 12,000
- 29.** A and B invest Rs 30,000 and Rs 20,000 respectively. A is the working partner and receives 10% of the profit for managing the business; the rest is divided in the ratio of their capitals. If the annual profit is Rs 20,000, what does A receive in all? **HARD**
- (A) Rs 12,000 (B) Rs 14,000
(C) Rs 12,800 (D) Rs 11,800
- 30.** A starts a business with Rs 8,000. After 4 months B joins him with Rs 9,000. At the end of the year the profit is Rs 21,000. What is A's share?
- (A) Rs 9,000 (B) Rs 11,200
(C) Rs 12,000 (D) Rs 14,000

- 31.** What is 15% of 640?
(A) 96 (B) 84
(C) 90 (D) 108
- 32.** If the price of a commodity rises by 25%, by what percentage must a family reduce its consumption so that its expenditure on the commodity remains unchanged?
(A) 25% (B) 20%
(C) 15% (D) 30%
- 33.** The price of an article is increased by 20% and the new price is then increased by a further 10%. What is the net percentage increase?
(A) 32% (B) 30%
(C) 28% (D) 31%
- 34.** The price of an article is first increased by 20% and then decreased by 20%. The net effect on the price is
(A) No change (B) A decrease of 4%
(C) An increase of 4% (D) A decrease of 2%
- 35.** A man's salary of Rs 25,000 is first increased by 10% and then the increased salary is reduced by 10%. What is his final salary?
(A) Rs 25,000 (B) Rs 24,750
(C) Rs 24,500 (D) Rs 25,250
- 36.** A's income is 25% more than B's income. By what percentage is B's income less than A's?
(A) 20% (B) 25%
(C) 30% (D) 16%
- 37.** In an election between two candidates, the winner secured 60% of the valid votes and won by a margin of 1,200 votes. What was the total number of valid votes polled?
(A) 4000 (B) 6000
(C) 12000 (D) 3000
- 38.** A student scored 30% of the maximum marks and failed by 20 marks. If the pass mark is 35% of the maximum, what are the maximum marks?
(A) 300 (B) 500
(C) 400 (D) 450
- 39.** The population of a town is 20,000 and it increases at the rate of 10% per annum. What will the population be after 2 years?
(A) 24200 (B) 24000
(C) 22000 (D) 26000
- 40.** The value of a machine costing Rs 50,000 depreciates at 10% per annum. What is its value after 2 years?
(A) Rs 40,500 (B) Rs 40,000
(C) Rs 41,000 (D) Rs 45,000
- 41.** If 60% of A is equal to 40% of B, then A : B is
(A) 3 : 2 (B) 2 : 3
(C) 2 : 5 (D) 5 : 3

- 42.** An article bought for Rs 400 is sold for Rs 500. What is the profit percentage?
- (A) 25% (B) 20%
(C) 30% (D) 10%
- 43.** An article costing Rs 250 is sold for Rs 200. What is the loss percentage?
- (A) 25% (B) 15%
(C) 20% (D) 10%
- 44.** By selling an article for Rs 690 a shopkeeper earns a profit of 15%. What is the cost price of the article?
- (A) Rs 600 (B) Rs 575
(C) Rs 620 (D) Rs 650
- 45.** An article whose cost price is Rs 1,200 is sold at a loss of 20%. What is the selling price?
- (A) Rs 1,000 (B) Rs 960
(C) Rs 940 (D) Rs 1,440
- 46.** The marked price of an article is Rs 1,000 and two successive discounts of 20% and 10% are allowed. What is the selling price?
- (A) Rs 700 (B) Rs 720
(C) Rs 730 (D) Rs 750
- 47.** A shopkeeper buys an article for Rs 300, marks it 25% above the cost price and then allows a discount of 10% on the marked price. What is his profit percentage?
- (A) 15% (B) 10%
(C) 12% (D) 12.5%
- 48.** A man sells two articles at Rs 1,200 each. On one he gains 20% and on the other he loses 20%. What is his overall gain or loss?
- (A) Neither gain nor loss (B) A loss of Rs 96
(C) A loss of Rs 100 (D) A gain of Rs 100
- 49.** If the cost price of 20 articles is equal to the selling price of 16 articles, what is the profit percentage?
- (A) 20% (B) 16%
(C) 30% (D) 25%
- 50.** A dishonest dealer sells his goods at the cost price but uses a weight of 800 g in place of 1 kg. What is his gain percentage?
- (A) 20% (B) 25%
(C) 12.5% (D) 33.33%
- 51.** By selling an article for Rs 900 a man loses 10%. At what price should he sell it to gain 10%?
- (A) Rs 1,000 (B) Rs 1,100
(C) Rs 1,080 (D) Rs 1,200
- 52.** A shopkeeper marks his goods 40% above the cost price and then allows a discount of 25% on the marked price. What is his profit percentage?
- (A) 15% (B) 10%
(C) 20% (D) 5%

- 53.** A trader sold an article at a profit of 15%. Had he sold it for Rs 90 more, his profit would have been 25%. What is the cost price of the article?
- (A) Rs 900 (B) Rs 800
(C) Rs 1,000 (D) Rs 750
- 54.** A profit of 25% on the cost price is equal to what percentage of the selling price?
- (A) 25% (B) 30%
(C) 15% (D) 20%
- 55.** What is the simple interest on Rs 6,000 at 8% per annum for 3 years?
- (A) Rs 1,400 (B) Rs 1,600
(C) Rs 1,480 (D) Rs 1,440
- 56.** At what rate of simple interest will a sum of money double itself in 8 years?
- (A) 10% (B) 8%
(C) 12.5% (D) 15%
- 57.** At what rate of simple interest will Rs 2,500 amount to Rs 3,000 in 4 years?
- (A) 4% (B) 6%
(C) 5% (D) 5.5%
- 58.** The simple interest on a certain sum for 5 years at 6% per annum is Rs 1,800. What is the sum?
- (A) Rs 5,000 (B) Rs 7,200
(C) Rs 6,600 (D) Rs 6,000
- 59.** In how many years will Rs 4,000 yield a simple interest of Rs 1,200 at 10% per annum?
- (A) 2 (B) 3
(C) 4 (D) 5
- 60.** A sum of money at simple interest amounts to Rs 2,400 in 3 years and to Rs 2,800 in 5 years. What is the sum?
- (A) Rs 2,000 (B) Rs 1,600
(C) Rs 2,100 (D) Rs 1,800
- 61.** At what rate per cent per annum of simple interest will a sum become three times itself in 20 years?
- (A) 12.5% (B) 15%
(C) 10% (D) 5%
- 62.** What is the compound interest on Rs 10,000 at 10% per annum for 2 years, compounded annually?
- (A) Rs 2,000 (B) Rs 2,200
(C) Rs 1,000 (D) Rs 2,100
- 63.** Find the amount on Rs 8,000 at 5% per annum compound interest for 2 years.
- (A) Rs 8,820 (B) Rs 8,800
(C) Rs 8,400 (D) Rs 9,000
- 64.** What is the compound interest on Rs 12,000 at 10% per annum for 3 years, compounded annually?
- (A) Rs 3,600 (B) Rs 3,972
(C) Rs 4,000 (D) Rs 3,960
- 65.** What is the compound interest on Rs 8,000 for 1 year at 10% per annum when the interest is compounded half-yearly?
- (A) Rs 820 (B) Rs 800
(C) Rs 840 (D) Rs 1,640

- 66.** A sum of money placed at compound interest doubles itself in 5 years. In how many years will it become four times itself?
- (A) 15 (B) 20
(C) 10 (D) 12
- 67.** The value of a machine bought for Rs 10,000 depreciates at 20% per annum. What is its value at the end of 2 years?
- (A) Rs 6,000 (B) Rs 6,400
(C) Rs 7,200 (D) Rs 6,600
- 68.** If a sum of Rs 6,000 fetches a compound interest of Rs 2,640 in 2 years, what is the rate of interest per annum?
- (A) 18% (B) 22%
(C) 25% (D) 20%
- 69.** The cash price of a television set is Rs 12,000. Under an instalment scheme a buyer pays Rs 4,000 at once and Rs 4,200 at the end of each of the next two years. How much more than the cash price does he pay in all?
- (A) Rs 200 (B) Rs 600
(C) Rs 800 (D) Rs 400
- 70.** A debt of Rs 2,100 is to be repaid in two equal annual instalments at 10% per annum compound interest. What is the amount of each instalment? HARD
- (A) Rs 1,155 (B) Rs 1,210
(C) Rs 1,100 (D) Rs 1,250

RM10

Numeracy II: Mensuration, Averages, Statistics, Data Interpretation, Probability

70 questions · 29 fact-sheet points

Must-know facts

1. Square: perimeter $4a$, area a^2 , diagonal $a\sqrt{2}$, area from diagonal $= d^2/2$. Rectangle: perimeter $2(l+b)$, area lb , diagonal $\sqrt{l^2+b^2}$.
2. Triangle: area $= \frac{1}{2} \times \text{base} \times \text{height}$; Heron's formula $= \sqrt{s(s-a)(s-b)(s-c)}$, $s = (a+b+c)/2$; equilateral of side a : area $= (\sqrt{3}/4)a^2$, height $= (\sqrt{3}/2)a$.
3. Parallelogram: area $= \text{base} \times \text{height}$ (NO $\frac{1}{2}$). Rhombus: area $= \frac{1}{2} d_1 d_2$. Trapezium: area $= \frac{1}{2} \times (\text{sum of parallel sides}) \times \text{distance between them}$.
4. Circle: circumference $2\pi r$, area πr^2 . Sector of angle θ : arc $= (\theta/360) \cdot 2\pi r$, area $= (\theta/360) \cdot \pi r^2 = \frac{1}{2} \times \text{arc} \times r$. Semicircle perimeter $= \pi r + 2r$.
5. Path w metres wide OUTSIDE a rectangle $l \times b$: area $= 2w(l+b) + 4w^2$. Path INSIDE: area $= 2w(l+b) - 4w^2$. Crossing roads of width w : $w(l+b) - w^2$.
6. Cube: volume a^3 , total surface area $6a^2$, lateral $4a^2$, diagonal $a\sqrt{3}$. Cuboid: volume lbh , TSA $2(lb+bh+hl)$, diagonal $\sqrt{l^2+b^2+h^2}$.
7. Cylinder: volume $\pi r^2 h$, curved SA $2\pi rh$, total SA $2\pi r(r+h)$. Cone: volume $\frac{1}{3}\pi r^2 h$, curved SA πrl , total SA $\pi r(l+r)$, slant $l = \sqrt{r^2+h^2}$.
8. Sphere: volume $(4/3)\pi r^3$, surface area $4\pi r^2$. Hemisphere: volume $(2/3)\pi r^3$, curved SA $2\pi r^2$, total SA $3\pi r^2$. Use $\pi = 22/7$ with radii that are multiples of 7.
9. Effect of change: if every linear dimension changes by k , length $\times k$, area $\times k^2$, volume $\times k^3$. Side $+20\% \rightarrow$ area $+44\%$; edge $+50\% \rightarrow$ surface $+125\%$, volume $+237.5\%$.
10. Two-dimension percentage rule: net change $= x + y + xy/100$ (use a minus sign for a decrease). $+20\%$ and -20% together give -4% , not zero.
11. Average $= \text{sum} \div \text{number of items}$. Average of first n natural numbers $= (n+1)/2$; of first n odd numbers $= n$; of first n even numbers $= n+1$.
12. Weighted average $= (n_1a_1 + n_2a_2)/(n_1 + n_2)$ — never average two group averages directly unless the groups are equal in size.
13. Replacement rule: if the average of n items changes by d when one item is replaced, new item $= \text{old item} \pm n \cdot d$. Inclusion rule: new member $= \text{new average} + n \times (\text{rise})$.
14. Average speed for EQUAL DISTANCES at speeds x and $y = 2xy/(x+y)$ (harmonic mean) — never $(x+y)/2$. For equal TIMES the average is $(x+y)/2$.
15. Mean of grouped data $= \Sigma fx/\Sigma f$. Median: arrange in order; n odd $\rightarrow ((n+1)/2)$ th value; n even $\rightarrow$ mean of $(n/2)$ th and $(n/2 + 1)$ th. Mode $=$ most frequent value.
16. Empirical relation for a moderately skewed distribution: Mode $= 3 \text{ Median} - 2 \text{ Mean}$. Range $= \text{maximum} - \text{minimum}$ (a measure of dispersion).
17. $nPr = n!/(n-r)!$ (order matters); $nCr = n!/[(r!(n-r)!)]$ (order does not); $nC0 = nCn = 1$, $nCr = nC(n-r)$, $nC1 = n$, $nC2 = n(n-1)/2$.
18. Word with repeated letters: arrangements $= n!/(p!q!...)$. BANANA $= 6!/(3!2!) = 60$. Handshakes/lines/chords among n points $= nC2$; number of diagonals of an n -gon $= n(n-3)/2$.
19. Probability $P(E) = \text{favourable} \div \text{total}$, $0 \leq P \leq 1$; $P(\text{not } E) = 1 - P(E)$. Coins: n coins $\rightarrow 2^n$ outcomes; exactly r heads $\rightarrow nCr/2^n$.

20. Two dice $\rightarrow$ 36 outcomes. Number of ways to get a sum: $2 \rightarrow 1$, $3 \rightarrow 2$, $4 \rightarrow 3$, $5 \rightarrow 4$, $6 \rightarrow 5$, $7 \rightarrow 6$, $8 \rightarrow 5$, $9 \rightarrow 4$, $10 \rightarrow 3$, $11 \rightarrow 2$, $12 \rightarrow 1$. Doublets = 6, so $P(\text{doublet}) = 1/6$; $P(\text{sum } 7) = 1/6$, the most likely sum.
21. Pack of 52 cards: 4 suits $\times$ 13; 26 red (hearts, diamonds) and 26 black (spades, clubs); 12 face cards (J, Q, K), 4 aces; red face cards = 6.
22. Drawing without replacement uses combinations: $P(\text{both black from } 4W, 5B) = 5C2 \div 9C2 = 10/36 = 5/18$. With replacement it would be $(5/9)^2$.
23. Speed conversion: $1 \text{ km/h} = 5/18 \text{ m/s}$; $1 \text{ m/s} = 18/5 \text{ km/h}$. Train past a pole $\rightarrow$ own length; past a platform/bridge $\rightarrow$ length + length of platform.
24. Relative speed: opposite directions $\rightarrow$ add the speeds; same direction $\rightarrow$ subtract. Boat: downstream = $u + v$, upstream = $u - v$.
25. Time and work: if A takes x days and B takes y days, together they take $xy/(x+y)$ days. Always work with ONE DAY'S work ($1/x$), never subtract the days themselves.
26. $M1D1H1/W1 = M2D2H2/W2$ (men-days-hours). Pipes: filling is $+1/x$, emptying is $-1/y$; with a leak the net rate is $1/x - 1/y$.
27. Calendar odd days: ordinary year 1, leap year 2; 100 years 5, 200 years 3, 300 years 1, 400 years 0. Count 0 = Sunday, 1 = Monday ... 6 = Saturday. A leap year is divisible by 4, but a century year must be divisible by 400.
28. Month odd-day codes (Jan-Dec): 0,3,3,6,1,4,6,2,5,0,3,5. The calendar of a year repeats after 400 years; 26 January 1950 was a Thursday, 15 August 1947 a Friday.
29. Clock: angle = $|30H - 5.5M|$ degrees (take $360 - x$ if the result exceeds 180). Hour hand $0.5^\circ/\text{min}$, minute hand $6^\circ/\text{min}$, relative gain $5.5^\circ/\text{min}$. Hands coincide 11 times in 12 hours, are opposite 11 times, and are at right angles 22 times.

1. The perimeter of a square field is 48 cm. What is its area?
 (A) 144 cm^2 (B) 96 cm^2
 (C) 576 cm^2 (D) 2304 cm^2
2. A rectangular field is 24 m long and 18 m broad. What will it cost to fence it once at the rate of ₹15 per metre?
 (A) ₹6,480 (B) ₹630
 (C) ₹1,440 (D) ₹1,260
3. The base of a triangle is 18 cm and the corresponding height is 14 cm. Its area is:
 (A) 252 cm^2 (B) 126 cm^2
 (C) 63 cm^2 (D) 32 cm^2
4. The sides of a triangle are 13 cm, 14 cm and 15 cm. Its area is:
 (A) 84 cm^2 (B) 42 cm^2
 (C) 168 cm^2 (D) 91 cm^2
5. The area of a parallelogram whose base is 15 cm and corresponding height 8 cm is:
 (A) 60 cm^2 (B) 120 cm^2
 (C) 240 cm^2 (D) 46 cm^2
6. The parallel sides of a trapezium are 22 cm and 14 cm and the perpendicular distance between them is 9 cm. Its area is:
 (A) 324 cm^2 (B) 81 cm^2
 (C) 198 cm^2 (D) 162 cm^2

7. The circumference of a circle of radius 7 cm (take $\pi = 22/7$) is:
- (A) 154 cm (B) 22 cm
(C) 88 cm (D) 44 cm
8. The diameter of a circle is 28 cm. Its area (take $\pi = 22/7$) is:
- (A) 2464 cm² (B) 88 cm²
(C) 616 cm² (D) 154 cm²
9. The area of a sector of a circle of radius 21 cm with central angle 120° is (take $\pi = 22/7$):
- (A) 1386 cm² (B) 44 cm²
(C) 462 cm² (D) 231 cm²
10. A rectangular field is 50 m long and 40 m wide. A path 3 m wide is laid all around it on the outside. The area of the path is: **HARD**
- (A) 540 m² (B) 1152 m²
(C) 2576 m² (D) 576 m²
11. If each side of a square is increased by 20%, by what percentage does its area increase?
- (A) 44% (B) 20%
(C) 40% (D) 21%
12. The volume of a cube whose edge is 7 cm is:
- (A) 294 cm³ (B) 49 cm³
(C) 343 cm³ (D) 147 cm³
13. A cuboid measures 12 cm × 8 cm × 5 cm. Its total surface area is:
- (A) 480 cm² (B) 196 cm²
(C) 784 cm² (D) 392 cm²
14. The volume of a cylinder of radius 7 cm and height 10 cm is (take $\pi = 22/7$):
- (A) 440 cm³ (B) 748 cm³
(C) 3080 cm³ (D) 1540 cm³
15. The curved surface area of a cylinder of radius 14 cm and height 20 cm is (take $\pi = 22/7$):
- (A) 12320 cm² (B) 1760 cm²
(C) 880 cm² (D) 3520 cm²
16. A cone has radius 7 cm and vertical height 24 cm. Its curved surface area is (take $\pi = 22/7$):
- (A) 704 cm² (B) 154 cm²
(C) 550 cm² (D) 1232 cm²
17. The volume of a cone of radius 3 cm and height 7 cm is (take $\pi = 22/7$):
- (A) 66 cm³ (B) 198 cm³
(C) 22 cm³ (D) 132 cm³
18. The surface area of a sphere of radius 21 cm is (take $\pi = 22/7$):
- (A) 1386 cm² (B) 5544 cm²
(C) 38808 cm² (D) 2772 cm²
19. The total surface area of a solid hemisphere of radius 7 cm is (take $\pi = 22/7$):
- (A) 308 cm² (B) 154 cm²
(C) 616 cm² (D) 462 cm²

- 20.** If each edge of a cube is increased by 50%, the total surface area of the cube increases by: HARD
- (A) 125% (B) 50%
(C) 100% (D) 225%
- 21.** The average of 5 numbers is 27. If one of the numbers, 35, is removed, the average of the remaining numbers is:
- (A) 27 (B) 26
(C) 25 (D) 28
- 22.** The average weight of 8 persons increases by 2.5 kg when a new person replaces one of them who weighs 56 kg. The weight of the new person is:
- (A) 58.5 kg (B) 76 kg
(C) 66 kg (D) 86 kg
- 23.** In a class of 50 students, the 20 boys score an average of 45 marks and the 30 girls an average of 55 marks. The average marks of the whole class are:
- (A) 50 (B) 49
(C) 52 (D) 51
- 24.** A car goes from A to B at 60 km/h and returns along the same road at 40 km/h. Its average speed for the whole journey is:
- (A) 48 km/h (B) 50 km/h
(C) 45 km/h (D) 52 km/h
- 25.** The average age of 30 students of a class is 14 years. When the teacher's age is included, the average rises by 1 year. The age of the teacher is:
- (A) 15 years (B) 31 years
(C) 45 years (D) 44 years
- 26.** A batsman has an average of 36 runs in 16 innings. He scores 70 runs in his 17th innings. His new average is:
- (A) 37 (B) 38
(C) 40 (D) 36
- 27.** The arithmetic mean of the numbers 12, 15, 18, 20, 22, 25 and 28 is:
- (A) 20 (B) 18
(C) 19 (D) 21
- 28.** Find the median of the following data: 7, 12, 15, 9, 21, 18, 11.
- (A) 12 (B) 15
(C) 11 (D) 14
- 29.** Find the mode of the following data: 3, 5, 7, 5, 9, 5, 3, 7, 5, 9.
- (A) 3 (B) 7
(C) 9 (D) 5
- 30.** The range of the data 45, 12, 78, 33, 60, 21 is:
- (A) 78 (B) 66
(C) 57 (D) 45

31. Find the arithmetic mean of the following frequency distribution:

Value (x) Frequency (f)

10 3

20 5

30 8

40 3

50 1

(A) 30

(B) 25

(C) 26

(D) 27

32. The table gives the number of children in 50 families:

Number of children Number of families

0 6

1 10

2 12

3 14

4 8

What is the MEDIAN number of children per family?

(A) 3

(B) 2

(C) 2.5

(D) 1

33. For a moderately skewed distribution the mean is 26 and the median is 28. Using the empirical relation, the mode is: **HARD**

(A) 32

(B) 30

(C) 24

(D) 27

34. The monthly expenditure of a family, totalling ₹36,000, is shown by a pie chart. The central angles of the sectors are given below:

Head of expenditure Central angle

Food 120°

House rent 90°

Education 60°

Transport 45°

Miscellaneous 45°

How much does the family spend on food? **RAS PRE 2013, 2018, 2024**

(A) ₹10,800

(B) ₹12,000

(C) ₹9,000

(D) ₹14,400

35. The monthly expenditure of a family, totalling ₹36,000, is shown by a pie chart. The central angles of the sectors are given below:

Head of expenditure Central angle

Food 120°

House rent 90°

Education 60°

Transport 45°

Miscellaneous 45°

By how much does the expenditure on house rent exceed that on education? RAS PRE 2013, 2018, 2024

(A) ₹2,400

(B) ₹3,600

(C) ₹6,000

(D) ₹3,000

36. The monthly expenditure of a family, totalling ₹36,000, is shown by a pie chart. The central angles of the sectors are given below:

Head of expenditure Central angle

Food 120°

House rent 90°

Education 60°

Transport 45°

Miscellaneous 45°

Expenditure on house rent and transport taken together is what percentage of the total monthly expenditure? RAS PRE 2013, 2018, 2024

(A) 35%

(B) 40%

(C) 33 $\frac{1}{3}$ %

(D) 37.5%

37. The following table gives the wheat production of a district (in lakh tonnes) over five years:

Year Production

2019 24

2020 30

2021 27

2022 36

2023 33

What is the average annual production over these five years?

(A) 30 lakh tonnes

(B) 27 lakh tonnes

(C) 32 lakh tonnes

(D) 33 lakh tonnes

38. The following table gives the wheat production of a district (in lakh tonnes) over five years:

Year Production

2019 24

2020 30

2021 27

2022 36

2023 33

The percentage increase in production from 2019 to 2020 is:

(A) 25%

(B) 6%

(C) 20%

(D) 24%

39. The following table gives the wheat production of a district (in lakh tonnes) over five years:

Year Production

2019 24

2020 30

2021 27

2022 36

2023 33

In how many of these years was the production ABOVE the five-year average?

(A) 2

(B) 1

(C) 3

(D) 4

40. The table below shows the number of boys and girls in five schools:

School Boys Girls

A 240 160

B 180 220

C 300 200

D 150 150

E 210 190

The ratio of the total number of boys to the total number of girls in all five schools is:

(A) 23 : 27

(B) 54 : 43

(C) 27 : 23

(D) 6 : 5

41. The table below shows the number of boys and girls in five schools:

School Boys Girls

A 240 160

B 180 220

C 300 200

D 150 150

E 210 190

In which school is the percentage of girls in the total strength the HIGHEST?

- (A) School A
- (B) School B
- (C) School C
- (D) School E

42. In how many different ways can the letters of the word 'DELHI' be arranged?

- (A) 120
- (B) 24
- (C) 60
- (D) 720

43. In how many different ways can the letters of the word 'BANANA' be arranged?

- (A) 120
- (B) 180
- (C) 60
- (D) 720

44. There are 12 persons in a party and each shakes hands with every other person exactly once. The total number of handshakes is:

- (A) 132
- (B) 144
- (C) 24
- (D) 66

45. A committee of 3 men and 2 women is to be formed from 6 men and 5 women. In how many ways can this be done?

HARD

- (A) 100
- (B) 300
- (C) 200
- (D) 150

46. Two fair coins are tossed together. What is the probability of getting at least one head?

- (A) $\frac{1}{2}$
- (B) $\frac{3}{4}$
- (C) $\frac{1}{4}$
- (D) $\frac{2}{3}$

47. Three fair coins are tossed simultaneously. The probability of getting exactly two heads is:

- (A) $\frac{1}{4}$
- (B) $\frac{3}{8}$
- (C) $\frac{1}{2}$
- (D) $\frac{1}{8}$

48. A fair die is thrown once. What is the probability of getting a prime number?

RAS PRE 2021, 2023, 2024

- (A) $\frac{1}{3}$
- (B) $\frac{2}{3}$
- (C) $\frac{1}{2}$
- (D) $\frac{1}{6}$

49. Two dice are thrown together. What is the probability that the sum of the numbers on them is 8?

RAS PRE 2021, 2023,

2024

- (A) $\frac{5}{36}$
- (B) $\frac{1}{6}$
- (C) $\frac{1}{9}$
- (D) $\frac{5}{18}$

50. Two dice are rolled together. What is the probability of getting a doublet (the same number on both dice)?

RAS PRE

2021, 2023, 2024

- (A) $\frac{1}{12}$
- (B) $\frac{1}{36}$
- (C) $\frac{1}{6}$
- (D) $\frac{1}{3}$

- 51.** Two dice are thrown simultaneously. The probability that the sum of the numbers obtained is a multiple of 3 is: RAS PRE
2021, 2023, 2024
- (A) $1/4$ (B) $5/12$
(C) $1/6$ (D) $1/3$
- 52.** One card is drawn at random from a well-shuffled pack of 52 playing cards. The probability that it is a king is:
- (A) $1/26$ (B) $1/13$
(C) $4/13$ (D) $1/4$
- 53.** One card is drawn at random from a pack of 52 cards. The probability that it is a red face card is:
- (A) $3/13$ (B) $1/26$
(C) $3/26$ (D) $6/13$
- 54.** A bag contains 5 red, 4 green and 3 blue balls. One ball is drawn at random. The probability that it is NOT red is:
- (A) $5/12$ (B) $7/12$
(C) $1/3$ (D) $1/4$
- 55.** A bag contains 4 white and 5 black balls. Two balls are drawn at random together. The probability that both are black is: HARD
- (A) $25/81$ (B) $5/9$
(C) $10/81$ (D) $5/18$
- 56.** A train 180 metres long is running at 54 km/h. How long will it take to cross an electric pole?
- (A) 12 seconds (B) 10 seconds
(C) 15 seconds (D) 20 seconds
- 57.** A train 240 metres long, running at 72 km/h, crosses a platform 360 metres long. The time taken is:
- (A) 12 seconds (B) 30 seconds
(C) 18 seconds (D) 20 seconds
- 58.** A car covers 210 km in 3 hours. At the same speed, how long will it take to cover 350 km?
- (A) 4 hours (B) 4 hours 30 minutes
(C) 6 hours (D) 5 hours
- 59.** Two men start walking from the same point in opposite directions at 5 km/h and 3 km/h. How far apart will they be after 4 hours?
- (A) 8 km (B) 16 km
(C) 20 km (D) 32 km
- 60.** Walking at 5 km/h a student reaches his school 6 minutes late; walking at 6 km/h he reaches 4 minutes early. The distance to the school is: HARD
- (A) 6 km (B) 5 km
(C) 4 km (D) 10 km
- 61.** A can do a piece of work in 12 days and B can do the same work in 24 days. Working together, they will finish it in:
- (A) 8 days (B) 6 days
(C) 18 days (D) 36 days
- 62.** A and B together can complete a work in 10 days, while A alone can complete it in 15 days. B alone will complete it in:
- (A) 30 days (B) 25 days
(C) 20 days (D) 5 days

- 63.** If 12 men can complete a work in 18 days, in how many days will 27 men complete the same work?
- (A) 12 days (B) 9 days
(C) 8 days (D) 6 days
- 64.** Pipe A can fill a tank in 6 hours and pipe B in 4 hours. If both are opened together, the tank will be full in:
- (A) 120 minutes (B) 144 minutes
(C) 150 minutes (D) 300 minutes
- 65.** A tap can fill a tank in 10 hours, but a leak at the bottom can empty the full tank in 15 hours. If both work together, the tank will be filled in:
- (A) 25 hours (B) 5 hours
(C) 30 hours (D) 6 hours
- 66.** 15 August 2026 falls on a Saturday. On which day of the week will 15 August 2027 fall?
- (A) Friday (B) Saturday
(C) Sunday (D) Monday
- 67.** The Constitution of India came into force on 26 January 1950. What day of the week was it?
- (A) Wednesday (B) Thursday
(C) Friday (D) Sunday
- 68.** The number of odd days in 400 consecutive years is:
- (A) 1 (B) 2
(C) 0 (D) 3
- 69.** What is the angle between the hour hand and the minute hand of a clock at 3:30?
- (A) 90° (B) 80°
(C) 75° (D) 105°
- 70.** At what time between 4 o'clock and 5 o'clock will the two hands of a clock coincide? HARD
- (A) 21 $\frac{9}{11}$ minutes past 4 (B) 22 minutes past 4
(C) 20 minutes past 4 (D) 23 $\frac{7}{11}$ minutes past 4

Answer key & explanations

RM01 – STATEMENT & ASSUMPTIONS / ARGUMENTS / CONCLUSIONS / COURSES OF ACTION

RM01.1 A	RM01.16 A	RM01.31 B	RM01.46 A
RM01.2 A	RM01.17 A	RM01.32 D	RM01.47 A
RM01.3 D	RM01.18 B	RM01.33 A	RM01.48 D
RM01.4 D	RM01.19 B	RM01.34 A	RM01.49 D
RM01.5 B	RM01.20 D	RM01.35 B	RM01.50 B
RM01.6 D	RM01.21 D	RM01.36 B	RM01.51 B
RM01.7 A	RM01.22 C	RM01.37 C	RM01.52 B
RM01.8 D	RM01.23 C	RM01.38 C	RM01.53 B
RM01.9 A	RM01.24 C	RM01.39 C	RM01.54 A
RM01.10 C	RM01.25 A	RM01.40 D	RM01.55 A
RM01.11 C	RM01.26 A	RM01.41 B	RM01.56 C
RM01.12 B	RM01.27 B	RM01.42 C	RM01.57 C
RM01.13 B	RM01.28 B	RM01.43 A	RM01.58 C
RM01.14 B	RM01.29 D	RM01.44 B	RM01.59 D
RM01.15 D	RM01.30 D	RM01.45 D	RM01.60 D

RM02 – ANALYTICAL AND CRITICAL REASONING; SYLLOGISM

RM02.1 A	RM02.14 C	RM02.27 D	RM02.40 C
RM02.2 A	RM02.15 C	RM02.28 A	RM02.41 D
RM02.3 B	RM02.16 A	RM02.29 B	RM02.42 B
RM02.4 B	RM02.17 B	RM02.30 A	RM02.43 D
RM02.5 C	RM02.18 D	RM02.31 C	RM02.44 A
RM02.6 C	RM02.19 B	RM02.32 D	RM02.45 C
RM02.7 C	RM02.20 A	RM02.33 A	RM02.46 A
RM02.8 D	RM02.21 C	RM02.34 B	RM02.47 D
RM02.9 A	RM02.22 D	RM02.35 B	RM02.48 C
RM02.10 C	RM02.23 B	RM02.36 C	RM02.49 B
RM02.11 D	RM02.24 A	RM02.37 D	RM02.50 A
RM02.12 D	RM02.25 B	RM02.38 C	
RM02.13 A	RM02.26 C	RM02.39 B	

RM03 – NUMBER AND LETTER SERIES

RM03.1 C	RM03.14 C	RM03.27 B	RM03.40 B
RM03.2 A	RM03.15 D	RM03.28 D	RM03.41 B
RM03.3 D	RM03.16 D	RM03.29 A	RM03.42 D
RM03.4 C	RM03.17 C	RM03.30 C	RM03.43 A
RM03.5 B	RM03.18 A	RM03.31 A	RM03.44 C
RM03.6 C	RM03.19 A	RM03.32 B	RM03.45 C
RM03.7 A	RM03.20 D	RM03.33 B	RM03.46 A
RM03.8 B	RM03.21 A	RM03.34 B	RM03.47 B
RM03.9 D	RM03.22 C	RM03.35 B	RM03.48 B
RM03.10 C	RM03.23 A	RM03.36 D	RM03.49 A
RM03.11 B	RM03.24 A	RM03.37 D	RM03.50 C
RM03.12 A	RM03.25 D	RM03.38 D	
RM03.13 B	RM03.26 C	RM03.39 D	

RM04 – CODING-DECODING

RM04.1 C	RM04.13 C	RM04.25 A	RM04.37 B
RM04.2 B	RM04.14 B	RM04.26 A	RM04.38 C
RM04.3 C	RM04.15 C	RM04.27 C	RM04.39 D
RM04.4 B	RM04.16 A	RM04.28 D	RM04.40 C
RM04.5 B	RM04.17 B	RM04.29 A	RM04.41 D
RM04.6 D	RM04.18 B	RM04.30 C	RM04.42 D
RM04.7 A	RM04.19 C	RM04.31 A	RM04.43 B
RM04.8 B	RM04.20 C	RM04.32 A	RM04.44 D
RM04.9 A	RM04.21 C	RM04.33 B	RM04.45 A
RM04.10 B	RM04.22 D	RM04.34 D	
RM04.11 D	RM04.23 D	RM04.35 A	
RM04.12 A	RM04.24 D	RM04.36 A	

RM05 – BLOOD RELATIONS

RM05.1 D	RM05.11 D	RM05.21 A	RM05.31 C
RM05.2 D	RM05.12 A	RM05.22 C	RM05.32 D
RM05.3 D	RM05.13 C	RM05.23 D	RM05.33 A
RM05.4 D	RM05.14 D	RM05.24 A	RM05.34 C
RM05.5 B	RM05.15 A	RM05.25 C	RM05.35 C
RM05.6 D	RM05.16 C	RM05.26 D	RM05.36 B
RM05.7 D	RM05.17 D	RM05.27 A	RM05.37 A
RM05.8 A	RM05.18 A	RM05.28 C	RM05.38 D
RM05.9 D	RM05.19 C	RM05.29 D	RM05.39 B
RM05.10 A	RM05.20 D	RM05.30 A	RM05.40 A

RM06 – DIRECTION SENSE

RM06.1 B	RM06.10 C	RM06.19 C	RM06.28 D
RM06.2 C	RM06.11 D	RM06.20 D	RM06.29 A
RM06.3 D	RM06.12 A	RM06.21 A	RM06.30 B
RM06.4 A	RM06.13 B	RM06.22 B	RM06.31 C
RM06.5 B	RM06.14 B	RM06.23 C	RM06.32 D
RM06.6 C	RM06.15 C	RM06.24 D	RM06.33 A
RM06.7 D	RM06.16 D	RM06.25 A	RM06.34 B
RM06.8 A	RM06.17 A	RM06.26 B	RM06.35 C
RM06.9 B	RM06.18 B	RM06.27 C	

RM07 – RANKING, SEATING ARRANGEMENT AND LOGICAL VENN DIAGRAMS

RM07.1 D	RM07.13 B	RM07.25 A	RM07.37 A
RM07.2 B	RM07.14 B	RM07.26 C	RM07.38 A
RM07.3 B	RM07.15 C	RM07.27 D	RM07.39 B
RM07.4 B	RM07.16 A	RM07.28 A	RM07.40 C
RM07.5 B	RM07.17 D	RM07.29 B	RM07.41 B
RM07.6 C	RM07.18 A	RM07.30 C	RM07.42 D
RM07.7 A	RM07.19 B	RM07.31 D	RM07.43 A
RM07.8 C	RM07.20 C	RM07.32 A	RM07.44 B
RM07.9 B	RM07.21 C	RM07.33 B	RM07.45 B
RM07.10 C	RM07.22 D	RM07.34 B	RM07.46 B
RM07.11 D	RM07.23 A	RM07.35 C	RM07.47 A
RM07.12 A	RM07.24 B	RM07.36 D	RM07.48 C

RM07.49 A	RM07.51 B	RM07.53 B	RM07.55 C
RM07.50 C	RM07.52 C	RM07.54 B	

RM08 – SHAPES, FIGURE COUNTING, MIRROR AND WATER IMAGES, EMBEDDED FIGURES

RM08.1 B	RM08.13 C	RM08.25 D	RM08.37 D
RM08.2 C	RM08.14 B	RM08.26 C	RM08.38 A
RM08.3 B	RM08.15 D	RM08.27 D	RM08.39 C
RM08.4 B	RM08.16 C	RM08.28 D	RM08.40 C
RM08.5 D	RM08.17 B	RM08.29 C	RM08.41 A
RM08.6 D	RM08.18 C	RM08.30 A	RM08.42 B
RM08.7 A	RM08.19 C	RM08.31 A	RM08.43 B
RM08.8 A	RM08.20 A	RM08.32 D	RM08.44 A
RM08.9 A	RM08.21 B	RM08.33 D	RM08.45 C
RM08.10 B	RM08.22 A	RM08.34 D	
RM08.11 A	RM08.23 A	RM08.35 B	
RM08.12 C	RM08.24 D	RM08.36 B	

RM09 – NUMERACY I: RATIO, PROPORTION, PARTNERSHIP, PERCENTAGE, PROFIT & LOSS, INTEREST

RM09.1 A	RM09.19 B	RM09.37 B	RM09.55 D
RM09.2 B	RM09.20 D	RM09.38 C	RM09.56 C
RM09.3 C	RM09.21 A	RM09.39 A	RM09.57 C
RM09.4 D	RM09.22 C	RM09.40 A	RM09.58 D
RM09.5 C	RM09.23 D	RM09.41 B	RM09.59 B
RM09.6 D	RM09.24 C	RM09.42 A	RM09.60 D
RM09.7 B	RM09.25 A	RM09.43 C	RM09.61 C
RM09.8 D	RM09.26 C	RM09.44 A	RM09.62 D
RM09.9 B	RM09.27 B	RM09.45 B	RM09.63 A
RM09.10 A	RM09.28 A	RM09.46 B	RM09.64 B
RM09.11 D	RM09.29 C	RM09.47 D	RM09.65 A
RM09.12 C	RM09.30 C	RM09.48 C	RM09.66 C
RM09.13 A	RM09.31 A	RM09.49 D	RM09.67 B
RM09.14 A	RM09.32 B	RM09.50 B	RM09.68 D
RM09.15 A	RM09.33 A	RM09.51 B	RM09.69 D
RM09.16 D	RM09.34 B	RM09.52 D	RM09.70 B
RM09.17 C	RM09.35 B	RM09.53 A	
RM09.18 C	RM09.36 A	RM09.54 D	

RM10 – NUMERACY II: MENSURATION, AVERAGES, STATISTICS, DATA INTERPRETATION, PROBABILITY

RM10.1 A	RM10.12 C	RM10.23 D	RM10.34 B
RM10.2 D	RM10.13 D	RM10.24 A	RM10.35 D
RM10.3 B	RM10.14 D	RM10.25 C	RM10.36 D
RM10.4 A	RM10.15 B	RM10.26 B	RM10.37 A
RM10.5 B	RM10.16 C	RM10.27 A	RM10.38 A
RM10.6 D	RM10.17 A	RM10.28 A	RM10.39 A
RM10.7 D	RM10.18 B	RM10.29 D	RM10.40 C
RM10.8 C	RM10.19 D	RM10.30 B	RM10.41 B
RM10.9 C	RM10.20 A	RM10.31 D	RM10.42 A
RM10.10 D	RM10.21 C	RM10.32 B	RM10.43 C
RM10.11 A	RM10.22 B	RM10.33 A	RM10.44 D

RM10·45 C	RM10·52 B	RM10·59 D	RM10·66 C
RM10·46 B	RM10·53 C	RM10·60 B	RM10·67 B
RM10·47 B	RM10·54 B	RM10·61 A	RM10·68 C
RM10·48 C	RM10·55 D	RM10·62 A	RM10·69 C
RM10·49 A	RM10·56 A	RM10·63 C	RM10·70 A
RM10·50 C	RM10·57 B	RM10·64 B	
RM10·51 D	RM10·58 D	RM10·65 C	

RM01 – EXPLANATIONS

- 1. (A) Only assumption I is implicit** — I is implicit: a body can announce a price rise only if it takes for granted that it may fix that price. II is a possible future effect, and an extreme one ('will stop'); an effect of the decision is never an assumption behind it. Hence only I.
- 2. (A) Only assumption I is implicit** — Every advertisement takes for granted that the advertised quality is wanted by buyers, so I is implicit. II contains the extreme word 'only' and is not needed for the advertisement to make sense. Only I.
- 3. (D) Both I and II are implicit** — 'In case of any difficulty' presupposes that difficulty is possible - I is implicit. Directing people to a helpline is pointless unless it can help, so II is also taken for granted. Both are implicit.
- 4. (D) Both I and II are implicit** — A penalty clause is framed only because late-coming is expected (I) and because the penalty is believed to work as a deterrent (II). Both assumptions are necessary for the notice to make sense.
- 5. (B) Only assumption II is implicit** — Advising Sunil to read the editorial page does not presuppose that he reads no newspaper - he may read it but skip the editorials, so I is not implicit. The advice is meaningless unless the editorial page is worth daily reading, so II is implicit.
- 6. (D) Both I and II are implicit** — The decision is worth taking only if the students will turn up (I) and only if extra teaching can raise performance (II). Both are taken for granted by the school.
- 7. (A) Only assumption I is implicit** — Offering a discount presupposes that discounts attract buyers - I is implicit. Nothing in the notice takes last-day buying for granted; a deadline is set precisely to prevent delay. Only I.
- 8. (D) Both I and II are implicit** — The advisory is issued because contamination is expected in the monsoon (I) and because boiling is believed to make the water safe (II). Both are unstated but necessary.
- 9. (A) Only assumption I is implicit** — An inauguration presupposes that the work is finished and the road is fit to be opened, so I is implicit. The Chief Minister's personal preference is irrelevant to the announcement. Only I.
- 10. (C) Neither I nor II is implicit** — The statement reports one student's marks and takes nothing for granted about other students. I and II are comparisons that the statement neither states nor presupposes, so neither is implicit.
- 11. (C) Neither I nor II is implicit** — Neither the cause of the announcement (a strike) nor its future consequence is taken for granted by the announcement. A past reason that is not required for the statement to make sense, and a future prediction, are both outside the statement.
- 12. (B) Only assumption II is implicit** — The construction 'in spite of X, Y happened' takes for granted that X normally prevents Y - so II is implicit. I says 'never', which is contradicted by the very use of 'in spite of'. Only II.
- 13. (B) Only assumption II is implicit** — 'Always' makes I an extreme claim that the instruction does not presuppose. Any instruction is issued in the expectation of compliance, so II is implicit. Only II.
- 14. (B) Only assumption II is implicit** — Explaining one absence does not presuppose that one is unwell frequently, so I is not implicit. Offering an explanation for absence takes for granted that one's presence was expected - II is implicit. Only II.
- 15. (D) Both I and II are implicit** — A safety rule is made only if the device is believed to protect (I) and only if the rule is believed to be enforceable (II). Both are taken for granted by the decision.
- 16. (A) Only argument I is strong** — I is relevant to the purpose of the proposal and deals with a major public gain - it is strong. II appeals to the private expenditure of owners, which does not weigh against a serious public problem - a weak argument. Only I.

- 17. (A) Only argument I is strong** — I addresses the central aim of the proposal with a significant benefit - strong. II is a sweeping generalisation ('never valued'), unsupported and irrelevant to school education - weak. Only I.
- 18. (B) Only argument II is strong** — I is a sweeping generalisation about all young people - weak. II gives a real cost of the proposal and points to a better-targeted remedy, which is both relevant and significant - strong. Only II.
- 19. (B) Only argument II is strong** — I states an unverified generalisation and, even if true, does not support a ban - weak. II identifies a foreseeable perverse effect and offers a proportionate alternative - strong. Only II.
- 20. (D) Both I and II are strong** — I cites major, documented harms; II cites a major livelihood cost and a workable middle path. Both are relevant to the question and deal with weighty aspects, so both are strong.
- 21. (D) Both I and II are strong** — Both arguments address the substantive consequences of the proposal - retention of experience and deferred pension outgo on one side, blocked recruitment on the other. Both are strong.
- 22. (C) Neither I nor II is strong** — I rests on a contested claim and, in any case, antiquity by itself is no ground for compulsion - weak. II is a sweeping assertion about students' interest, not a reasoned objection - weak. Neither is strong.
- 23. (C) Neither I nor II is strong** — I is mere imitation - it does not say why the other system is better for us. II is an appeal to tradition. Both fail the relevance-and-significance test, so neither is strong.
- 24. (C) Neither I nor II is strong** — I offers a negligible effect on city population - trivial, hence weak. II is a defeatist remark that neither examines the causes of begging nor the merits of rehabilitation - weak. Neither is strong.
- 25. (A) Only argument I is strong** — I gives a concrete fiscal and equity gain that goes to the heart of the proposal - strong. II is a slogan; besides, a subsidy is itself an intervention, so the argument is self-defeating - weak. Only I.
- 26. (A) Only argument I is strong** — I identifies a direct threat to the integrity of the examination - strong. II is the classic weak form: a measure is not useless merely because it does not remove the whole problem. Only I.
- 27. (B) Only argument II is strong** — I overstates the case with 'forever' and gives no evidence - weak. II points to serious, foreseeable ecological and human costs of the very proposal under discussion - strong. Only II.
- 28. (B) Only argument II is strong** — I appeals to fashion, not to any advantage of the mode - weak. II shows that the proposal would exclude a large class of eligible candidates, a decisive objection - strong. Only II.
- 29. (D) Both I and II are strong** — I addresses a real avenue of tax evasion; II addresses feasibility and the administrative cost-benefit of the measure. Both are relevant and weighty, hence both strong.
- 30. (D) Both I and II are strong** — I states established environmental harm; II states a serious implementation difficulty that decides whether the ban will work. Both are strong.
- 31. (B) Only conclusion II follows** — A rate is not a number: if District B has a far larger population, 62 per cent of it can exceed 78 per cent of District A, so I does not follow. Since B's literacy is 62 per cent, the remaining 38 per cent are illiterate, so II must be true. Only II follows.
- 32. (D) Both I and II follow** — 'All trains were cancelled' entails that none ran, so I follows. The statement itself gives fog as the cause of the cancellation, which establishes that fog can disrupt operations, so II follows. Both follow.
- 33. (A) Only conclusion I follows** — $85 < 90$, so I follows. The statement makes scoring above 90 SUFFICIENT for a scholarship, not necessary - scholarships may also have been given on other grounds, so II does not follow. Only I.
- 34. (A) Only conclusion I follows** — 'Fourth time' entails 'more than once', so I follows. II is a prediction about people's behaviour for which the statement supplies no evidence. Only I.
- 35. (B) Only conclusion II follows** — The total number of women in the company is not given, so we cannot say what proportion of women are graduates - I does not follow. But 150 women graduates are themselves employees, so the company has at least 150 women. Only II.
- 36. (B) Only conclusion II follows** — A warning does not guarantee universal compliance, so I does not follow. A threat of prosecution presupposes and therefore establishes that the act is an offence, so II follows. Only II.

- 37. (C) Neither I nor II follows** — The statement reports winter arrivals only; it says nothing about summer (I) and makes no comparison with other states (II). Both conclusions add information not contained in the statement, so neither follows.
- 38. (C) Neither I nor II follows** — A rise in sales tells us nothing about relative prices (I), and the market may have expanded so that petrol vehicle sales also rose (II). Neither conclusion follows.
- 39. (C) Either I or II follows** — Neither conclusion can be established on its own, but the statement guarantees that exactly one of the two is true and that no third possibility exists. When two conclusions are complementary and jointly exhaustive, the answer is 'Either I or II follows'.
- 40. (D) Both I and II follow** — Closure on Monday is stated, so I follows. The statement gives the festival as the reason for the closure, which shows the school does close for festivals, so II follows. Both follow.
- 41. (B) 40 households own neither a two-wheeler nor a car** — Households owning at least one vehicle = $120 + 60 - 20 = 160$. Hence $200 - 160 = 40$ own neither. (Only-two-wheeler = $120 - 20 = 100$, only-car = $60 - 20 = 40$, both = 20, so the other three options are wrong.)
- 42. (C) Some members of the music club are members of the drama club** — Drama club is contained in the music club and is non-empty, so some music-club members are drama-club members - that is certain. The debating-club overlap may lie entirely outside the drama club, so neither an overlap nor its absence can be asserted.
- 43. (A) The bus left Jodhpur before 6 a.m. today** — 'P only if Q' means P implies Q. Here P = reaches on time, Q = left before 6 a.m. Since P is true today, Q must be true today. The converse (option 4) and the generalisation (option 3) do not follow.
- 44. (B) It did not rain** — This is modus tollens: from 'Rain implies cancellation' and 'not cancelled', we get 'not rain'. If it had rained, the match would necessarily have been cancelled.
- 45. (D) Mohit has passed the written test** — 'Only those who passed are called' makes passing a necessary condition for being called. Mohit was called, so he must have passed. It does not follow that all who passed were called, nor anything about his rank or final selection.
- 46. (A) Only I follows** — I attacks the stated cause (overcrowding) and is within the authority's power - it follows. II is disproportionate and would deprive commuters of transport instead of correcting the fault. Only I.
- 47. (A) Only I follows** — I addresses the problem at its source and is practicable. II punishes the victims of poor results and does not remedy the teaching problem. Only I.
- 48. (D) Both I and II follow** — I is the direct vector-control measure and II is the accompanying awareness measure; both are practicable and both attack the same problem. Both follow.
- 49. (D) Both I and II follow** — I removes the cause and II deals with the damage already done; both are within the competence of the state agencies. Both follow.
- 50. (B) Only II follows** — Demolition before any technical assessment is premature and extreme. An inspection followed by fixing of responsibility is the standard, proportionate response. Only II.
- 51. (B) Only II follows** — Abolishing recruitment examinations because a few candidates cheated is an extreme reaction that destroys the system instead of protecting it. Debarring the guilty and prosecuting them under the applicable law is the proper action. Only II.
- 52. (B) Only II follows** — Asking a whole city to move out is impracticable. Enforcing dust and waste-burning restrictions attacks identified sources of the pollution and lies within the board's powers. Only II.
- 53. (B) Only II follows** — Withdrawing children penalises the students, not the school, and is not a remedy for an illegality. Enforcement of the law by the competent department is the appropriate action. Only II.
- 54. (A) Only I follows** — A survey followed by relief is the established and practicable response to a natural calamity. Banning the crop punishes farmers for a weather event they did not cause. Only I.
- 55. (A) Only I follows** — I increases the capacity to serve and removes the bottleneck. II denies treatment to patients, which defeats the purpose of the hospital. Only I.
- 56. (C) Neither I nor II follows** — Neither action is aimed at the stated problem. Banning vehicle sales does not stop drunken driving by existing drivers, and arresting everyone who drinks is neither lawful nor practicable. Neither follows. (Breath-analyser checking and licence suspension would.)

57. (C) Neither I nor II follows — Closing the school deprives the locality of education instead of improving it; making the children private candidates removes teaching altogether. Neither addresses the cause of poor results, so neither follows.

58. (C) Neither I nor II follows — Discontinuing the scheme punishes the children it is meant to feed, and silencing complainants suppresses the information needed to fix the problem. Neither follows; inspection of kitchens and action against defaulting suppliers would.

59. (D) Both I and II follow — I removes the danger to life through the competent authority; II takes care of the consequence of I for the displaced families. Both are practicable and both follow.

60. (D) Both I and II follow — Both actions address the stated mismatch between degrees and employability - one on the supply side of skills, the other on the demand-linkage side. Both follow.

RM02 – EXPLANATIONS

1. (A) Both I and II follow — Pens lie inside books, and books lie inside papers, so pens lie inside papers - I follows. Since the pen circle is non-empty and lies within the paper circle, some papers are pens - II follows (A + A = A, and 'All A are B' converts to 'Some B are A'). Both follow.

2. (A) Only conclusion I follows — The cats that are dogs must be animals, since all dogs are animals - so I follows (I + A = I). II reverses a universal without warrant; the animal class may contain many non-cats. Only I.

3. (B) Only conclusion I follows — Flowers lie inside roses and the rose circle is wholly separate from the leaf circle, so no flower can be a leaf - I follows (A + E = E). II directly contradicts I and therefore cannot follow. Only I.

4. (B) Only conclusion II follows — The books that are pens cannot be pencils, so at least some books are not pencils - II follows (I + E = O). But books that are not pens may well be pencils, so the universal negative in I does not follow. Only II.

5. (C) Only conclusion II follows — The dancers who are poets need not be singers, so I does not follow (the middle term 'dancers' is undistributed). II is only the conversion of 'Some dancers are poets', which is always valid. Only II.

6. (C) Either I or II follows — From the statements only 'Some pens are copies' is certain; the remaining pens may or may not be copies. So neither I nor II follows separately. But I and II are contradictories that between them cover every possibility, so 'Either I or II follows'.

7. (C) Either I or II follows — The spoons among the plates may lie inside the cup circle or entirely outside it, so neither conclusion is certain. As the two conclusions are contradictory and exhaust all possibilities, either I or II must be true - the 'Either or' case.

8. (D) Neither I nor II follows — Two particular (Some + Some) statements never yield a definite conclusion: the mango-apple overlap and the apple-grape overlap may be different parts of the apple circle, so I is not certain. II is a universal that goes far beyond 'some'. Neither follows.

9. (A) Neither I nor II follows — Doctors lie inside engineers, and engineers are wholly outside teachers; hence no doctor can be a teacher, which refutes I. 'No engineer is a teacher' converts to 'No teacher is an engineer', which refutes II. Neither follows.

10. (C) Both I and II follow — Lazy persons exist, are all sleepy, and none of them is a student; hence some sleepy persons are not students - I follows. 'All lazy are sleepy' converts to 'Some sleepy are lazy' - II follows. Both follow.

11. (D) Both I and II follow — 'All teachers are graduates' converts to 'Some graduates are teachers' - I follows. For II, place the whole lawyer group inside the teacher circle: this keeps 'some graduates are lawyers' true and 'all teachers are graduates' true, so the possibility is not ruled out. Both follow.

12. (D) Only conclusion II follows — Apples lie inside fruits and no fruit is a vegetable, so definitely no apple is a vegetable - II follows. Because that is definite, the possibility asserted in I is ruled out. Only II.

13. (A) Both I and II follow — The chairs that are tables are of wood, so 'some wooden things are chairs' is certain - II follows. Nothing prevents the remaining chairs from also being wooden, so the possibility in I stands. Both follow.

14. (C) Only conclusion I follows — Keys lie inside locks and locks are wholly outside doors, so no key is a door - I follows. The windows that are doors are certainly not locks, and the other windows may or may not be locks, so II is not certain. Only I.

15. (C) Both I and II follow — Erasers lie inside sharpeners, and no sharpener is a scale, so no eraser is a scale - II follows. The pencils that are erasers are therefore not scales, giving 'some pencils are not scales' - I follows. Both follow.

16. (A) Only conclusion I follows — Some paths are lanes, and no lane is a street, so those paths are not streets - I follows. The lanes among the paths need not lie in the road circle, so II is not certain. Only I.

17. (B) Only conclusion I follows — The journals that are magazines cannot be books, since no book is a magazine - so some journals are definitely not books, and I follows. That definite conclusion rules out the possibility asserted in II. Only I.

18. (D) Both I and II follow — The cars that are buses cannot be trucks, so I follows. Trucks exist, all of them are vehicles and none of them is a bus, so some vehicles are not buses - II follows. Both follow.

19. (B) Statement II is the cause and statement I is its effect — Losses were reported first and the tariff was raised to cover them, so the earlier and independent event (II) is the cause and the tariff rise (I) is its effect.

20. (A) Statement I is the cause and statement II is its effect — Rain in the catchment is what fills a dam, so I is the cause and II its effect. The reverse is impossible - a rising reservoir cannot produce rainfall in its catchment.

21. (C) Both the statements are effects of a common cause — Flight cancellations do not close schools, and school closure does not cancel flights. Both are the usual consequences of one unmentioned event - severe weather such as dense fog or a heavy storm - so they are effects of a common cause.

22. (D) Both the statements are effects of independent causes — The two markets are unconnected: international bullion prices move with global demand and currency factors, local vegetable prices with local supply and weather. Neither causes the other and no single event explains both, so they are effects of independent causes.

23. (B) Statement II is the cause and statement I is its effect — The announcement came before the sowing season and changed the expected returns from the two crops; the change in the sowing pattern followed it. So II is the cause and I is the effect.

24. (A) A statement of fact — The statement reports a published, verifiable figure from a named source. It draws no conclusion and expresses no opinion, so it is a statement of fact.

25. (B) An inference — The second half is a conclusion drawn from an observed fact (lighter traffic) - the words 'must have' mark it. A conclusion drawn from given facts is an inference, not a fact or an opinion.

26. (C) A judgement — 'Must ban' expresses what the speaker thinks ought to be done. A recommendation or an approval/disapproval of something is a judgement, not a verifiable fact or a derived inference.

27. (D) A statement of fact — This is a checkable geographical statement about an existing state of affairs, so it is a statement of fact. (It is also correct: Rajasthan's boundary with Madhya Pradesh is its longest inter-state boundary.)

28. (A) A judgement — 'More peaceful and satisfying' is an evaluative comparison that cannot be verified from data and rests on the speaker's values. Hence it is a judgement.

29. (B) Factors other than local production also influence local wheat prices — A record crop would normally push prices down; since prices held, something else must also be acting on them - demand, procurement, storage or transport costs. Options 1 and 4 are extreme ('no relation', 'always') and option 3 invents a fact.

30. (A) Declaring a village 'covered' does not ensure that its households actually get connections — The gap between village-level coverage (95 per cent) and household-level connections (30 per cent) shows that the two measures are different - coverage of a village does not translate into connections in every house. The other options either overstate ('completely', 'no source at all') or allege fraud that the data do not establish.

31. (C) Employees will have to work longer on each working day — The same weekly hours spread over four days instead of five must mean more hours per day (for example, 40 hours as 4 x 10 instead of 5 x 8). Nothing is stated about salary or output, and total weekly hours are expressly unchanged.

32. (D) Clearing the physical test is necessary but not sufficient for selection — Since every selected candidate cleared it, clearing is necessary; since some who cleared it were not selected, clearing alone is not enough - so it is necessary but not sufficient. Option 2 contradicts the first half of the statement.

33. (A) Many drivers merely shifted to roads just outside the central zone, and the total number of car trips in the city was unchanged — The mayor's claim is about car use in the CITY, but his evidence is only about one zone. If the trips were merely displaced to other roads, the evidence no longer supports the conclusion. The other options say nothing about the level of car use.

34. (B) In a trial where patients were assigned at random either to daily walking or to bed rest, the walking group recovered faster — The original evidence is observational: the patients chose to walk, so healthier patients may simply have walked more. Random assignment removes that self-selection and isolates walking as the cause, which is the strongest possible support. Option 4 actually weakens the claim by supplying a rival explanation.

35. (B) The awarded capacity will actually be built and will generate power by 2030 — The argument jumps from 'contracts awarded' to 'target met'. That step works only if the contracted projects are actually commissioned and start generating in time - the unstated link the argument takes for granted. The other options are irrelevant to that link.

36. (C) It treats a characteristic shared by successful entrepreneurs as if it were enough to guarantee success — The premise makes risk-taking a common feature of those who succeeded (a necessary-looking condition); the conclusion treats it as sufficient for success. Confusing a necessary condition with a sufficient one is the classic flaw here.

37. (D) During the same year the highway was closed for repairs for four months and the traffic on it fell by half — If traffic itself halved, the fall in accidents is explained by fewer vehicles rather than by the cameras. Supplying an alternative explanation for the evidence is the standard way to weaken a causal claim; cost and foreign practice are irrelevant to it.

38. (C) The association between breakfast and higher scores may be explained by another factor — Two things vary together with the scores - breakfast and longer sleep - so the breakfast-score link may be produced by the sleep difference. That is exactly what the data permit; a causal claim in either direction (options 1 and 2) is not established, and option 4 is contradicted by the passage.

39. (B) 7 — Students playing at least one game = $35 + 30 - 12 = 53$. Therefore students playing neither = $60 - 53 = 7$.

40. (C) 50 per cent — Failing both means passing neither, so candidates passing at least one subject = $100 - 15 = 85$ per cent. Using $n(A \text{ or } B) = n(A) + n(B) - n(\text{both})$: $85 = 70 + 65 - n(\text{both})$, so $n(\text{both}) = 50$ per cent.

41. (D) Three concentric circles, the innermost representing Rajasthan and the outermost representing Asia — Rajasthan is wholly contained in India, and India is wholly contained in Asia. Complete inclusion in this chain is shown by concentric circles, smallest class innermost.

42. (B) Two separate circles, both lying inside a larger third circle — Aunts and uncles are mutually exclusive classes - no one is both - so their circles must not overlap; but every aunt and every uncle is a relative, so both circles lie inside the 'Relative' circle.

43. (D) D — From the clues: $C > A > B > E$, and $D > C$. Combining, $D > C > A > B > E$, so D is the heaviest and E the lightest.

44. (A) Suresh — The chain is Kamal > Suresh > Ravi > Mohan > Naveen. Kamal is the tallest, so Suresh is the second tallest.

45. (C) Chitra — Chitra is eldest and Anita is older than Bina, so Bina is the youngest and therefore the engineer (this also satisfies 'Bina is not the doctor'). Anita is not the teacher, so Anita is the doctor and Chitra is the teacher.

46. (A) Box A — Test each case. Prize in A: A's label false, B's false, C's true - exactly one true, which fits. Prize in B: A's true and B's true - two true, rejected. Prize in C: A's true and C's true - two true, rejected. Hence the prize is in Box A.

47. (D) 65 — Only A = $60 - 20 = 40$; only B = $45 - 20 = 25$. Exactly one = $40 + 25 = 65$. (Equivalently $60 + 45 - 2 \times 20 = 65$.)

48. (C) Fifth — Positions 1 and 6 are fixed as U and R, leaving 2, 3, 4, 5 for P, Q, T and S. P-Q must be adjacent (P above Q) and T-S must be adjacent (T above S), and Q must be above S, so the order is P, Q, T, S in positions 2, 3, 4, 5. Hence S is fifth from the top.

49. (B) Qadir — Ravi takes Geography, so Pooja - who is barred from Physics and Economics - must take History. Sunita is barred from Economics, so she takes Physics, leaving Economics for Qadir.

50. (A) 32 24 09 68 51 — At each step the smallest number not yet moved is taken out and placed at the extreme left, the remaining numbers keeping their relative order. Smallest 09 moved first, then 24; the next smallest is 32, so Step III = 32 24 09 68 51.

RM03 — EXPLANATIONS

- 1. (C) 19** — The series has a constant difference of 4: 3, 7, 11, 15, 19, 23. So the missing term is $15 + 4 = 19$.
- 2. (A) 55** — Differences are +6, +8, +10, +12 — increasing by 2 each time. Next difference is +14, so $41 + 14 = 55$.
- 3. (D) 42** — Each term is $n(n+1)$: $1 \times 2 = 2$, $2 \times 3 = 6$, $3 \times 4 = 12$, $4 \times 5 = 20$, $5 \times 6 = 30$, $6 \times 7 = 42$. (Differences are +4, +6, +8, +10, +12.)
- 4. (C) 159** — Each term = previous $\times 2 + 1$: $4 \times 2 + 1 = 9$, $9 \times 2 + 1 = 19$, $19 \times 2 + 1 = 39$, $39 \times 2 + 1 = 79$, $79 \times 2 + 1 = 159$.
- 5. (B) 100** — Each term = previous $\times 2 - 4$: $7 \times 2 - 4 = 10$, $10 \times 2 - 4 = 16$, $16 \times 2 - 4 = 28$, $28 \times 2 - 4 = 52$, $52 \times 2 - 4 = 100$.
- 6. (C) 49** — The terms are perfect squares: 1^2 , 2^2 , 3^2 , 4^2 , 5^2 , 6^2 . Next is $7^2 = 49$.
- 7. (A) 216** — The terms are perfect cubes: 1^3 , 2^3 , 3^3 , 4^3 , 5^3 . Next is $6^3 = 216$.
- 8. (B) 17** — These are consecutive prime numbers: 2, 3, 5, 7, 11, 13. The next prime is 17 ($15 = 3 \times 5$ is not prime).
- 9. (D) 21** — Fibonacci type — each term is the sum of the two preceding terms: $5 + 8 = 13$, $8 + 13 = 21$.
- 10. (C) 360** — The multipliers increase by 1 each step: $3 \times 2 = 6$, $6 \times 3 = 18$, $18 \times 4 = 72$, $72 \times 5 = 360$.
- 11. (B) 1** — Each term is the previous term divided by 5: $625/5 = 125$, $125/5 = 25$, $25/5 = 5$, $5/5 = 1$.
- 12. (A) 37** — Each term is $n^2 + 1$: $1 + 1 = 2$, $4 + 1 = 5$, $9 + 1 = 10$, $16 + 1 = 17$, $25 + 1 = 26$, $36 + 1 = 37$. (Differences are the odd numbers 3, 5, 7, 9, 11.)
- 13. (B) 215** — Each term is $n^3 - 1$: $1 - 1 = 0$, $8 - 1 = 7$, $27 - 1 = 26$, $64 - 1 = 63$, $125 - 1 = 124$, $216 - 1 = 215$.
- 14. (C) 81** — Differences are +5, +10, +15, +20 — an AP with common difference 5. Next difference is +25, so $56 + 25 = 81$.
- 15. (D) 35** — Differences are -21, -19, -17, -15 — decreasing odd numbers. Next difference is -13, so $48 - 13 = 35$.
- 16. (D) 36** — Two alternating series. Terms at odd places: 4, 7, 10, 13 (+3 each). Terms at even places: 9, 18, 27, ? (+9 each), so the next term is $27 + 9 = 36$.
- 17. (C) 20** — Alternate series. Odd places: 1, 4, 9, 16 (perfect squares). Even places: 2, 6, 12, ? which are $n(n+1) = 1 \times 2$, 2×3 , 3×4 , $4 \times 5 = 20$.
- 18. (A) 47** — Each term is the sum of the two preceding terms (Fibonacci type): $3 + 4 = 7$, $4 + 7 = 11$, $7 + 11 = 18$, $11 + 18 = 29$, $18 + 29 = 47$.
- 19. (A) 28** — Triangular numbers — differences are +2, +3, +4, +5, +6. Next difference is +7, so $21 + 7 = 28$.
- 20. (D) 252** — Each term is $n^3 + n^2$: $1 + 1 = 2$, $8 + 4 = 12$, $27 + 9 = 36$, $64 + 16 = 80$, $125 + 25 = 150$, $216 + 36 = 252$.
- 21. (A) 96** — The operations alternate $\times 3$ and -3 : $5 \times 3 = 15$, $15 - 3 = 12$, $12 \times 3 = 36$, $36 - 3 = 33$, $33 \times 3 = 99$, $99 - 3 = 96$.
- 22. (C) 342** — Each term is $n^3 - 1$ starting from $n = 2$: $8 - 1 = 7$, $27 - 1 = 26$, $64 - 1 = 63$, $125 - 1 = 124$, $216 - 1 = 215$, $343 - 1 = 342$.
- 23. (A) 34** — The series is $n^2 - 1$: $4 - 1 = 3$, $9 - 1 = 8$, $16 - 1 = 15$, $25 - 1 = 24$, $36 - 1 = 35$, $49 - 1 = 48$, $64 - 1 = 63$. The fifth term should be 35, not 34.
- 24. (A) 38** — The series is $n^2 + 1$: 2, 5, 10, 17, 26, 37, 50. The sixth term should be $6^2 + 1 = 37$, not 38.
- 25. (D) 75** — Each term should be double the previous one: 5, 10, 20, 40, 80, 160. The fifth term should be 80, not 75.
- 26. (C) 24** — The series is of perfect squares: 1, 4, 9, 16, 25, 36, 49. The fifth term should be $5^2 = 25$, not 24.
- 27. (B) U** — Positions are A=1, C=3, F=6, J=10, O=15 — gaps of +2, +3, +4, +5. The next gap is +6, giving position 21 = U.
- 28. (D) R** — Every second letter is skipped, moving backwards: Z(26), X(24), V(22), T(20), R(18).
- 29. (A) Y** — The positions are perfect squares: A=1, D=4, I=9, P=16, so the next is 25 = Y.

- 30. (C) O** — Each letter advances by 3: C(3), F(6), I(9), L(12), O(15).
- 31. (A) EV** — The first letters move forward A, B, C, D, E while the second letters move backward Z, Y, X, W, V. So the answer is EV. (Note B and Y, C and X are 'opposite' letters, sum of positions 27.)
- 32. (B) DFH** — Within each group the letters rise by 2 (A, C, E). Each successive group starts one letter later: A, B, C, D. So the fourth group is D, F, H.
- 33. (B) MNOPQ** — The groups have 2, 3, 4 and then 5 consecutive letters, and exactly one letter is skipped between groups (C, G, L). After HIJK we skip L, so the next group is M, N, O, P, Q.
- 34. (B) Z** — Positions are 2, 5, 10, 17 — that is $n^2 + 1$. The next is $5^2 + 1 = 26 = Z$. (Gaps are +3, +5, +7, +9.)
- 35. (B) I9** — Each letter is followed by its own position number, and only alternate letters are used: A(1), C(3), E(5), G(7), I(9).
- 36. (D) V5** — Letters run backwards Z, Y, X, W, V while numbers run forwards 1, 2, 3, 4, 5. So the answer is V5.
- 37. (D) M32** — The letters advance by 3 (A, D, G, J, M) and the numbers double (2, 4, 8, 16, 32).
- 38. (D) 26M** — The numbers follow $n^2 + 1$ (2, 5, 10, 17, 26) and the letters advance by 3 (A, D, G, J, M). So the answer is 26M.
- 39. (D) 125** — $8 = 2^3$ and $27 = 3^3$, i.e. cubes of consecutive numbers. $64 = 4^3$, so the answer is $5^3 = 125$.
- 40. (B) 64** — $25 = 5^2$ and $36 = 6^2$ — squares of consecutive numbers. $49 = 7^2$, so the answer is $8^2 = 64$.
- 41. (B) 72** — The relation is $n : n(n+1)$. Here $6 \times 7 = 42$, so $8 \times 9 = 72$.
- 42. (D) NPR** — Each letter moves one place forward: B→C, D→E, F→G. Applying the same to M, O, Q gives N, P, R.
- 43. (A) QSU** — Each letter of the first group is advanced by one place. So P→Q, R→S, T→U, giving QSU.
- 44. (C) Cut** — The relation is tool to its function. A pen is used to write, a knife is used to cut.
- 45. (C) Heart** — The relation is specialist to the organ treated. An ophthalmologist treats the eye; a cardiologist treats the heart. (Nephrologist-kidney, neurologist-brain, pulmonologist-lung.)
- 46. (A) 100** — $27 = 3^3$, $64 = 4^3$ and $125 = 5^3$ are perfect cubes. $100 = 10^2$ is a perfect square but not a perfect cube.
- 47. (B) 9** — 3, 5 and 11 are prime numbers. $9 = 3 \times 3$ is composite.
- 48. (B) Mango** — Rose, lotus and lily are flowers; mango is a fruit.
- 49. (A) Brass** — Copper, zinc and iron are pure metals (elements). Brass is an alloy of copper and zinc.
- 50. (C) RTW** — In ACE, GIK and MOQ the gap between successive letters is +2 and +2. In RTW the gaps are +2 and +3, so it is the odd one out.

RM04 — EXPLANATIONS

- 1. (C) CPNCBZ** — Each letter moves one place forward: M→N, A→B, D→E, R→S, A→B, S→T. Applying +1 to BOMBAY gives C, P, N, C, B, Z = CPNCBZ.
- 2. (B) ECPFNG** — Each letter moves two places forward: F→H, R→T, I→K, E→G, N→P, D→F. Applying +2 to CANDLE gives E, C, P, F, N, G = ECPFNG.
- 3. (C) VWXGHQW** — Every letter is shifted three places forward: T→W, E→H, A→D, C→F, H→K, E→H, R→U. So STUDENT becomes V, W, X, G, H, Q, W = VWXGHQW.
- 4. (B) CHRONRD** — Each letter moves one place backward: T→S, R→Q, I→H, P→O, P→O, L→K, E→D. Applying -1 to DISPOSE gives C, H, R, O, N, R, D = CHRONRD.
- 5. (B) CFOB** — Each letter moves three places backward: W→T, A→X (wrapping round the alphabet), T→Q, E→B, R→O. Applying -3 to FIRE gives C, F, O, B = CFOB.
- 6. (D) XZG** — Each letter is replaced by its opposite letter (position 27 minus its own position): D(4)→W(23), O(15)→L(12), G(7)→T(20). For CAT: C(3)→X(24), A(1)→Z(26), T(20)→G(7) = XZG.
- 7. (A) IABMUM** — The word is simply written in reverse order: D-E-L-H-I read backwards is I-H-L-E-D. MUMBAI read backwards is I-A-B-M-U-M.

- 8. (B) HPE** — First reverse the word, then move each letter one place forward. CAT reversed is TAC; T→U, A→B, C→D gives UBD. DOG reversed is GOD; G→H, O→P, D→E gives HPE.
- 9. (A) UZCKF** — The shift alternates +1 and -1: C+1=D, H-1=G, A+1=B, I-1=H, R+1=S. For TABLE: T+1=U, A-1=Z, B+1=C, L-1=K, E+1=F = UZCKF.
- 10. (B) QSU** — Each letter is shifted forward by its own position in the word: A+1=B, B+2=D, C+3=F. For PQR: P+1=Q, Q+2=S, R+3=U = QSU.
- 11. (D) HCIASR** — The letters are interchanged in successive pairs: MO→OM, NK→KN, EY→YE. For CHAIRS: CH→HC, AI→IA, RS→SR, giving HCIASR.
- 12. (A) RUMBEN** — Only the first and the last letters are interchanged; the middle letters stay in place. SYSTEM → MYSTES. So NUMBER becomes RUMBEN.
- 13. (C) 654** — Replace each letter by its alphabet position: C=3, A=1, B=2 gives 312. So F=6, E=5, D=4 gives 654.
- 14. (B) 57** — The code is the sum of the alphabet positions: S(19) + U(21) + N(14) = 54. For MOON: M(13) + O(15) + O(15) + N(14) = 57.
- 15. (C) 26** — The code is the sum of alphabet positions: C(3) + A(1) + T(20) = 24. For DOG: D(4) + O(15) + G(7) = 26.
- 16. (A) 32** — Add the alphabet positions: L(12) + O(15) + V(22) + E(5) = 54. For LIFE: L(12) + I(9) + F(6) + E(5) = 32.
- 17. (B) 12-2-6-10** — Each letter is coded as twice its alphabet position: F = 2 × 6 = 12, A = 2 × 1 = 2, C = 2 × 3 = 6, E = 2 × 5 = 10. So FACE = 12-2-6-10.
- 18. (B) 24-26-25** — Reverse position = 27 minus the forward position. C: 27 - 3 = 24; A: 27 - 1 = 26; B: 27 - 2 = 25. So CAB = 24-26-25.
- 19. (C) 24153** — From PALE = 2134: P=2, A=1, L=3, E=4. From EARTH = 41590: E=4, A=1, R=5, T=9, H=0. So PEARL = P(2) E(4) A(1) R(5) L(3) = 24153.
- 20. (C) 214673** — Building the map: R=6, O=8, S=2, E=1 (from ROSE); C=7, H=3, A=4, I=5 (from CHAIR); P=9 (from PREACH). So SEARCH = S(2) E(1) A(4) R(6) C(7) H(3) = 214673.
- 21. (C) 1-16-16-12-5** — Each letter is replaced by its alphabet position: M=13, A=1, N=14, G=7, O=15. For APPLE: A=1, P=16, P=16, L=12, E=5.
- 22. (D) 2120** — The alphabet positions are written one after another: C(3) A(1) T(20) = 3120; D(4) O(15) G(7) = 4157. For BAT: B(2) A(1) T(20) = 2120.
- 23. (D) Air** — Fish actually live in water; but in this language 'water' is called 'air'. Hence the answer is 'air'.
- 24. (D) Yellow** — Blood is really red; in this language 'red' is called 'yellow'. Hence the answer is 'yellow'.
- 25. (A) Book** — We write with a pen; in this language 'pen' is called 'book'. Hence the answer is 'book'.
- 26. (A) Blue** — A person normally dances to music, and 'dance' is expressed by the word 'blue'. Hence the answer is 'blue'.
- 27. (C) hai** — 'are' is the only English word common to the two sentences, and 'hai' is the only code word common to the two codes. Therefore hai = are.
- 28. (D) bap** — Sentences 1 and 2 share only 'nap'/'is'; sentences 1 and 3 share only 'sok'/'intelligent'; sentences 2 and 3 share only 'lin'/'smart'. In sentence 3 the remaining code word 'bap' must therefore stand for the remaining English word 'boy'.
- 29. (A) pit** — 'na' is common to sentences 1 and 2 as 'are'; 'tod' is common to sentences 1 and 3 as 'good'; 'rok' is common to sentences 2 and 3 as 'bad'. The remaining word of sentence 1, 'pit', must therefore mean 'you'.
- 30. (C) mip** — 'sud' = work (common to sentences 1 and 2); 'pal' = pays (common to 1 and 3); 'tor' = is (common to 2 and 3). The word left over in sentence 2, 'mip', therefore means 'worship'.
- 31. (A) lin** — 'po' = eat (common to sentences 1 and 2); 'ri' = fish (common to 1 and 3); 'sa' = bones (common to 2 and 3). The remaining word of sentence 2, 'lin', therefore means 'dogs'.
- 32. (A) SUNDAY** — Each letter of the message is written one place backward, so to decode we move each letter one place forward: R→S, T→U, M→N, C→D, Z→A, X→Y. The word is SUNDAY.

- 33. (B) \$795\$** — A is a vowel and it is the first letter, so condition (i) applies: the first and the last letters are both coded as \$. The middle letters keep their digits M=7, K=9, R=5. Hence the code is \$795\$.
- 34. (D) 4729#** — The first letter P is not a vowel, so condition (i) does not apply. The last letter E is a vowel, so by condition (ii) it is coded as #. The rest keep their digits: P=4, M=7, A=2, K=9. Hence 4729#.
- 35. (A) 95274** — Neither the first letter K nor the last letter P is a vowel, so no condition applies and each letter simply takes its own digit: K=9, R=5, A=2, M=7, P=4, giving 95274.
- 36. (A) 67493** — The first letter U is a vowel and the last letter B is a consonant, so condition (ii) applies: their codes are interchanged. Normally U=3, D=7, N=4, S=9, B=6; after interchanging the first and last codes we get 6, 7, 4, 9, 3 = 67493.
- 37. (B) *4671** — The first letter S and the last letter J are both consonants, so by condition (i) the first letter is coded as *. The remaining letters keep their digits: N=4, B=6, D=7, J=1. Hence *4671.
- 38. (C) 02-00-34** — C lies in row 0, column 2 = 02. A lies in row 0, column 0 = 00. T lies in row 3, column 4 = 34. So CAT = 02-00-34.
- 39. (D) FIRE** — 10 = row 1, column 0 = F; 13 = row 1, column 3 = I; 32 = row 3, column 2 = R; 04 = row 0, column 4 = E. The word is FIRE.
- 40. (C) 22-24-24-23** — M = row 2, column 2 = 22; O = row 2, column 4 = 24; O = 24 again; N = row 2, column 3 = 23. So MOON = 22-24-24-23.
- 41. (D) MPDL** — Each letter of the first word is advanced by one place: B->C, O->P, O->P, K->L. Applying +1 to LOCK gives M, P, D, L = MPDL.
- 42. (D) EPPS** — The rule is +1 on every letter: G->H, A->B, T->U, E->F. So D->E, O->P, O->P, R->S, giving EPPS.
- 43. (B) WLT** — Each letter is replaced by its opposite letter (27 minus its position): C(3)->X(24), A(1)->Z(26), T(20)->G(7). For DOG: D(4)->W(23), O(15)->L(12), G(7)->T(20) = WLT.
- 44. (D) IDWKHU** — Every letter moves three places forward: M->P, O->R, T->W, H->K, E->H, R->U. Applying +3 to FATHER gives I, D, W, K, H, U = IDWKHU.
- 45. (A) QMSLB** — Every letter moves two places backward: L->J, I->G, G->E, H->F, T->R. Applying -2 to SOUND gives Q, M, S, L, B = QMSLB.

RM05 — EXPLANATIONS

- 1. (D) Maternal uncle** — The man's mother's only daughter is the man's own sister, and she is Reena's mother. So the man is the brother of Reena's mother, i.e. Reena's maternal uncle.
- 2. (D) Brother** — The grandfather's only child must be the girl's own father. The son of the girl's father is the girl's brother.
- 3. (D) Maternal grandfather** — A is B's father; B is C's mother; C and D are siblings, so B is D's mother too. A is therefore the father of D's mother, i.e. D's maternal grandfather.
- 4. (D) Son** — Q is R's son and P is Q's brother, so P is also R's son. R is S's wife, hence S is P's father and P is S's son.
- 5. (B) P + R + Q** — 'P + R' makes P the father of R and 'R + Q' makes R the father of Q. The father of a father is the paternal grandfather, so P + R + Q is correct.
- 6. (D) Mother** — M × N makes M the mother of N; N - O makes N the sister of O, so N and O have the same mother. Hence M is the mother of O.
- 7. (D) Maternal uncle** — K ÷ L makes K the brother of L; L × M makes L the mother of M. K is therefore the brother of M's mother, i.e. M's maternal uncle.
- 8. (A) Maternal uncle** — D + E makes D the brother of E; E ÷ F makes E the mother of F. So D is the brother of F's mother, i.e. F's maternal uncle.
- 9. (D) Father** — X × Y makes X the father of Y; Y + Z makes Y the brother of Z, so Y and Z share the same father. Hence X is the father of Z.
- 10. (A) Wife** — S - T makes S the mother of T; T × U makes T the son of U, so U is also a parent of T. As U is male he is T's father, and therefore S is U's wife.

- 11. (D) Son-in-law** — $P + Q$ makes P the husband of Q ; $Q \div R$ makes Q the daughter of R . The husband of a person's daughter is that person's son-in-law.
- 12. (A) Granddaughter** — $A - B$ makes A the daughter of B ; $B + C$ makes B the son of C . So B is C 's child and A is B 's daughter, making A the granddaughter of C .
- 13. (C) Cousin sister** — The only sister of the man's father is his paternal aunt. Her daughter is the man's cousin sister.
- 14. (D) Brother** — The only son of Meena's grandmother is Meena's father. The son of Meena's father is Meena's brother.
- 15. (A) Granddaughter** — The wife's only son is the man's own son. The daughter of one's son is one's granddaughter.
- 16. (C) Paternal aunt** — Her mother's daughter is the woman herself; her only brother is the boy's father. The sister of a boy's father is his paternal aunt (bua).
- 17. (D) Sister-in-law** — The only son of his mother other than himself is his brother. His brother's wife is his sister-in-law (bhabhi).
- 18. (A) Cousin brother** — The only brother of the lady's mother is her maternal uncle (mama). His son is her cousin brother.
- 19. (C) Great-grandfather** — $A \rightarrow B \rightarrow C \rightarrow D$ covers three parent steps: A is B 's father, B is C 's mother and C is D 's father. Three steps up from D is the great-grandfather.
- 20. (D) Grandmother** — T is R 's son, and since S is R 's wife, T is also S 's son. V is T 's son, so V is the son of S 's son, i.e. S is V 's grandmother.
- 21. (A) Son-in-law** — D is the daughter of A and B . E , being D 's husband, is the son-in-law of both A and B .
- 22. (C) Brother-in-law** — S is R 's sister and T is S 's husband. The husband of one's sister is one's brother-in-law (jija).
- 23. (D) Grandmother** — Z is the daughter of X and Y ; V is the daughter of Z and W . So V is the daughter of Y 's daughter, making Y her (maternal) grandmother.
- 24. (A) Cousin sister** — D is A 's brother, so E (D 's daughter) and C (A 's son) are children of two brothers, i.e. cousins. E is C 's cousin sister.
- 25. (C) Paternal aunt** — E 's father is D and D 's brother is A , whose wife is B . The wife of a father's brother is the paternal aunt (chachi).
- 26. (D) Cousin brother** — M and N are brother and sister; O is M 's son and P is N 's daughter. Children of a brother and a sister are cousins, so O is P 's cousin brother.
- 27. (A) Grandmother** — M is the son of K and L ; O is M 's son. So O is the son of L 's son, and L is his grandmother.
- 28. (C) Niece** — F is A 's mother (B is her daughter-in-law). C is A 's brother, and D and E are C 's children. So E is the daughter of A 's brother, i.e. A 's niece.
- 29. (D) Grandmother** — F is the mother of A and therefore also of A 's brother C . D is C 's son, so F is D 's grandmother.
- 30. (A) Brother** — R is not Q 's mother, so R is Q 's father and P is Q 's mother. S is P 's daughter, hence Q and S are brother and sister; Q is S 's brother.
- 31. (C) Paternal uncle** — R is S 's father and T is R 's brother. The brother of one's father is the paternal uncle (chacha).
- 32. (D) Paternal aunt** — C is G 's father and D is C 's sister. The sister of a child's father is his paternal aunt (bua).
- 33. (A) Daughter-in-law** — C is the son of A and B ; E is C 's wife. The wife of one's son is one's daughter-in-law.
- 34. (C) Paternal aunt** — B is C 's father and A is B 's sister. The sister of one's father is the paternal aunt (bua). Note that C is one generation below A and D two generations below her.
- 35. (C) Five** — The chain $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$ adds one level at each step: A (1st), B (2nd), C (3rd), D (4th), E (5th). So five generations are covered.
- 36. (B) Three** — A forms the first generation; B and C (her children) the second; D and E (C 's children) the third. Hence three generations.

37. (A) S is the daughter-in-law of Q — R is the son of P and Q, and S is R's wife. The wife of a person's son is that person's daughter-in-law, so S is the daughter-in-law of both P and Q.

38. (D) Maternal aunt — The only brother of his mother is his maternal uncle (mama). The wife of the maternal uncle is the maternal aunt (mami).

39. (B) Three — The family is A (husband), B (wife), C (son), D (daughter), G (C's wife) and E (C's son). The males are A, C and E - three in all.

40. (A) Cousin brother — Sharing the same maternal grandmother without being brothers means their mothers are sisters. Children of two sisters are cousins, so the other man is his cousin brother.

RM06 – EXPLANATIONS

1. (B) 6 km — Taking the start as origin: 10 km North gives (0, 10); a right turn faces East, 6 km gives (6, 10); a right turn faces South, 10 km gives (6, 0). The displacement is 6 km due East, so the distance is 6 km.

2. (C) East — The path ends at (6, 0) with the start at (0, 0). The y-coordinates cancel, leaving a positive x-displacement, so he is due East of his starting point.

3. (D) 5 km — The two legs are at right angles, so the shortest distance is the hypotenuse: $\sqrt{4^2 + 3^2} = \sqrt{25} = 5$ km.

4. (A) 3 km East — 5 km South gives (0, -5); a left turn from South faces East, 3 km gives (3, -5); a left turn from East faces North, 5 km gives (3, 0). He is 3 km due East of the start.

5. (B) 13 km — The legs are perpendicular, so the shortest distance is $\sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$ km.

6. (C) 10 km South-West — Her position is (-6, -8). The distance is $\sqrt{6^2 + 8^2} = 10$ km, and since x is negative and y is negative she is South-West of the start.

7. (D) 25 metres — The right turn from North faces East, so the two legs are perpendicular. Distance = $\sqrt{15^2 + 20^2} = \sqrt{625} = 25$ metres.

8. (A) 14.14 km — The final position is (25 - 15, 10) = (10, 10). The distance is $\sqrt{10^2 + 10^2} = 10\sqrt{2}$, which is about 14.14 km.

9. (B) North-East — His net displacement is 10 km East and 10 km North, i.e. the point (10, 10). Both components are positive, so he is North-East of the start.

10. (C) 2 km — The moves are N3 -> (0,3), E4 -> (4,3), S3 -> (4,0), W2 -> (2,0). He is 2 km due East of the start.

11. (D) West — Facing North, a 90-degree clockwise turn makes him face East. A 180-degree anti-clockwise turn from East makes him face West.

12. (A) South — From South-West, 135 degrees clockwise is three steps of 45 degrees: South-West -> West -> North-West -> North. A 180-degree anti-clockwise turn from North gives South.

13. (B) South — East, turn left -> North; turn right -> East; turn right -> South. He is finally facing South.

14. (B) South-East — South-East moving to North is a rotation of 135 degrees anti-clockwise. Applying the same 135-degree anti-clockwise rotation to West gives South-East.

15. (C) South — After sunrise the Sun is in the East, so every shadow falls towards the West. If West is on his right, he must be facing South.

16. (D) South — Before sunset the Sun is in the West, so shadows fall towards the East. If East is on her left, she is walking towards the South.

17. (A) West — At sunrise the Sun is in the East and shadows fall towards the West. A shadow falling behind the first friend means West is behind him, so he faces East. The second friend, standing face to face with him, must therefore face West.

18. (B) South-East — The path is N7 -> (0,7), E5 -> (5,7), S9 -> (5,-2), E3 -> (8,-2). With x positive and y negative he is South-East of the start.

19. (C) 15 km — The legs are perpendicular, so the shortest distance is $\sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$ km.

- 20. (D) 25 km** — His position is 24 km South and 7 km East of the start, so the direct distance back is $\sqrt{24^2 + 7^2} = \sqrt{576 + 49} = \sqrt{625} = 25$ km.
- 21. (A) 10 km** — The net displacement is 8 km East and 6 km North, giving the point (8, 6). Distance = $\sqrt{64 + 36} = \sqrt{100} = 10$ km.
- 22. (B) At the starting point** — Net North-South movement = $20 - 5 - 15 = 0$ m; net East-West movement = $10 - 10 = 0$ m. Both components vanish, so he is exactly back at the starting point.
- 23. (C) 8 km West** — The two 6 km legs (North and South) cancel, leaving only 8 km West. She is 8 km due West of the start.
- 24. (D) East** — North, right $\rightarrow$ East, right $\rightarrow$ South, left $\rightarrow$ East. His final direction of movement is East.
- 25. (A) 10 km** — The path is N5 $\rightarrow$ (0,5), E5 $\rightarrow$ (5,5), S5 $\rightarrow$ (5,0), E5 $\rightarrow$ (10,0). He is 10 km due East of the starting point.
- 26. (B) East** — Taking B as the origin: A is at (0, 5), C at (5, 0) and D at (5, 5). D and A have the same y-coordinate, and D lies 5 km to the east, so D is due East of A.
- 27. (C) 120 km** — The outward legs total $30 + 40 = 70$ km. The return is the hypotenuse, $\sqrt{30^2 + 40^2} = 50$ km. The whole journey is $70 + 50 = 120$ km.
- 28. (D) 9 km** — S10 $\rightarrow$ (0,-10); left from South is East, 5 km $\rightarrow$ (5,-10); left from East is North, 10 km $\rightarrow$ (5,0); right from North is East, 4 km $\rightarrow$ (9,0). She is 9 km from the start.
- 29. (A) South-East** — Ninety degrees clockwise means two steps of 45 degrees on the eight-point compass: North-East $\rightarrow$ East $\rightarrow$ South-East.
- 30. (B) North** — North-West, 45 degrees anti-clockwise $\rightarrow$ West; 180 degrees clockwise $\rightarrow$ East; 90 degrees anti-clockwise $\rightarrow$ North.
- 31. (C) 5 km; grandson** — Put B at the origin: A is at (0, 3) and C at (4, 0). The distance is $\sqrt{4^2 + 3^2} = 5$ km. Since A is B's father and B is C's father, C is A's grandson.
- 32. (D) 10 km; maternal uncle** — With Q at the origin, P is at (-8, 0) and R at (0, 6); the distance is $\sqrt{64 + 36} = 10$ km. P is the brother of R's mother, i.e. R's maternal uncle.
- 33. (A) mother; North-East** — X is Z's father and Y is X's wife, so Y is Z's mother. Taking Z's home as origin, the temple is at (5, 5), which is North-East of her home.
- 34. (B) 6 km West** — N12 $\rightarrow$ (0,12), E9 $\rightarrow$ (9,12), S12 $\rightarrow$ (9,0), W15 $\rightarrow$ (-6,0). He is 6 km due West of the starting point.
- 35. (C) 10 km East** — The 5 km South and 5 km North cancel each other; the two eastward legs add to 10 km. He is 10 km due East of the start.

RM07 — EXPLANATIONS

- 1. (D) 29th** — Rank from the bottom = total - rank from the top + 1 = $40 - 12 + 1 = 29$ th. Remember that the person himself is counted once from each end, hence the +1.
- 2. (B) 24** — When both ranks are of the same person, total = (rank from front) + (rank from back) - 1 = $10 + 15 - 1 = 24$. Meena is counted twice, so one is subtracted.
- 3. (B) 10** — Rakesh's position from the left = $40 - 18 + 1 = 23$ rd. Suresh is 12th from the left. Boys between them = $23 - 12 - 1 = 10$.
- 4. (B) 15th** — After the shift his position from the left = $30 - 12 + 1 = 19$ th. He had moved 4 places to the right, so originally he was $19 - 4 = 15$ th from the left.
- 5. (B) 34** — Total = $15 + 20 - 1 = 34$, subtracting one because Rahul himself is counted in both the counts.
- 6. (C) 19** — After the exchange Nita occupies Radha's original seat, which is 11th from the left and 9th from the right. Total = $11 + 9 - 1 = 19$.
- 7. (A) 28th** — Naveen's rank from the top = $12 + 6 = 18$ th. His rank from the bottom = $45 - 18 + 1 = 28$ th.
- 8. (C) 12th** — Her new position is 8th from the left, so her earlier position was $8 + 2 = 10$ th from the left. From the right end that is $21 - 10 + 1 = 12$ th.

- 9. (B) D** — The conditions give $E < B < A < C < D$, a single consistent order. D, being above C who is above everyone else, is the tallest.
- 10. (C) P** — The order from lightest to heaviest is $T < S < P < R < Q$. Counting from the heaviest: Q (1st), R (2nd), P (3rd). So P is the third heaviest.
- 11. (D) F** — Working through the clues fixes exactly one arrangement, from left to right: C, E, A, G, B, D, F. The person at the extreme right end is therefore F.
- 12. (A) E** — The unique arrangement from left to right is C, E, A, G, B, D, F. B occupies the fifth seat, so the third seat to his left is the second seat, occupied by E.
- 13. (B) T** — Since all face South, the person's right corresponds to the printed left. The clues yield one arrangement, printed left to right: T, Q, U, P, S, R. Hence the extreme right end (of the persons) is occupied by T.
- 14. (B) Two** — The unique arrangement, printed left to right, is T, Q, U, P, S, R. S and Q occupy the 5th and 2nd printed seats, so exactly two persons sit between them.
- 15. (C) A** — The clues admit only one seating, from left to right: D, G, B, H, A, F, C, E. H is in seat 4 and F in seat 6, so the student between them is the one in seat 5, namely A.
- 16. (A) D and E** — The unique arrangement from left to right is D, G, B, H, A, F, C, E, so the two ends are held by D and E.
- 17. (D) V** — Only one arrangement satisfies every clue, from left to right: X, Z, V, Y, W. The middle (third) seat is occupied by V.
- 18. (A) W** — The unique order from left to right is X, Z, V, Y, W. As everybody faces North, the right of a person is the next seat towards the printed right, so the person immediately to Y's right is W.
- 19. (B) P** — The clues admit only one seating. First row (left to right): C, A, E, B, D; second row (left to right): S, Q, T, P, R. B is the fourth from the printed left in row 1, so the person facing him is the fourth from the printed left in row 2, i.e. P.
- 20. (C) S** — The unique seating is: First row (left to right): C, A, E, B, D; second row (left to right): S, Q, T, P, R. The second row faces South, so their right is the printed left; the person at their extreme right end is S.
- 21. (C) Three** — In the unique arrangement the first row reads C, A, E, B, D from left to right. C is first and D is fifth, so three persons sit between them.
- 22. (D) S** — The second row reads S, Q, T, P, R from the printed left. Since this row faces South, a person's right is towards the printed left, so the person immediately to Q's right is S.
- 23. (A) P** — The clues fix a single seating. Clockwise order: S, P, W, T, Q, V, R, U (and back to S). For a person facing the centre the right side is the anti-clockwise direction, so three seats anti-clockwise from Q brings us to P.
- 24. (B) R** — With eight persons evenly placed, the seat opposite any person is four seats away. Clockwise order: S, P, W, T, Q, V, R, U (and back to S). Four seats from W gives R.
- 25. (A) One** — In the unique seating the clockwise order is S, P, W, T, Q, V, R, U. Moving clockwise from T we pass exactly one person before reaching V.
- 26. (C) C** — The clues give exactly one seating. Clockwise order: A, D, F, B, E, C (and back to A). Facing the centre, the left neighbour is the next person clockwise, which is C.
- 27. (D) A** — For six persons round a table the seat opposite any person is three seats away. Clockwise order: A, D, F, B, E, C (and back to A), so the person opposite B is A.
- 28. (A) M** — Only one seating fits. Clockwise order: K, N, L, O, M (and back to K). Facing the centre, moving to a person's left means moving clockwise, so two seats clockwise from L is M.
- 29. (B) L** — Clockwise order: K, N, L, O, M (and back to K) is the unique solution. The right of a person facing the centre is the anti-clockwise neighbour, which for O is L.
- 30. (C) S** — The clues give one seating only. Clockwise order: P, T, R, U, Q, S (and back to P). Facing outward, a person's left is the anti-clockwise side, so three seats anti-clockwise from R is S.
- 31. (D) U** — Clockwise order: P, T, R, U, Q, S (and back to P) is the unique arrangement. Facing outward, the right is the clockwise side, so two seats clockwise from T gives U.

- 32. (A) U** — Whether the persons face inward or outward, the seat opposite a person in a circle of six is three seats away. Clockwise order: P, T, R, U, Q, S (and back to P), so U sits opposite P.
- 33. (B) A and B** — At a square table the person diagonally opposite is two corners away. Clockwise order: A, C, B, D, so A and B sit diagonally opposite each other (and so do the remaining two).
- 34. (B) A, C, B, D** — Applying the clues to the four corners leaves exactly one possibility. Clockwise order: A, C, B, D.
- 35. (C) Q** — Going round the table there is only one possible order. Clockwise order: L, P, N, Q, M, O. Facing the centre, a person's right is the anti-clockwise neighbour, so the person immediately to M's right is Q.
- 36. (D) D** — The clues leave only one possible allotment: floor 1 - C; floor 2 - A; floor 3 - E; floor 4 - B; floor 5 - D. Hence D lives on floor 5.
- 37. (A) E** — The unique allotment is floor 1 - C; floor 2 - A; floor 3 - E; floor 4 - B; floor 5 - D, so the third floor is occupied by E.
- 38. (A) One** — The allotment is fixed uniquely as floor 1 - R; floor 2 - U; floor 3 - P; floor 4 - T; floor 5 - Q; floor 6 - S. P is on floor 3 and Q on floor 5, so exactly one person lives between them.
- 39. (B) Q** — The unique allotment is floor 1 - R; floor 2 - U; floor 3 - P; floor 4 - T; floor 5 - Q; floor 6 - S. T is on floor 4, so the person immediately above him, on floor 5, is Q.
- 40. (C) A** — There is exactly one arrangement consistent with the clues: floor 1 - F; floor 2 - B; floor 3 - G; floor 4 - D; floor 5 - A; floor 6 - E; floor 7 - C. Floor 5 is occupied by A.
- 41. (B) Three** — In the unique arrangement floor 1 - F; floor 2 - B; floor 3 - G; floor 4 - D; floor 5 - A; floor 6 - E; floor 7 - C, D lives on floor 4, so three persons live on the floors above him.
- 42. (D) T** — Only one schedule satisfies all the clues: Monday - S; Tuesday - P; Wednesday - T; Thursday - R; Friday - Q. Wednesday belongs to T.
- 43. (A) Thursday** — The unique schedule is Monday - S; Tuesday - P; Wednesday - T; Thursday - R; Friday - Q, so R visits on Thursday.
- 44. (B) April** — The clues fix a single schedule: January - D; February - A; March - F; April - C; May - E; June - B. C's examination therefore falls in April.
- 45. (B) Two** — The unique schedule is January - D; February - A; March - F; April - C; May - E; June - B. A appears in February and E in May, so two candidates (March and April) fall between them.
- 46. (B) Two intersecting circles for Doctors and Men, both lying wholly inside a larger circle for Human beings** — Some doctors are men and some are women, so Doctors and Men only partly overlap. Both classes are entirely contained in the class of Human beings.
- 47. (A) A circle for Pigeon inside a circle for Bird, with a separate circle for Dog outside both** — Every pigeon is a bird, so Pigeon lies wholly within Bird. No dog is a bird and no dog is a pigeon, so Dog is a wholly separate circle.
- 48. (C) A circle for Jaipur inside a circle for Rajasthan, which in turn lies inside a circle for India** — Jaipur is entirely within Rajasthan and Rajasthan is entirely within India, so the three form completely nested circles from smallest to largest.
- 49. (A) Two separate circles for Table and Chair, both lying wholly inside a larger circle for Furniture** — No table is a chair, so the two classes are disjoint; but both are kinds of furniture, so both lie completely inside the class Furniture.
- 50. (C) Two separate circles for Boys and Girls, each of which partly overlaps a third circle for Students** — No boy is a girl, so those two circles are disjoint. Some boys and some girls are students while others are not, so Students overlaps each of them only partly.
- 51. (B) 10** — Students playing at least one game = $35 + 25 - 10 = 50$. Therefore those playing neither = $60 - 50 = 10$.
- 52. (C) 90** — By the inclusion-exclusion rule, $n(A \text{ or } B) = n(A) + n(B) - n(A \text{ and } B) = 65 + 45 - 20 = 90$.
- 53. (B) 10** — Since nobody is left out, $n(A \text{ or } B) = 80$. Then $n(A \text{ and } B) = n(A) + n(B) - n(A \text{ or } B) = 50 + 40 - 80 = 10$.

54. (B) 15 — By inclusion-exclusion for three sets: $100 + 90 + 80 - 40 - 35 - 30 + 20 = 185$ study at least one subject. Hence $200 - 185 = 15$ study none of them.

55. (C) 61 — Number using exactly one = (sum of singles) - 2(sum of pairwise) + 3(all three) = $165 - 2(67) + 3(10) = 165 - 134 + 30 = 61$.

RM08 — EXPLANATIONS

1. (B) 8 — The two diagonals cut each other at the centre O. Smallest triangles: AOB, BOC, COD, DOA = 4. Triangles made of two of these, i.e. the halves of the rectangle: ABC, BCD, CDA, DAB = 4. Total = $4 + 4 = 8$.

2. (C) 16 — All four lines pass through the centre O, cutting the square into 8 smallest triangles. Adding pairs that merge across a corner gives the 4 triangles AOB, BOC, COD, DOA; adding further gives the 4 half-squares ABC, BCD, CDA, DAB. Total = $8 + 4 + 4 = 16$.

3. (B) 12 — The mid-line passes through the centre O, so it splits only the top and bottom triangles. Smallest triangles = 6 (two on top, two at the bottom, one on each side). Adding pairs gives the 2 triangles AOB and COD; the four half-rectangles ABC, ABD, BCD, ACD give 4 more. Total = $6 + 2 + 4 = 12$.

4. (B) 16 — The three medians meet at the centroid G and cut the triangle into 6 smallest triangles of equal area. Larger ones: ABG, BCG and CAG (each made of 2 small ones) = 3; the six halves ABD, ADC, BCE, BEA, CAF, CFB (each made of 3 small ones) = 6; and the whole triangle = 1. Total = $6 + 3 + 6 + 1 = 16$.

5. (D) 6 — The diagonal AC and both mid-lines pass through the centre O. On one side of AC the mid-lines cut off the two small triangles A-P-O and C-Q-O (P, Q being midpoints of AB and BC); on the other side they cut off two more. That is 4 small triangles. The diagonal also splits the square into the two halves ABC and ACD. Total = $4 + 2 = 6$.

6. (D) 12 — This is the same configuration as a rectangle with both diagonals plus one mid-line: 6 smallest triangles + 2 (the two halves that the mid-line splits) + 4 half-squares = 12.

7. (A) 6 — The opposite side now carries 4 points (2 end-points + 2 interior points). Every pair of these points, joined to the apex, gives one triangle, so the count is $C(4,2) = 6$.

8. (A) 10 — The opposite side carries $2 + 3 = 5$ points. Number of triangles = $C(5,2) = 10$.

9. (A) 5 — Joining the three midpoints cuts the triangle into 4 small congruent triangles. Adding the original triangle itself gives $4 + 1 = 5$. (The medial triangle's own three sides create no further triangle.)

10. (B) 13 — Upward-pointing triangles: 6 of unit size + 3 of side 2 + 1 of side 3 = 10. Downward-pointing triangles: 3 of unit size. Total = $10 + 3 = 13$.

11. (A) 14 — Squares of side 1 unit = 9; of side 2 units = 4; of side 3 units = 1. Total = $9 + 4 + 1 = 14$. (General rule: for an $n \times n$ grid, total squares = $1^2 + 2^2 + \dots + n^2$.)

12. (C) 30 — For a 4×4 grid: $4^2 + 3^2 + 2^2 + 1^2 = 16 + 9 + 4 + 1 = 30$.

13. (C) 204 — Total = $1^2 + 2^2 + \dots + 8^2 = 8 \times 9 \times 17 / 6 = 204$. (The 1296 is the count of rectangles, not squares.)

14. (B) 18 — There are 4 vertical lines and 3 horizontal lines. Any 2 verticals and any 2 horizontals bound one rectangle: $C(4,2) \times C(3,2) = 6 \times 3 = 18$.

15. (D) 1296 — A chessboard has 9 horizontal and 9 vertical lines. Choose 2 of each: $C(9,2) \times C(9,2) = 36 \times 36 = 1296$.

16. (C) T — A letter is unchanged by lateral inversion only if it has a vertical axis of symmetry. T does; F, G and J do not. The full set of such capitals is A, H, I, M, O, T, U, V, W, X, Y.

17. (B) W — W has a vertical axis of symmetry, so lateral inversion leaves it unaltered. B, C and E have a horizontal axis of symmetry instead, so they are unchanged in a water image, not in a mirror image.

18. (C) 6 — Mirror-invariant capitals are A, H, I, M, O, T, U, V, W, X, Y. In M-O-U-N-T-A-I-N the letters M, O, U, T, A and I qualify (6 letters); the two N's do not.

19. (C) GNINOSAER — Lateral inversion reverses the order of the letters. Reading REASONING backwards letter by letter gives G-N-I-N-O-S-A-E-R.

20. (A) 8:20 — Mirror time and true time add up to 11:60 (i.e. 12:00). So true time = $11:60 - 3:40 = 8:20$.

21. (B) 7:10 — Mirror image of a clock time = 11:60 minus the time. $11:60 - 4:50 = 7:10$.

- 22. (A) TOOT** — For a word to be unchanged, every letter must be mirror-symmetric AND the word must read the same backwards. TOOT uses only T and O and is a palindrome, so it qualifies. MOTOR, HIM and AXE fail on one or both counts.
- 23. (A) D** — A water image is a reflection about a horizontal axis, so only letters with a horizontal axis of symmetry survive: B, C, D, E, H, I, K, O, X. D qualifies; A, V and T are mirror-symmetric instead.
- 24. (D) 6** — Water-invariant capitals are B, C, D, E, H, I, K, O, X. Every letter of C-H-O-I-C-E belongs to this set, so all 6 letters are unchanged and the water image also reads CHOICE.
- 25. (D) 4** — Water-invariant capitals are B, C, D, E, H, I, K, O, X. From the given six, B, H, O and X qualify; A and M are mirror-symmetric, not water-symmetric. Hence 4.
- 26. (C) H, I, O, X** — Mirror-invariant set: A H I M O T U V W X Y. Water-invariant set: B C D E H I K O X. Their intersection (letters with both a vertical and a horizontal axis of symmetry) is H, I, O, X.
- 27. (D) 4** — Each fold doubles the number of layers: after 2 folds there are $2^2 = 4$ layers, and one punch pierces every layer, giving 4 holes.
- 28. (D) 8** — Number of layers after n folds = 2^n . Here $n = 3$, so $2^3 = 8$ layers and hence 8 holes.
- 29. (C) 8** — Two folds give $2^2 = 4$ layers. Each punch makes 4 holes, so 2 punches make $2 \times 4 = 8$ holes.
- 30. (A) 6** — A face of a cube touches exactly four faces and is opposite to the fifth. Since 1 touches 2, 3, 4 and 5, the only number left is 6, so 6 is opposite 1.
- 31. (A) 5** — Opposite faces sum to 7, so the face opposite 2 bears $7 - 2 = 5$.
- 32. (D) 4** — In both positions 1 is on top, so 2, 3, 4 and 5 all lie on the four side faces around 1 and 6 must be opposite 1. Rotating from Position I to Position II by a quarter turn about the vertical axis takes 2 to 4's place and 3 to 5's place, so 2 and 4 lie on opposite side faces, and so do 3 and 5. Hence 4 is opposite 2.
- 33. (D) 4** — The two positions are consistent with exactly one dice, in which the opposite pairs are (1, 5), (2, 6) and (3, 4). Hence 4 lies opposite 3.
- 34. (D) 5** — These two views are consistent with exactly one dice, whose opposite pairs are (1, 5), (2, 4) and (3, 6). Hence 5 lies opposite 1.
- 35. (B) E** — A is on top in both positions, so F must be at the bottom, opposite A. The quarter turn about the vertical axis that carries B to D's place carries C to E's place, so B is opposite D and C is opposite E.
- 36. (B) 24** — $64 = 4^3$, so the cube is 4 units on a side. Cubes with exactly two painted faces lie along the edges but not at the corners: $12 \text{ edges} \times (4 - 2) = 12 \times 2 = 24$.
- 37. (D) 54** — $125 = 5^3$, so the side is 5 units. One-face cubes are the interior of each face: $6 \times (5 - 2)^2 = 6 \times 9 = 54$.
- 38. (A) 8** — Only the corner cubes show three painted faces, and a cube always has 8 corners, whatever the number of cuts. Hence 8.
- 39. (C) 8** — $64 = 4^3$. Unpainted cubes form the inner block of side $(4 - 2) = 2$, so their number is $2^3 = 8$.
- 40. (C) 26** — Only the single cube at the centre of the $3 \times 3 \times 3$ block is untouched by paint, so $27 - 1 = 26$ smaller cubes have at least one painted face.
- 41. (A) 32** — The cube is $4 \times 4 \times 4$. Only the top layer and the bottom layer touch paint, and since the two painted faces are opposite, no small cube can get two painted faces. Each layer has $4 \times 4 = 16$ cubes, so $2 \times 16 = 32$ cubes have exactly one painted face.
- 42. (B) 20** — Number of diagonals of an n -sided polygon = $n(n - 3)/2$. For $n = 8$ this is $8 \times 5 / 2 = 20$.
- 43. (B) 720 degrees** — Sum of interior angles = $(n - 2) \times 180$ degrees. For $n = 6$ this is $4 \times 180 = 720$ degrees.
- 44. (A) 10** — Each exterior angle = $180 - 144 = 36$ degrees. Number of sides = $360 / 36 = 10$.
- 45. (C) Trapezium** — A rhombus, a square and a rectangle are all parallelograms, i.e. both pairs of opposite sides are parallel. A trapezium has only one pair of parallel sides, so it is the odd one out.

RM09 — EXPLANATIONS

1. **(A) Rs 25** — $SI = 10000 \times 5 \times 2 / 100 = \text{Rs } 1,000$. Amount at CI = $10000 \times (1.05)^2 = \text{Rs } 11,025$, so CI = Rs 1,025. Difference = $1025 - 1000 = \text{Rs } 25$. Shortcut: difference for 2 years = $P(R/100)^2 = 10000 \times (0.05)^2 = \text{Rs } 25$.
2. **(B) Rs 80** — $SI = 8000 \times 10 \times 2 / 100 = \text{Rs } 1,600$. CI = $8000 \times (1.1)^2 - 8000 = 9680 - 8000 = \text{Rs } 1,680$. Difference = Rs 80. By formula, $P(R/100)^2 = 8000 \times 0.01 = \text{Rs } 80$.
3. **(C) Rs 122** — $SI = 16000 \times 5 \times 3 / 100 = \text{Rs } 2,400$. Amount at CI = $16000 \times (1.05)^3 = \text{Rs } 18,522$, so CI = Rs 2,522. Difference = Rs 122. Formula for 3 years: $P \times R^2 \times (300 + R) / 10^6 = 16000 \times 25 \times 305 / 1000000 = \text{Rs } 122$.
4. **(D) Rs 310** — $SI = 10000 \times 10 \times 3 / 100 = \text{Rs } 3,000$. Amount at CI = $10000 \times (1.1)^3 = \text{Rs } 13,310$, so CI = Rs 3,310. Difference = Rs 310. Formula: $P \times R^2 \times (300 + R) / 10^6 = 10000 \times 100 \times 310 / 1000000 = \text{Rs } 310$.
5. **(C) Rs 20,000** — For 2 years, difference = $P(R/100)^2$. So $32 = P \times (4/100)^2 = P \times 0.0016$, giving $P = 32 / 0.0016 = \text{Rs } 20,000$.
6. **(D) 12%** — Difference for 2 years = $P(R/100)^2$, so $144 = 10000 \times (R/100)^2$. Hence $(R/100)^2 = 0.0144$, $R/100 = 0.12$ and $R = 12\%$ per annum.
7. **(B) Rs 2,000** — For 3 years, difference = $P \times R^2 \times (300 + R) / 10^6 = P \times 100 \times 310 / 1000000 = 0.031P$. So $0.031P = 62$ and $P = \text{Rs } 2,000$.
8. **(D) Rs 160** — $SI = 25000 \times 8 \times 2 / 100 = \text{Rs } 4,000$. CI = $25000 \times (1.08)^2 - 25000 = 29160 - 25000 = \text{Rs } 4,160$. Difference = Rs 160, which equals $P(R/100)^2 = 25000 \times 0.0064$.
9. **(B) 64 litres** — After each replacement the milk left = previous milk $\times (1 - 20/100)$. So milk after 2 operations = $100 \times (4/5)^2 = 100 \times 16/25 = 64$ litres. General rule: milk left = $x(1 - y/x)^n$.
10. **(A) 9 : 7** — Milk left = $64 \times (1 - 16/64)^2 = 64 \times (3/4)^2 = 36$ litres. Water = $64 - 36 = 28$ litres. Ratio = $36 : 28 = 9 : 7$.
11. **(D) 1 : 3** — Water costs Rs 0 per litre. By alligation, water : milk = $(32 - 24) : (24 - 0) = 8 : 24 = 1 : 3$.
12. **(C) 1 : 4** — Since he gains 25% while selling at Rs 40, the cost of one litre of the mixture = $40 / 1.25 = \text{Rs } 32$. By alligation between water (Rs 0) and milk (Rs 40) at a mean of Rs 32, water : milk = $(40 - 32) : (32 - 0) = 8 : 32 = 1 : 4$.
13. **(A) 7 : 3** — Take 5 litres from each vessel. From A: 4 litres milk, 1 litre water. From B: 3 litres milk, 2 litres water. Total milk = 7, total water = 3, so the ratio is 7 : 3.
14. **(A) 20 litres** — Milk = $60 \times 2/3 = 40$ litres, water = 20 litres. For a 1 : 1 ratio the water must also become 40 litres, so $40 - 20 = 20$ litres of water must be added.
15. **(A) 10 litres** — Milk = $40 \times 3/4 = 30$ litres, water = 10 litres. Milk stays 30 litres, so for a 3 : 2 ratio the water must be $30 \times 2/3 = 20$ litres. Water to be added = $20 - 10 = 10$ litres.
16. **(D) 3 : 2** — By the rule of alligation, (quantity of cheaper) : (quantity of dearer) = $(45 - 36) : (36 - 30) = 9 : 6 = 3 : 2$.
17. **(C) 6** — Let the numbers be $3x$ and $5x$. Then $(3x + 4)/(5x + 4) = 5/7$, so $21x + 28 = 25x + 20$, giving $4x = 8$ and $x = 2$. The numbers are 6 and 10; the smaller is 6.
18. **(C) Rs 700** — Total number of parts = $2 + 3 + 7 = 12$. One part = $1200/12 = \text{Rs } 100$. C gets 7 parts = Rs 700.
19. **(B) 96** — Let the numbers be $5x$ and $8x$. Then $8x - 5x = 3x = 36$, so $x = 12$. The numbers are 60 and 96; the larger is 96.
20. **(D) 24** — The mean proportional between a and b is the square root of ab . Here it is the square root of $9 \times 64 = 576$, which is 24.
21. **(A) 18** — If x is the third proportional then $8 : 12 :: 12 : x$, so $8x = 144$ and $x = 18$. In general the third proportional to a and b is b^2/a .
22. **(C) 27** — If x is the fourth proportional then $4 : 9 :: 12 : x$, so $4x = 108$ and $x = 27$.
23. **(D) 8 : 15** — Make b common: $a : b = 2 : 3 = 8 : 12$ and $b : c = 4 : 5 = 12 : 15$. Hence $a : b : c = 8 : 12 : 15$ and $a : c = 8 : 15$.
24. **(C) Rs 1,500** — Make B common: $A : B = 15 : 20$ and $B : C = 20 : 24$, so $A : B : C = 15 : 20 : 24$ with 59 parts in all. One part = $5900/59 = \text{Rs } 100$, so A gets 15 parts = Rs 1,500.

- 25. (A) Rs 4,000** — Since the time is the same, profit is shared in the ratio of capitals: $12000 : 16000 = 3 : 4$. B's share = $7000 \times \frac{4}{7} = \text{Rs } 4,000$.
- 26. (C) Rs 10,000** — Profit is shared in the ratio of capital $\times$ time: $A = 5000 \times 12 = 60,000$ and $B = 6000 \times 8 = 48,000$, i.e. $5 : 4$. A's share = $18000 \times \frac{5}{9} = \text{Rs } 10,000$.
- 27. (B) Rs 8,000** — Capital $\times$ time: $A = 3 \times 6 = 18$ units, $B = 2 \times 12 = 24$ units, so the profit ratio is $18 : 24 = 3 : 4$. B's share = $14000 \times \frac{4}{7} = \text{Rs } 8,000$.
- 28. (A) Rs 10,000** — The capital ratio is $20000 : 15000 : 25000 = 4 : 3 : 5$, a total of 12 parts. One part = $24000/12 = \text{Rs } 2,000$, so C gets 5 parts = $\text{Rs } 10,000$.
- 29. (C) Rs 12,800** — A's management fee = 10% of $20000 = \text{Rs } 2,000$. The remaining $\text{Rs } 18,000$ is divided in the ratio $30000 : 20000 = 3 : 2$, so A gets $18000 \times \frac{3}{5} = \text{Rs } 10,800$. A's total = $2000 + 10800 = \text{Rs } 12,800$.
- 30. (C) Rs 12,000** — Capital $\times$ time: $A = 8000 \times 12 = 96,000$ and $B = 9000 \times 8 = 72,000$, i.e. $4 : 3$. A's share = $21000 \times \frac{4}{7} = \text{Rs } 12,000$.
- 31. (A) 96** — 15% of $640 = 640 \times \frac{15}{100} = 6.4 \times 15 = 96$.
- 32. (B) 20%** — Required reduction = $100 \times \frac{R}{100 + R} = 100 \times \frac{25}{125} = 20\%$. Check: if 100 units at Re 1 cost Rs 100, then at Rs 1.25 the family can buy only 80 units, a 20% cut.
- 33. (A) 32%** — Net change = $a + b + \frac{ab}{100} = 20 + 10 + \frac{(20 \times 10)}{100} = 32\%$. Check: 100 becomes 120, and 120 becomes 132.
- 34. (B) A decrease of 4%** — Net change = $a + b + \frac{ab}{100}$ with $a = +20$ and $b = -20$, giving $20 - 20 - \frac{400}{100} = -4\%$, i.e. a 4% decrease. (Rule: an $x\%$ rise followed by an $x\%$ fall always gives a net fall of $\frac{x^2}{100}$ per cent.)
- 35. (B) Rs 24,750** — After the rise: $25000 \times 1.10 = \text{Rs } 27,500$. After the cut: $27500 \times 0.90 = \text{Rs } 24,750$. (Equivalently, a net fall of $\frac{10^2}{100} = 1\%$, i.e. $\text{Rs } 250$.)
- 36. (A) 20%** — Let $B = 100$, then $A = 125$. B is less than A by 25, which as a percentage of A is $\frac{25}{125} \times 100 = 20\%$. Formula: $\frac{100x}{100 + x}$.
- 37. (B) 6000** — The winner got 60% and the loser 40%, so the margin is 20% of the total. Thus 20% of the total = 1,200, and the total = $1200 \times \frac{100}{20} = 6,000$ votes.
- 38. (C) 400** — The shortfall of 20 marks equals $35\% - 30\% = 5\%$ of the maximum. So 5% of the maximum = 20, and the maximum = $20 \times \frac{100}{5} = 400$ marks. (Pass mark = 140.)
- 39. (A) 24200** — Population after n years = $P(1 + \frac{r}{100})^n = 20000 \times (1.1)^2 = 20000 \times 1.21 = 24,200$.
- 40. (A) Rs 40,500** — Depreciated value = $P(1 - \frac{r}{100})^n = 50000 \times (0.9)^2 = 50000 \times 0.81 = \text{Rs } 40,500$.
- 41. (B) 2 : 3** — $\frac{60A}{100} = \frac{40B}{100}$ gives $60A = 40B$, so $A/B = \frac{40}{60} = \frac{2}{3}$, i.e. $A : B = 2 : 3$.
- 42. (A) 25%** — Profit = $500 - 400 = \text{Rs } 100$. Profit % = $(\text{Profit} / \text{CP}) \times 100 = \frac{100}{400} \times 100 = 25\%$. (Profit percentage is always taken on the cost price.)
- 43. (C) 20%** — Loss = $250 - 200 = \text{Rs } 50$. Loss % = $(\text{Loss} / \text{CP}) \times 100 = \frac{50}{250} \times 100 = 20\%$.
- 44. (A) Rs 600** — $\text{CP} = \text{SP} \times \frac{100}{100 + \text{profit \%}} = 690 \times \frac{100}{115} = \text{Rs } 600$. Check: 15% of $600 = 90$ and $600 + 90 = 690$.
- 45. (B) Rs 960** — $\text{SP} = \text{CP} \times \frac{100 - \text{loss \%}}{100} = 1200 \times \frac{80}{100} = \text{Rs } 960$.
- 46. (B) Rs 720** — $\text{SP} = 1000 \times 0.80 \times 0.90 = \text{Rs } 720$. (Equivalent single discount = $20 + 10 - \frac{200}{100} = 28\%$.)
- 47. (D) 12.5%** — Marked price = $300 \times 1.25 = \text{Rs } 375$. Selling price = $375 \times 0.90 = \text{Rs } 337.50$. Profit = $\text{Rs } 37.50$, so profit % = $\frac{37.5}{300} \times 100 = 12.5\%$.
- 48. (C) A loss of Rs 100** — CP of the first = $1200 \times \frac{100}{120} = \text{Rs } 1,000$; CP of the second = $1200 \times \frac{100}{80} = \text{Rs } 1,500$. Total CP = $\text{Rs } 2,500$ against a total SP of $\text{Rs } 2,400$, so there is a loss of $\text{Rs } 100$. (Rule: selling two items at the same price with equal % gain and loss always gives a loss of $\frac{x^2}{100}$ per cent, here 4%.)
- 49. (D) 25%** — Let the CP of one article be Re 1. Then SP of 16 articles = CP of 20 articles = $\text{Rs } 20$, so SP of one article = $\frac{20}{16} = \text{Rs } 1.25$. Profit % = $\frac{0.25}{1} \times 100 = 25\%$. Formula: $\frac{(x - y)}{y} \times 100 = \frac{(20 - 16)}{16} \times 100$.
- 50. (B) 25%** — He pays for 800 g but charges for 1000 g. Gain % = $\frac{\text{error}}{(\text{true value} - \text{error})} \times 100 = \frac{200}{800} \times 100 = 25\%$.
- 51. (B) Rs 1,100** — $\text{CP} = 900 \times \frac{100}{90} = \text{Rs } 1,000$. For a 10% gain, $\text{SP} = 1000 \times \frac{110}{100} = \text{Rs } 1,100$.

- 52. (D) 5%** — Take CP = Rs 100. Marked price = Rs 140 and SP = $140 \times 0.75 = \text{Rs } 105$. Profit = Rs 5 on Rs 100, i.e. 5%.
- 53. (A) Rs 900** — The extra Rs 90 corresponds to the extra profit of $25\% - 15\% = 10\%$ of the cost price. So 10% of CP = 90 and CP = Rs 900.
- 54. (D) 20%** — Take CP = Rs 100, then profit = Rs 25 and SP = Rs 125. As a percentage of the selling price the profit is $25/125 \times 100 = 20\%$.
- 55. (D) Rs 1,440** — $SI = P \times R \times T/100 = 6000 \times 8 \times 3 / 100 = \text{Rs } 1,440$.
- 56. (C) 12.5%** — To double, the interest must equal the principal, so $SI = P$. From $P = P \times R \times 8/100$ we get $R = 100/8 = 12.5\%$ per annum.
- 57. (C) 5%** — $SI = 3000 - 2500 = \text{Rs } 500$. $R = SI \times 100/(P \times T) = 500 \times 100/(2500 \times 4) = 5\%$ per annum.
- 58. (D) Rs 6,000** — $P = SI \times 100/(R \times T) = 1800 \times 100/(6 \times 5) = 180000/30 = \text{Rs } 6,000$.
- 59. (B) 3** — $T = SI \times 100/(P \times R) = 1200 \times 100/(4000 \times 10) = 120000/40000 = 3$ years.
- 60. (D) Rs 1,800** — The interest for 2 years = $2800 - 2400 = \text{Rs } 400$, so the interest for 1 year = Rs 200 and for 3 years = Rs 600. Principal = $2400 - 600 = \text{Rs } 1,800$.
- 61. (C) 10%** — To become three times, the interest must be twice the principal, so $SI = 2P$. From $2P = P \times R \times 20/100$ we get $R = 200/20 = 10\%$ per annum.
- 62. (D) Rs 2,100** — Amount = $10000 \times (1 + 10/100)^2 = 10000 \times 1.21 = \text{Rs } 12,100$. CI = $12100 - 10000 = \text{Rs } 2,100$.
- 63. (A) Rs 8,820** — Amount = $P(1 + R/100)^n = 8000 \times (1.05)^2 = 8000 \times 1.1025 = \text{Rs } 8,820$.
- 64. (B) Rs 3,972** — Amount = $12000 \times (1.1)^3 = 12000 \times 1.331 = \text{Rs } 15,972$. CI = $15972 - 12000 = \text{Rs } 3,972$. (Simple interest would have been only Rs 3,600.)
- 65. (A) Rs 820** — For half-yearly compounding, halve the rate and double the number of periods: 5% for 2 periods. Amount = $8000 \times (1.05)^2 = \text{Rs } 8,820$, so CI = Rs 820 (against Rs 800 if compounded annually).
- 66. (C) 10** — In 5 years the sum becomes 2P; in the next 5 years that 2P again doubles to 4P. So it takes $5 + 5 = 10$ years. (In general, becoming 2^k times takes 5k years.)
- 67. (B) Rs 6,400** — Value = $P(1 - R/100)^n = 10000 \times (0.8)^2 = 10000 \times 0.64 = \text{Rs } 6,400$.
- 68. (D) 20%** — Amount = $6000 + 2640 = \text{Rs } 8,640$, so $(1 + R/100)^2 = 8640/6000 = 1.44$. Hence $1 + R/100 = 1.2$ and $R = 20\%$ per annum.
- 69. (D) Rs 400** — Total paid = $4000 + 4200 + 4200 = \text{Rs } 12,400$. Excess over the cash price = $12400 - 12000 = \text{Rs } 400$.
- 70. (B) Rs 1,210** — If each instalment is x, its present value is $x/1.1$ for the first year and $x/(1.1)^2$ for the second. So $x/1.1 + x/1.21 = 2100$, i.e. $x(110 + 100)/121 = 2100$, giving $210x = 2100 \times 121$ and $x = \text{Rs } 1,210$. (Check: $1210/1.1 = 1100$ and $1210/1.21 = 1000$, which add to 2,100.)

RM10 – EXPLANATIONS

- 1. (A) 144 cm²** — Perimeter of a square = $4 \times \text{side}$, so side = $48 \div 4 = 12$ cm. Area = $\text{side}^2 = 12 \times 12 = 144$ cm². (Taking side = $48 \div 2 = 24$ cm by mistake gives 576 cm², and squaring the perimeter itself gives 2304 cm².)
- 2. (D) ₹1,260** — Fencing is done along the perimeter. Perimeter = $2(l + b) = 2(24 + 18) = 2 \times 42 = 84$ m. Cost = $84 \times 15 = ₹1,260$. (Using the area $24 \times 18 = 432$ m² instead of the perimeter gives ₹6,480.)
- 3. (B) 126 cm²** — Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 18 \times 14 = \frac{1}{2} \times 252 = 126$ cm². (Forgetting the $\frac{1}{2}$ gives 252 cm²; adding the two numbers gives 32.)
- 4. (A) 84 cm²** — Use Heron's formula. $s = (13 + 14 + 15)/2 = 42/2 = 21$. Area = $\sqrt{s(s-a)(s-b)(s-c)} = \sqrt{21 \times 8 \times 7 \times 6} = \sqrt{7056} = 84$ cm².
- 5. (B) 120 cm²** — Area of a parallelogram = $\text{base} \times \text{height} = 15 \times 8 = 120$ cm². Unlike a triangle there is NO $\frac{1}{2}$ here; halving gives the wrong value 60 cm². (46 cm is the perimeter-type trap $2(15+8)$.)
- 6. (D) 162 cm²** — Area of a trapezium = $\frac{1}{2} \times (\text{sum of parallel sides}) \times \text{distance between them} = \frac{1}{2} \times (22 + 14) \times 9 = \frac{1}{2} \times 36 \times 9 = 162$ cm². (Omitting the $\frac{1}{2}$ gives 324 cm²; using the difference $22 - 14$ gives 36 cm².)

- 7. (D) 44 cm** — Circumference = $2\pi r = 2 \times (22/7) \times 7 = 44$ cm. (154 cm^2 is the AREA πr^2 of the same circle, and 88 cm is the circumference of a circle of radius 14 cm, i.e. what you get if you double the radius by confusing it with the diameter.)
- 8. (C) 616 cm^2** — Radius = diameter $\div 2 = 28/2 = 14$ cm. Area = $\pi r^2 = (22/7) \times 14 \times 14 = 22 \times 2 \times 14 = 616 \text{ cm}^2$. (Using the diameter 28 as the radius gives 2464 cm^2 — the commonest error; 88 cm is the circumference.)
- 9. (C) 462 cm^2** — Area of a sector = $(\theta/360) \times \pi r^2 = (120/360) \times (22/7) \times 21 \times 21 = (1/3) \times 1386 = 462 \text{ cm}^2$. (1386 cm^2 is the whole circle; 44 cm is the arc LENGTH $(\theta/360) \times 2\pi r$.)
- 10. (D) 576 m^2** — Outer rectangle = $(50 + 2 \times 3) \times (40 + 2 \times 3) = 56 \times 46 = 2576 \text{ m}^2$. Inner field = $50 \times 40 = 2000 \text{ m}^2$. Path = $2576 - 2000 = 576 \text{ m}^2$. Shortcut: $2w(l + b) + 4w^2 = 6(90) + 4(9) = 540 + 36 = 576 \text{ m}^2$ — dropping the four corner squares $4w^2$ gives the classic wrong answer 540 m^2 .
- 11. (A) 44%** — Take side = 100, so area = 10000. New side = 120, new area = 14400. Increase = 4400, i.e. $4400/10000 \times 100 = 44\%$. Shortcut for two dimensions: $x + y + xy/100 = 20 + 20 + 400/100 = 44\%$.
- 12. (C) 343 cm^3** — Volume of a cube = $a^3 = 7 \times 7 \times 7 = 343 \text{ cm}^3$. (294 is the total surface area $6a^2 = 6 \times 49$, and 49 is only a^2 .)
- 13. (D) 392 cm^2** — Total surface area = $2(lb + bh + hl) = 2(12 \times 8 + 8 \times 5 + 5 \times 12) = 2(96 + 40 + 60) = 2 \times 196 = 392 \text{ cm}^2$. (480 cm^3 is the VOLUME $l \times b \times h$; 196 is the bracket without doubling.)
- 14. (D) 1540 cm^3** — Volume = $\pi r^2 h = (22/7) \times 7 \times 7 \times 10 = 22 \times 7 \times 10 = 1540 \text{ cm}^3$. (440 cm^2 is the curved surface area $2\pi rh$ and 748 cm^2 is the total surface area $2\pi r(r + h)$.)
- 15. (B) 1760 cm^2** — Curved surface area = $2\pi rh = 2 \times (22/7) \times 14 \times 20 = 2 \times 44 \times 20 = 1760 \text{ cm}^2$. (12320 cm^3 is the volume $\pi r^2 h$; forgetting the 2 gives 880.)
- 16. (C) 550 cm^2** — First the slant height: $l = \sqrt{(r^2 + h^2)} = \sqrt{(49 + 576)} = \sqrt{625} = 25$ cm. Curved surface area = $\pi rl = (22/7) \times 7 \times 25 = 550 \text{ cm}^2$. (704 cm^2 is the TOTAL surface area $\pi r(l + r) = 550 + 154$; 1232 cm^3 is the volume.)
- 17. (A) 66 cm^3** — Volume of a cone = $(1/3)\pi r^2 h = (1/3) \times (22/7) \times 9 \times 7 = (1/3) \times 198 = 66 \text{ cm}^3$. (Forgetting the $1/3$ — i.e. using the cylinder formula — gives 198 cm^3 ; a cone is exactly one-third of the cylinder on the same base and height.)
- 18. (B) 5544 cm^2** — Surface area of a sphere = $4\pi r^2 = 4 \times (22/7) \times 21 \times 21 = 4 \times 22 \times 63 = 5544 \text{ cm}^2$. (1386 cm^2 is only πr^2 , and 38808 cm^3 is the volume $(4/3)\pi r^3$.)
- 19. (D) 462 cm^2** — A solid hemisphere has a curved part $2\pi r^2$ plus a flat circular face πr^2 , so total = $3\pi r^2 = 3 \times (22/7) \times 49 = 3 \times 154 = 462 \text{ cm}^2$. (Leaving out the flat face gives 308 cm^2 ; 616 cm^2 is the full sphere $4\pi r^2$.)
- 20. (A) 125%** — Let the edge be 2 units, so surface area = $6 \times 2^2 = 24$. New edge = 3 units, new surface area = $6 \times 3^2 = 54$. Increase = $54 - 24 = 30$, i.e. $30/24 \times 100 = 125\%$. (Surface area varies as the SQUARE of the edge: $1.5^2 = 2.25$, i.e. 225% of the old — which is an increase of 125%, not 225%.)
- 21. (C) 25** — Sum of the 5 numbers = $5 \times 27 = 135$. After removing 35 the sum = $135 - 35 = 100$ and 4 numbers remain. New average = $100 \div 4 = 25$.
- 22. (B) 76 kg** — Total weight rises by $8 \times 2.5 = 20$ kg. That whole rise is caused by the swap, so the new person = $56 + 20 = 76$ kg. (Adding 2.5 only once gives the trap 58.5 kg.)
- 23. (D) 51** — Total marks = $20 \times 45 + 30 \times 55 = 900 + 1650 = 2550$. Class average = $2550 \div 50 = 51$. (Averaging 45 and 55 directly gives 50 — wrong, because the two groups are of unequal size; this is a weighted average.)
- 24. (A) 48 km/h** — For equal distances the average speed is the harmonic mean: $2xy/(x + y) = (2 \times 60 \times 40)/(60 + 40) = 4800/100 = 48 \text{ km/h}$. (The arithmetic mean 50 km/h is the classic mistake — more time is spent at the slower speed, so the true average must be below 50.)
- 25. (C) 45 years** — Total age of students = $30 \times 14 = 420$ years. With the teacher there are 31 people whose average is 15, so total = $31 \times 15 = 465$. Teacher's age = $465 - 420 = 45$ years. Shortcut: new age = new average + (number of old members $\times$ rise) = $15 + 30 = 45$.
- 26. (B) 38** — Runs in 16 innings = $16 \times 36 = 576$. After the 17th innings total = $576 + 70 = 646$. New average = $646 \div 17 = 38$. Shortcut: the excess over the old average is $70 - 36 = 34$, spread over 17 innings = 2, so the average rises by 2 to 38.
- 27. (A) 20** — Sum = $12 + 15 + 18 + 20 + 22 + 25 + 28 = 140$. There are 7 observations, so mean = $140 \div 7 = 20$.

- 28. (A) 12** — First arrange in ascending order: 7, 9, 11, 12, 15, 18, 21. $n = 7$ (odd), so the median is the value at position $(n+1)/2 = 4$ th, which is 12. Never take the middle of the UNSORTED list — that would wrongly give 9.
- 29. (D) 5** — Count the frequencies: 3 occurs 2 times, 5 occurs 4 times, 7 occurs 2 times, 9 occurs 2 times. The mode is the observation with the highest frequency, i.e. 5.
- 30. (B) 66** — Range = largest value – smallest value = $78 - 12 = 66$. (78 alone is only the maximum; the range is a measure of dispersion, not of position.)
- 31. (D) 27** — $\Sigma f = 3 + 5 + 8 + 3 + 1 = 20$. $\Sigma fx = 10 \times 3 + 20 \times 5 + 30 \times 8 + 40 \times 3 + 50 \times 1 = 30 + 100 + 240 + 120 + 50 = 540$. Mean = $\Sigma fx / \Sigma f = 540 / 20 = 27$. (Averaging the values 10...50 without weights gives 30 — that ignores the frequencies.)
- 32. (B) 2** — Write the cumulative frequencies: 6, 16, 28, 42, 50. $N = 50$ (even), so the median is the average of the 25th and 26th observations. Both the 25th and 26th fall in the class 'cumulative 17 to 28', i.e. the value 2. Median = $(2 + 2)/2 = 2$. (3 is the MODE — it has the highest frequency, 14 — a standard trap.)
- 33. (A) 32** — Empirical relation: Mode = $3 \times \text{Median} - 2 \times \text{Mean} = 3(28) - 2(26) = 84 - 52 = 32$. (The equivalent form is Mean – Mode = $3(\text{Mean} - \text{Median})$.)
- 34. (B) ₹12,000** — A full pie chart is 360° , standing for ₹36,000, so $1^\circ = 36000/360 = ₹100$. Food has 120° , hence $120 \times 100 = ₹12,000$. (Equivalently, $120/360 = 1/3$ of ₹36,000.)
- 35. (D) ₹3,000** — $1^\circ = 36000/360 = ₹100$. House rent = $90 \times 100 = ₹9,000$; education = $60 \times 100 = ₹6,000$. Difference = $9000 - 6000 = ₹3,000$. (Or directly: $(90 - 60) \times 100 = ₹3,000$.)
- 36. (D) 37.5%** — House rent + transport = $90^\circ + 45^\circ = 135^\circ$. As a percentage of the whole pie: $(135/360) \times 100 = 37.5\%$. (In rupees this is ₹13,500 out of ₹36,000, the same 37.5%.)
- 37. (A) 30 lakh tonnes** — Total = $24 + 30 + 27 + 36 + 33 = 150$ lakh tonnes over 5 years. Average = $150 / 5 = 30$ lakh tonnes.
- 38. (A) 25%** — Increase = $30 - 24 = 6$ lakh tonnes. Percentage increase = $(\text{increase} \div \text{ORIGINAL value}) \times 100 = (6/24) \times 100 = 25\%$. (Dividing by the new value 30 gives the wrong 20%.)
- 39. (A) 2** — The five-year average is $150 / 5 = 30$ lakh tonnes. Comparing: 24 (below), 30 (equal, not above), 27 (below), 36 (above), 33 (above). Hence exactly 2 years — 2022 and 2023.
- 40. (C) 27 : 23** — Total boys = $240 + 180 + 300 + 150 + 210 = 1080$. Total girls = $160 + 220 + 200 + 150 + 190 = 920$. Ratio = $1080 : 920$; dividing both by 40 gives $27 : 23$.
- 41. (B) School B** — Compute the share of girls in each school: A = $160/400 = 40\%$, B = $220/400 = 55\%$, C = $200/500 = 40\%$, D = $150/300 = 50\%$, E = $190/400 = 47.5\%$. The highest is School B with 55%. (School C has the largest NUMBER of girls after B, but the largest number is not the largest percentage.)
- 42. (A) 120** — The word has 5 distinct letters, so the number of arrangements = $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$. (24 is $4!$ and 720 is $6!$.)
- 43. (C) 60** — There are 6 letters with A repeated 3 times and N repeated 2 times. Arrangements = $6!/(3! \times 2!) = 720/(6 \times 2) = 720/12 = 60$. (Treating all letters as distinct gives 720 — the standard error.)
- 44. (D) 66** — A handshake needs an unordered pair, so the count is $12C2 = (12 \times 11)/(2 \times 1) = 132/2 = 66$. (Using $12 \times 11 = 132$ counts each handshake twice, i.e. it is the permutation $12P2$.)
- 45. (C) 200** — Choose the men: $6C3 = (6 \times 5 \times 4)/(3 \times 2 \times 1) = 20$ ways. Choose the women: $5C2 = (5 \times 4)/(2 \times 1) = 10$ ways. The two choices are independent, so multiply: $20 \times 10 = 200$ ways. (Adding instead of multiplying gives 30.)
- 46. (B) 3/4** — Sample space = {HH, HT, TH, TT}, so $n(S) = 4$. 'At least one head' happens in HH, HT, TH, i.e. 3 cases. $P = 3/4$. (Alternatively $P(\text{at least one head}) = 1 - P(\text{no head}) = 1 - 1/4 = 3/4$.)
- 47. (B) 3/8** — With 3 coins $n(S) = 2^3 = 8$: HHH, HHT, HTH, THH, HTT, THT, TTH, TTT. Exactly two heads occurs in HHT, HTH, THH = 3 cases. $P = 3/8$. ($1/8$ is the probability of exactly three heads.)
- 48. (C) 1/2** — The faces are 1, 2, 3, 4, 5, 6, so $n(S) = 6$. The primes among them are 2, 3 and 5 — that is 3 outcomes (1 is NOT prime). $P = 3/6 = 1/2$. (Counting only 2 and 3 gives the wrong $1/3$.)
- 49. (A) 5/36** — Two dice give $n(S) = 6 \times 6 = 36$ equally likely outcomes. Sum 8 arises from (2,6), (3,5), (4,4), (5,3), (6,2) = 5 outcomes. $P = 5/36$. (Counting the unordered pairs 2+6, 3+5, 4+4 as only 3 cases gives the wrong $1/12$.)

- 50. (C) 1/6** — $n(S) = 36$. The doublets are (1,1), (2,2), (3,3), (4,4), (5,5), (6,6) = 6 outcomes. $P = 6/36 = 1/6$. (1/36 is the probability of one particular doublet, such as (5,5).)
- 51. (D) 1/3** — $n(S) = 36$. Sums that are multiples of 3 are 3, 6, 9 and 12. Number of ways: sum 3 → 2 ways, sum 6 → 5 ways, sum 9 → 4 ways, sum 12 → 1 way; total $2 + 5 + 4 + 1 = 12$. $P = 12/36 = 1/3$.
- 52. (B) 1/13** — A pack has 4 kings out of 52 cards, so $P = 4/52 = 1/13$. (1/26 would be the probability of a particular king of a particular colour pair, e.g. a red king is $2/52 = 1/26$.)
- 53. (C) 3/26** — Face cards are the Jack, Queen and King. The red suits are hearts and diamonds, giving $3 \times 2 = 6$ red face cards. $P = 6/52 = 3/26$. ($3/13 = 12/52$ counts ALL face cards, ignoring the colour condition.)
- 54. (B) 7/12** — Total balls = $5 + 4 + 3 = 12$. Balls that are not red = $4 + 3 = 7$. $P(\text{not red}) = 7/12$. (Equivalently $1 - 5/12 = 7/12$; 5/12 is the probability that it IS red.)
- 55. (D) 5/18** — Total ways of drawing 2 balls from 9 = ${}^9C_2 = (9 \times 8)/2 = 36$. Ways of drawing 2 black from 5 = ${}^5C_2 = (5 \times 4)/2 = 10$. $P = 10/36 = 5/18$. ($25/81$ comes from treating the draws as independent WITH replacement, i.e. $(5/9)^2$.)
- 56. (A) 12 seconds** — Convert: $54 \text{ km/h} = 54 \times 5/18 = 15 \text{ m/s}$. To cross a pole the train covers only its own length, 180 m. Time = $180 \div 15 = 12$ seconds.
- 57. (B) 30 seconds** — $72 \text{ km/h} = 72 \times 5/18 = 20 \text{ m/s}$. To cross a platform the train covers its own length plus the platform: $240 + 360 = 600 \text{ m}$. Time = $600 \div 20 = 30$ seconds. (Using only the train's length, 240 m, gives the wrong 12 seconds.)
- 58. (D) 5 hours** — Speed = $210 \div 3 = 70 \text{ km/h}$. Time for 350 km = $350 \div 70 = 5$ hours. (Since distance and time are directly proportional: $3 \times 350/210 = 5$ hours.)
- 59. (D) 32 km** — Walking in opposite directions, the gap grows at the SUM of the speeds: $5 + 3 = 8 \text{ km/h}$. In 4 hours the gap = $8 \times 4 = 32 \text{ km}$. (If they walked in the same direction the gap would grow at $5 - 3 = 2 \text{ km/h}$, giving 8 km.)
- 60. (B) 5 km** — Let the distance be $d \text{ km}$. The two walking times differ by $6 + 4 = 10$ minutes = $1/6$ hour. So $d/5 - d/6 = 1/6 \rightarrow (6d - 5d)/30 = 1/6 \rightarrow d/30 = 1/6 \rightarrow d = 5 \text{ km}$. (Check: $5/5 = 1 \text{ h}$ and $5/6 \text{ h} = 50 \text{ min}$ — a difference of 10 minutes.)
- 61. (A) 8 days** — In one day A does $1/12$ of the work and B does $1/24$. Together per day = $1/12 + 1/24 = 2/24 + 1/24 = 3/24 = 1/8$. Hence they finish in 8 days. (Averaging 12 and 24 to get 18 days is wrong — the joint time must be less than either time alone.)
- 62. (A) 30 days** — One day's work: $(A + B) = 1/10$ and $A = 1/15$. So $B = 1/10 - 1/15 = 3/30 - 2/30 = 1/30$. B alone takes 30 days. (Subtracting the DAYS, $15 - 10 = 5$, is a standard error — always subtract the one-day work, not the times.)
- 63. (C) 8 days** — Men and days are inversely proportional, so $M_1 \times D_1 = M_2 \times D_2$. Total work = $12 \times 18 = 216$ man-days. Days for 27 men = $216 \div 27 = 8$ days.
- 64. (B) 144 minutes** — In one hour A fills $1/6$ and B fills $1/4$ of the tank; together $1/6 + 1/4 = 2/12 + 3/12 = 5/12$ per hour. Time = $12/5$ hours = 2.4 hours = 2 hours 24 minutes = 144 minutes.
- 65. (C) 30 hours** — Filling is positive, emptying negative. Net work in one hour = $1/10 - 1/15 = 3/30 - 2/30 = 1/30$ of the tank. Hence the tank fills in 30 hours. (Adding instead of subtracting gives 6 hours.)
- 66. (C) Sunday** — From 15 August 2026 to 15 August 2027 the period contains February 2027, and 2027 is not a leap year, so it is an ordinary year of 365 days. $365 = 52 \text{ weeks} + 1 \text{ day}$, i.e. 1 odd day. Saturday + 1 = Sunday.
- 67. (B) Thursday** — Odd days: 1600 years → 0, 300 years → 1, then 49 completed years (1901-1949) contain 12 leap years and 37 ordinary years → $12 \times 2 + 37 \times 1 = 61 \text{ days} = 8 \text{ weeks} + 5 \text{ odd days}$. January up to the 26th → 26 days = 3 weeks + 5 odd days. Total odd days = $0 + 1 + 5 + 5 = 11 \rightarrow 11 - 7 = 4$, and 4 odd days from Sunday gives Thursday. 26 January 1950 was a Thursday.
- 68. (C) 0** — 100 years contain 24 leap years and 76 ordinary years = $24 \times 2 + 76 \times 1 = 124 \text{ days} = 17 \text{ weeks} + 5 \text{ odd days}$. So 200 years → 10 → 3 odd days, 300 years → 15 → 1 odd day, 400 years → 20 + 1 (because the 400th year is itself a leap year) = 21 days = exactly 3 weeks → 0 odd days. Hence the calendar repeats every 400 years.
- 69. (C) 75°** — Use the formula $|30H - 5.5M|$, where $H = 3$ and $M = 30$: $|30(3) - 5.5(30)| = |90 - 165| = 75^\circ$. (Reading it as 90° forgets that by 3:30 the hour hand has itself moved half-way from 3 towards 4.)

70. (A) 21 9/11 minutes past 4 — At 4 o'clock the hour hand is at 120° and the minute hand at 0° , so the minute hand must close a gap of 120° . The minute hand moves at 6° per minute and the hour hand at 0.5° per minute, so the gap closes at $6 - 0.5 = 5.5^\circ$ per minute. Time = $120 \div 5.5 = 240/11 = 21 \frac{9}{11}$ minutes past 4. (In minute-spaces: the minute hand must gain 20 spaces, needing $20 \times 60/55 = 240/11$ minutes.)
